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Introduction

Lab overview

Welcome to the watsonx Orchestrate flow and document processing lab guide. This guide is designed to help both beginners and experienced users alike, providing a step-by-step overview of the current capabilities of watsonx Orchestrate's flow and document processing capabilities.

When complete, the agent created will be able to classify documents, distinguish invoices from contracts and extract specific fields from these documents depending on their type.

What you will learn

Once you have completed this lab you will have developed skills enabling you to complete the following tasks:

- How to set up agents that use document classification
- How to identify and extract key data fields from uploaded documents
- Prompt the Model to identify key data in contracts and invoices
- Create flows inside your agent to manage complex task routing
- Test the agent you have created

How to get support

If you encounter an issue with this lab, you can request assistance through the following channels:

- IBMers can use the #ba-techlcd-support Slack channel.
- If you are an IBM Business Partner and require assistance, please open a support case at IBM Technology Zone Help.

Prerequisites

You should have completed the other labs in this learning plan:

- Completed the preparation guide activities
- Have access to a watsonx Orchestrate SaaS instance.
- You have downloaded the following files on your local computer:
 - [Contract.pdf](#)
 - [Utility bill training.pdf](#)
 - [Utility bill 2.png](#)
 - [Utility bill 4.png](#)
 - [Utility bill 5.png](#)

Known Issues and Workarounds

As of the time of writing this lab, we have identified several issues that may affect your experience. Please read the following carefully to understand how to work around them:

- **Adding a node to a link:** In certain cases, the '+' icon used to add a node to a link may not appear—especially after adding the first extractor to the workflow. Instead, small moving arrows may be displayed, preventing you from adding any elements to the link.
Workaround: Exit the workflow editor and reopen it.
- **Extracting fields from documents:** After creating a field or updating its definition, the UI may take time to display the newly extracted value—or may not show it at all. The same issue can occur when adding examples and clicking the *Show in document* button.
Workaround: Wait a few seconds for the screen to refresh or click your browser's refresh button.
- **Fields showing as “Not found”:** Occasionally, fields may appear as “Not found in the document” even though they were previously extracted correctly.
Workaround: Wait a few seconds to allow the LLM to return the correct values and for the UI to update.

Use case

This lab focuses on a practical use case involving document classification, followed by key data extraction from both unstructured documents (primarily contracts) and structured ones (such as invoices). At the outset, the guide outlines the sequence of activities required to support the user journey of extracting essential information from uploaded documents.

In this lab scenario, a payment recovery organization processes numerous contracts and invoices weekly, extracting critical information like payment terms, amounts, and due dates. Currently, this is done manually, consuming significant employee time and increasing the risk of errors. With growing document volumes, the firm faces scalability challenges and inefficiencies in their processes.

Challenges:

- **Manual Process:** Employees spend hours reviewing documents and entering data, which slows down operations.
- **Errors:** Human mistakes lead to discrepancies, delays, and potential client issues.
- **Scalability:** As the business expands, the volume of documents grows, requiring more manual labor, which is costly and unsustainable.

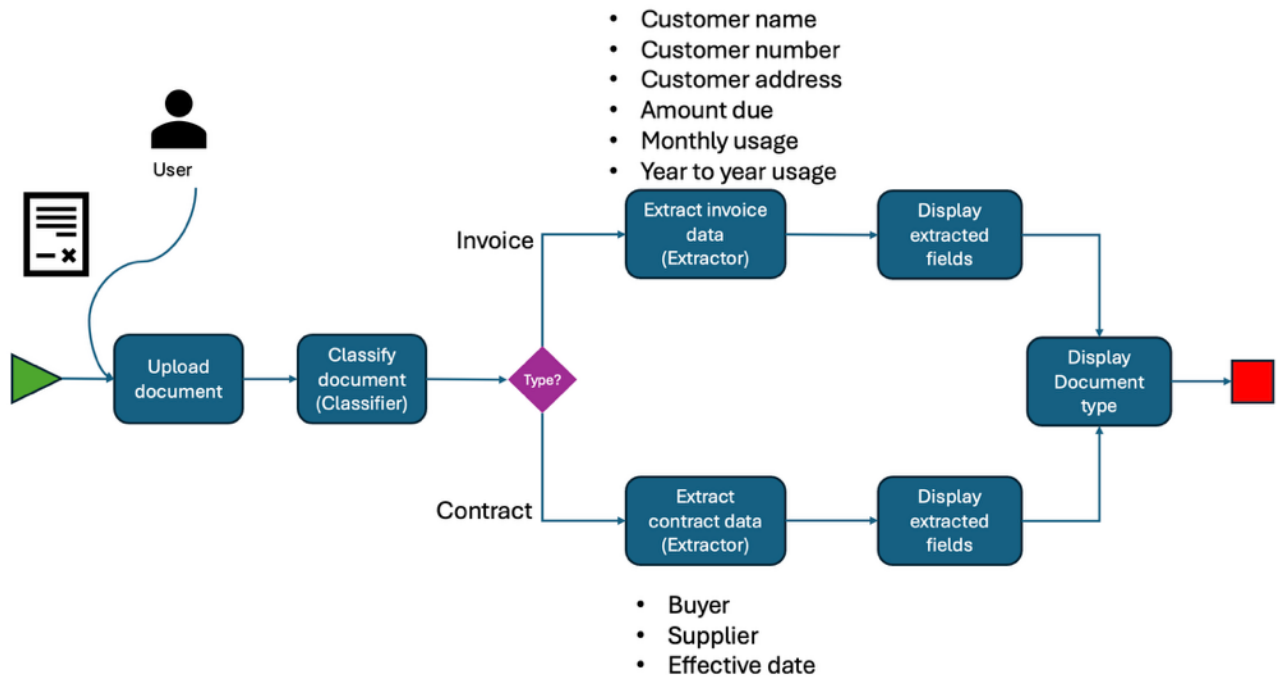
Solution: Watsonx Orchestrate

The firm implements Watsonx Orchestrate to automate the extraction of key data from contracts and invoices, streamlining the process and reducing errors.

- **Automated Extraction:** Watsonx Orchestrate automatically extracts relevant data from documents, eliminating manual entry.
- **Workflow Integration:** The extracted data is seamlessly integrated into the firm's systems (e.g., billing and accounting), speeding up the process.
- **Confidence Review:** If the extraction confidence is low, a review step ensures data accuracy without delays.

The process begins with configuring the document classifier and extractor tools to identify the type of document being ingested. Based on the classification, the system then determines which key-value pairs to extract. Finally, the flow concludes with a summary of the main fields retrieved by the agent during the extraction process, completing the end-to-end user journey.

The diagram below presents the workflow supporting this use case:



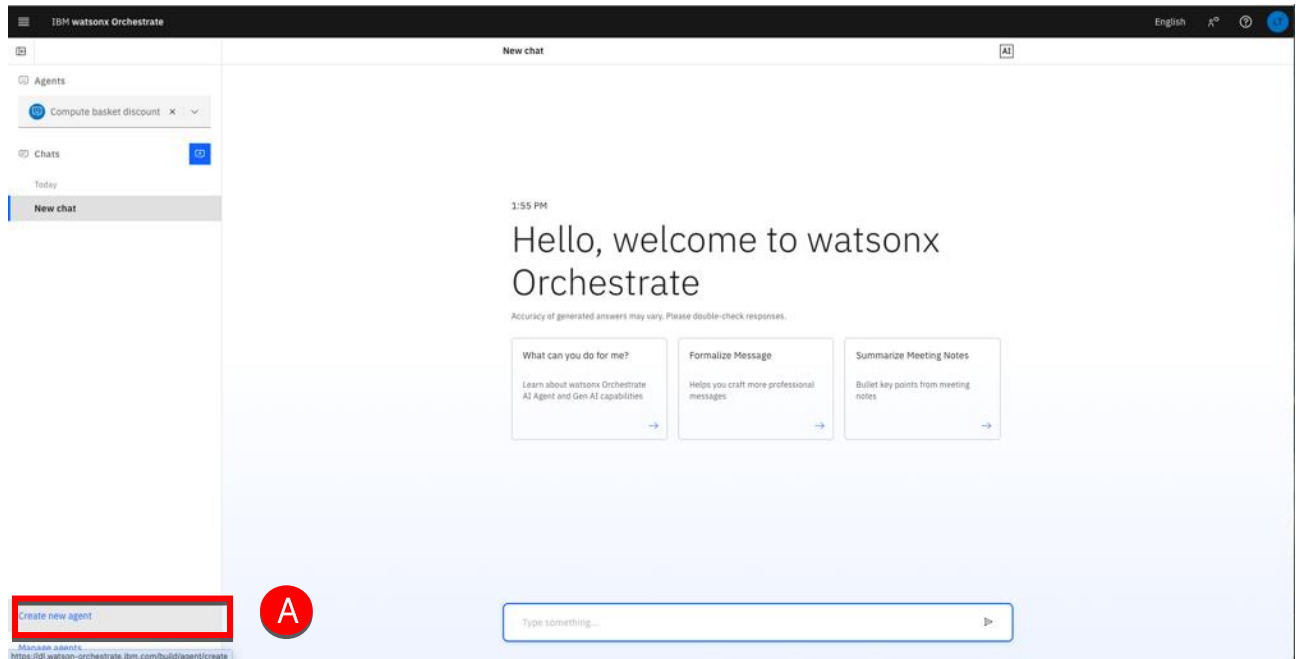
This lab guide will guide you through the different steps to create this flow.

Let's get started.

Create an Agent

In this section, you will create a document processing agent. This agent will implement the process previously described using the watsonx Orchestrate flow capabilities, as well as a Document Classifier and Document extractors.

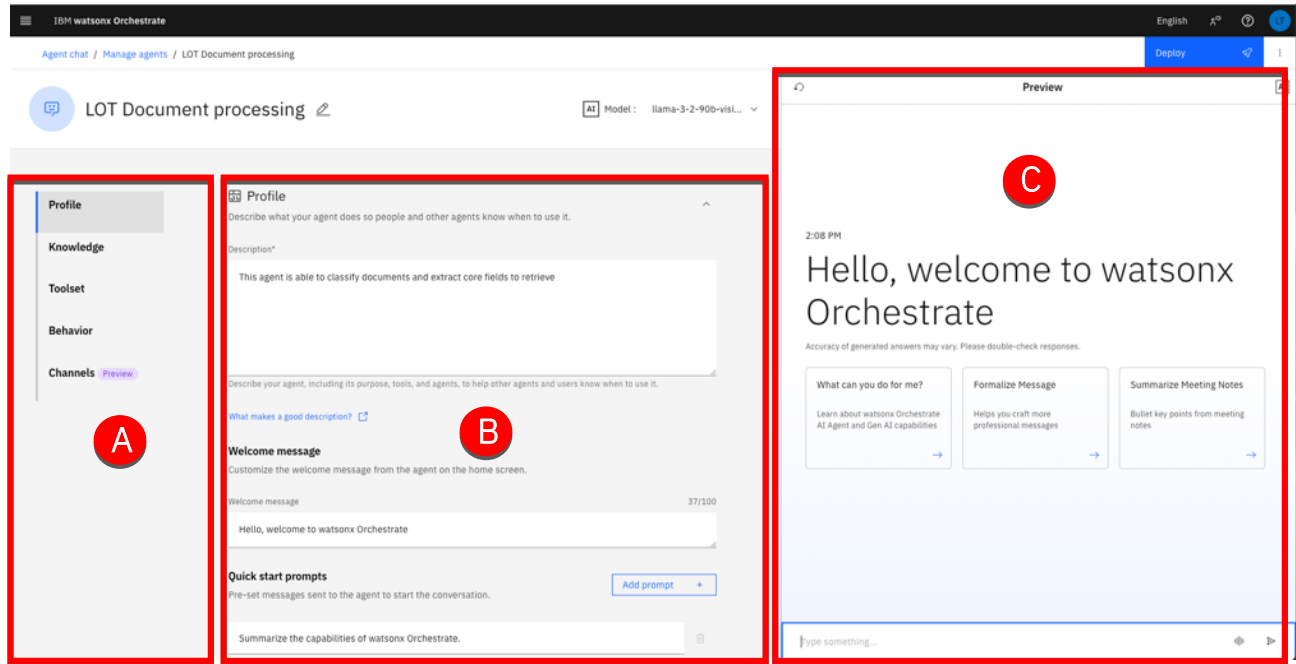
1. Launch your watsonx Orchestrate SaaS instance.
2. Click **Create new Agent (A)**.



3. Select **Create from scratch** (A). Type “[YOUR INITIALS] Document processing” in the **Name** field (B). Type “This agent is able to classify documents and extract core fields to retrieve” in the **Description** field (C). Click **Create** (D).

The screenshot shows the 'Create an agent' interface in IBM watsonx Orchestrate. The form has two main sections: 'Create from scratch' and 'Create from template'. The 'Create from scratch' section is selected and highlighted with a red box and a red circle labeled 'A'. Below this, there are two text input fields. The first field, labeled 'Name' and highlighted with a red box and a red circle labeled 'B', contains the text 'LOT Document processing'. The second field, labeled 'Description' and highlighted with a red box and a red circle labeled 'C', contains the text 'This agent is able to classify documents and extract core fields to retrieve'. At the bottom right of the form, there is a blue 'Create' button highlighted with a red box and a red circle labeled 'D'. A 'Cancel' button is also visible at the bottom left. The top of the interface shows the 'IBM watsonx Orchestrate' logo and a language selector set to 'English'.

4. Note the different sections of the Agent builder tool.



The Agent Builder opens; the screen is divided into three key areas:

- (A) **Navigation Panel** – move between different sections of the Agent Builder.
- (B) **Configuration** – set up and customize the core functionality of your agent.
- (C) **Preview** – preview and test your agent.

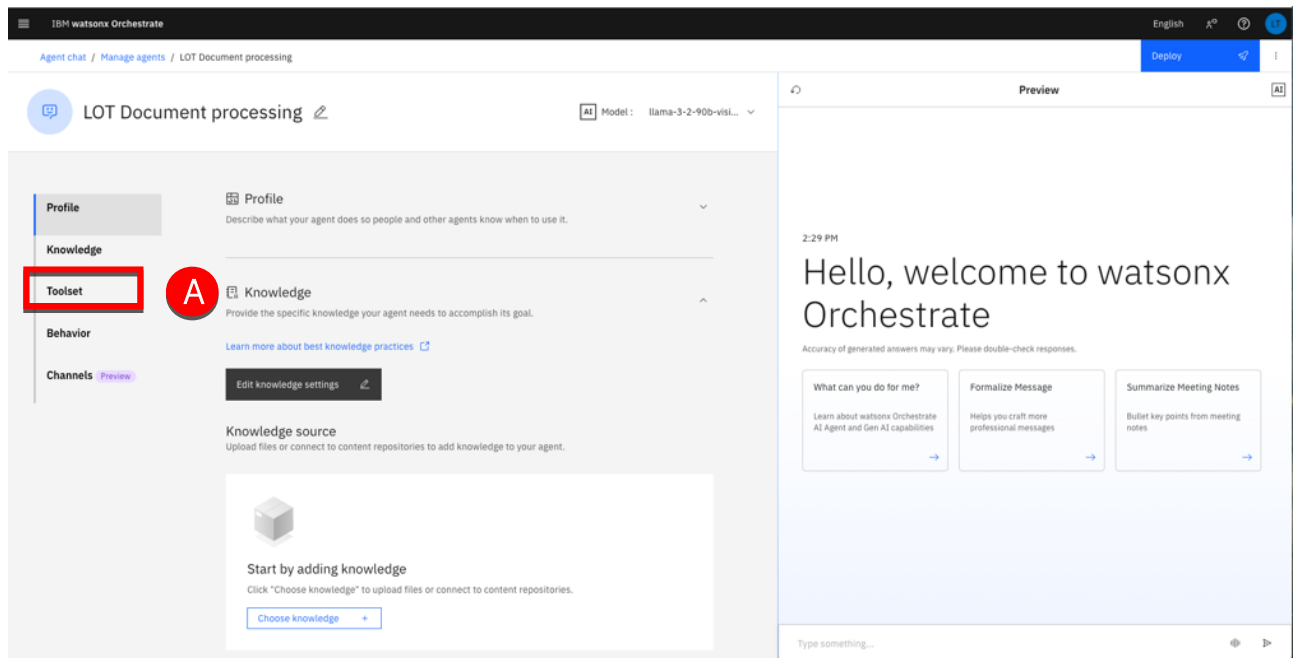
Below is a description of each section:

- **Profile:** Define your agent's purpose, usage scenarios, and interaction style. Describe what the agent does, when it should be used (especially in multi-agent configurations), and choose its style—Default or ReAct—to guide how it interprets requests, plans and uses tools.
- **Knowledge:** This is where you equip your agent with knowledge by uploading files or connecting to conversational search platforms such as Milvus, Elasticsearch, or custom-built sources. This ensures the agent can generate accurate, contextual responses by drawing on relevant content.
- **Toolset:** Provide your agent with tools to perform tasks. Tools can be added from the Catalog, imported from OpenAPI specification files and MCP servers, or built with custom flows. Tools extend the agent's capabilities, enabling it to automate actions such as retrieving data or sending emails.
- **Behavior:** Here, you define how the agent interacts with users, formats data and how it handles requests. Add rules and instructions to shape its tone, response style, and overall behavior during interactions.
- **Channels:** This is where you connect your agent to communication platforms like Slack or embed the agent in a website.

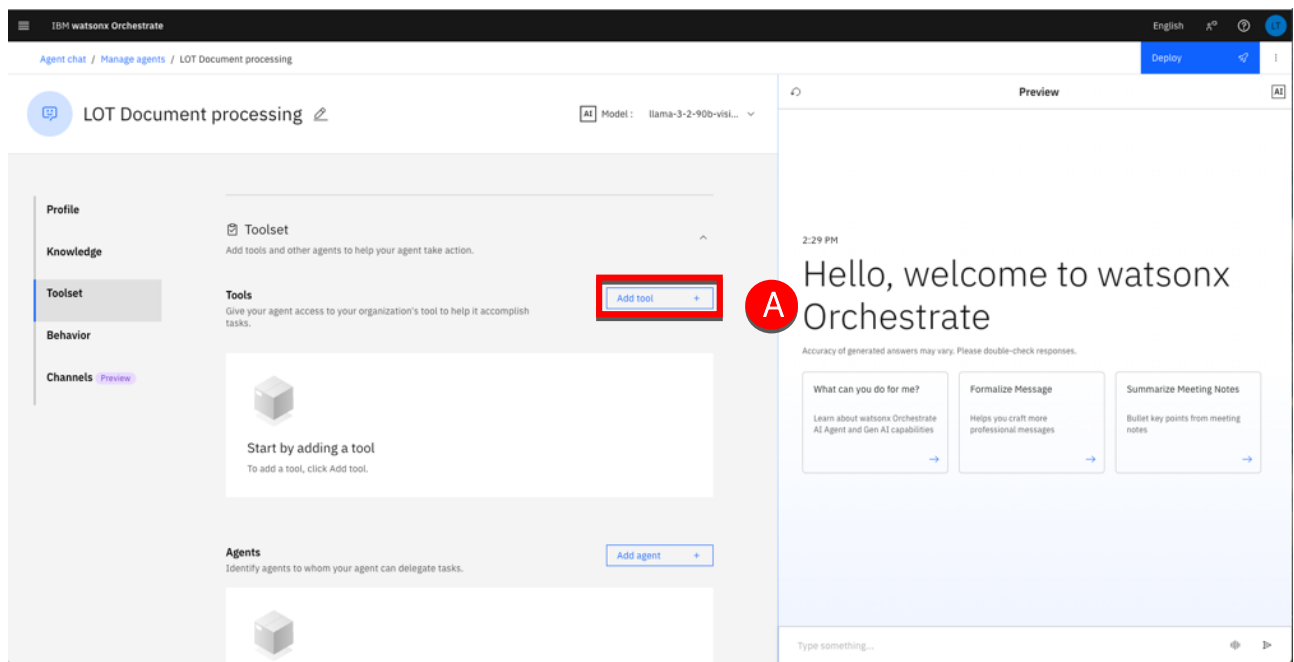
Add tools

In this section you will add a tool to your agent. This tool will contain the process described in the introduction of this lab. You will start by creating the tool and implementing the tool behavior as a flow.

1. Click **Toolset** (A).

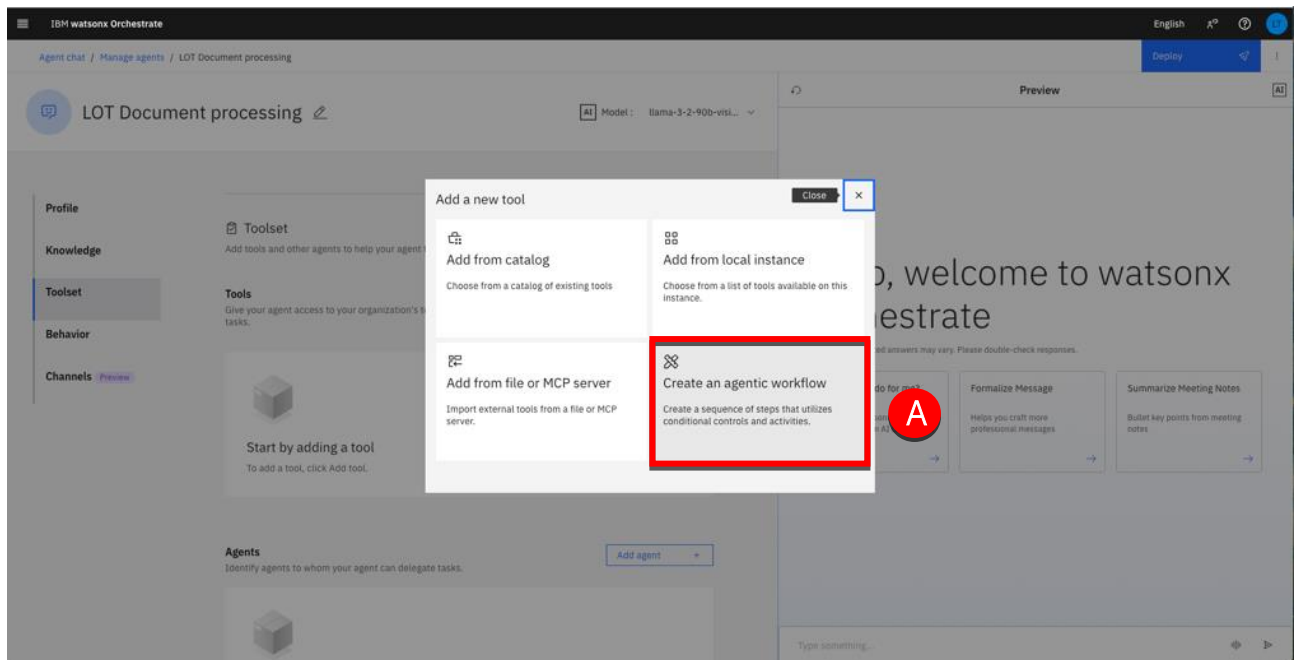


2. Click **Add tool** (A).

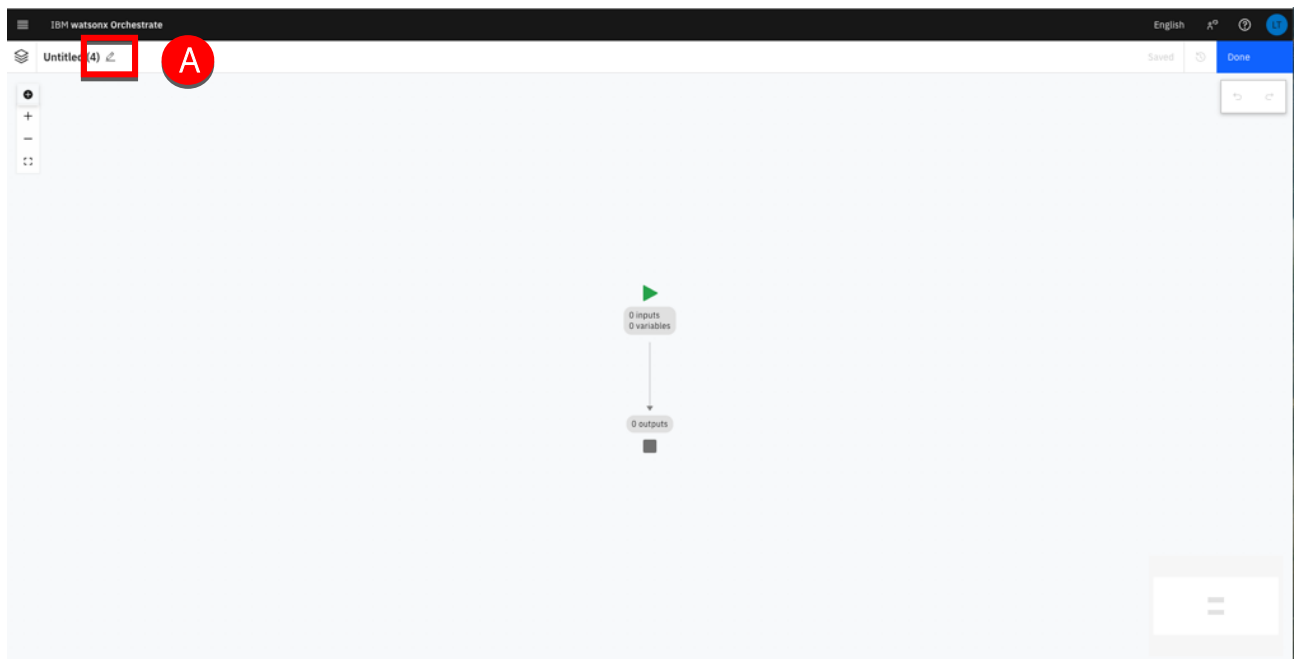


Create the workflow

1. Click **Create an agentic workflow** (A).



2. Click the pencil icon (A) to rename the workflow.



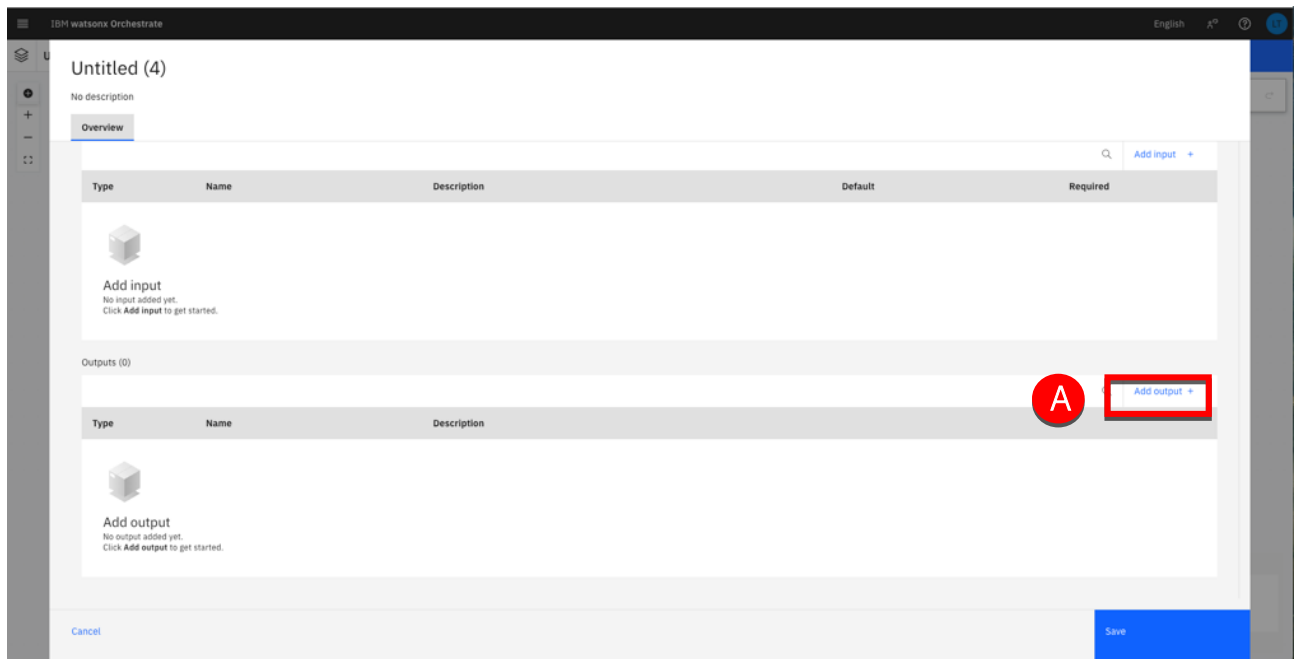
3. Type 'Document processing' in the **Name** (A). Type "This flow classifies and extracts values from documents." In the **Description** (B). Scroll down to the **Output** section (C).

The screenshot shows the IBM watsonx Orchestrate interface for a new flow titled 'Untitled (4)'. The 'Name' field (A) is highlighted with a red box and contains the text 'Document processing'. The 'Description' field (B) is also highlighted with a red box and contains the text 'This flow classifies and extracts values from documents.' A red arrow (C) points down towards the 'Inputs (0)' section, which is currently empty. The interface includes a sidebar on the left with icons for 'Overview', 'Inputs', and 'Outputs'. The 'Overview' tab is selected. The 'Inputs (0)' section has a table with columns: Type, Name, Description, Default, and Required. The table is currently empty. At the bottom right, there is a 'Save' button.

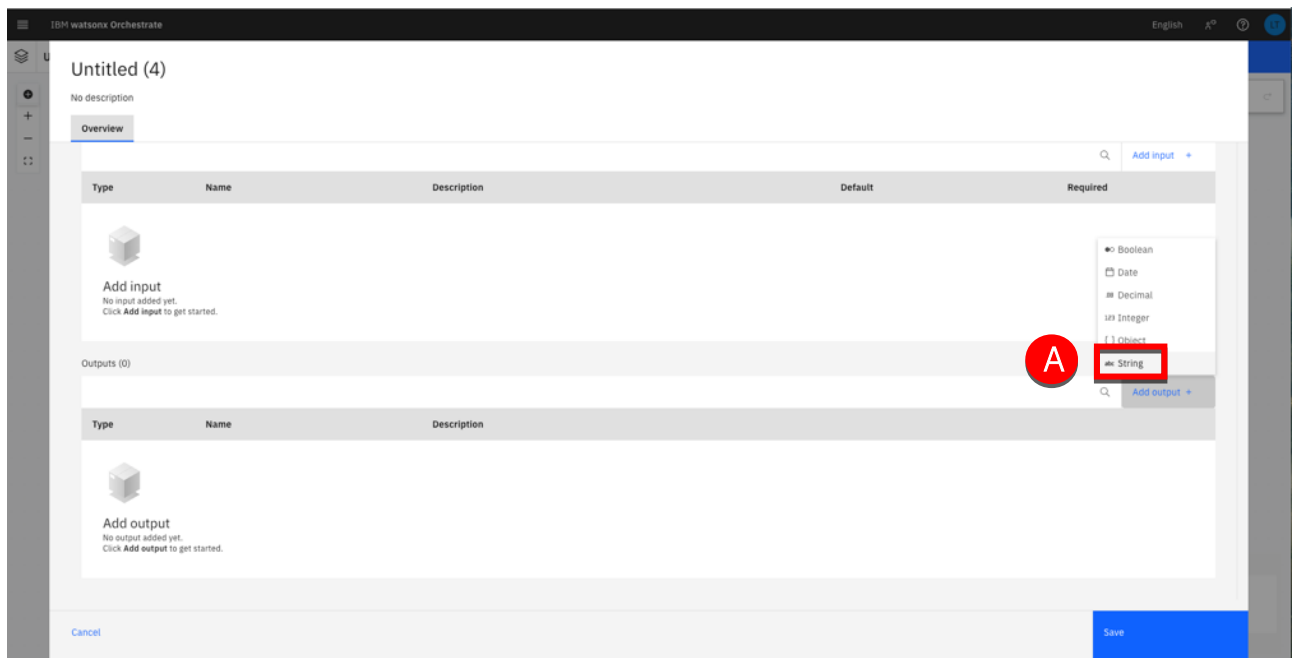
Define the process output

Scrolling down, you'll find the Input and Output sections, which apply to the entire flow. In this case, no inputs are required since they are already handled within the flow itself through the user activity file upload. For the output, instead of using a text output node, the flow can return the classification result, defined as a string.

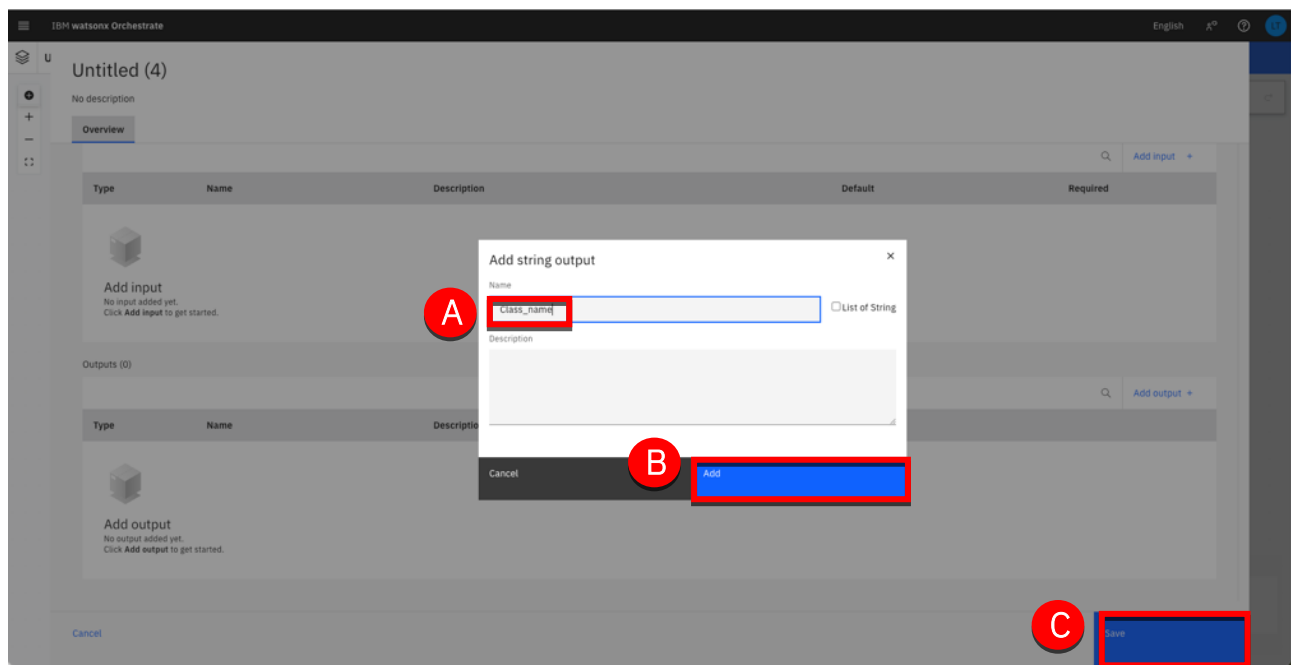
1. Click **Add output** (A).



2. Select **String** (A).



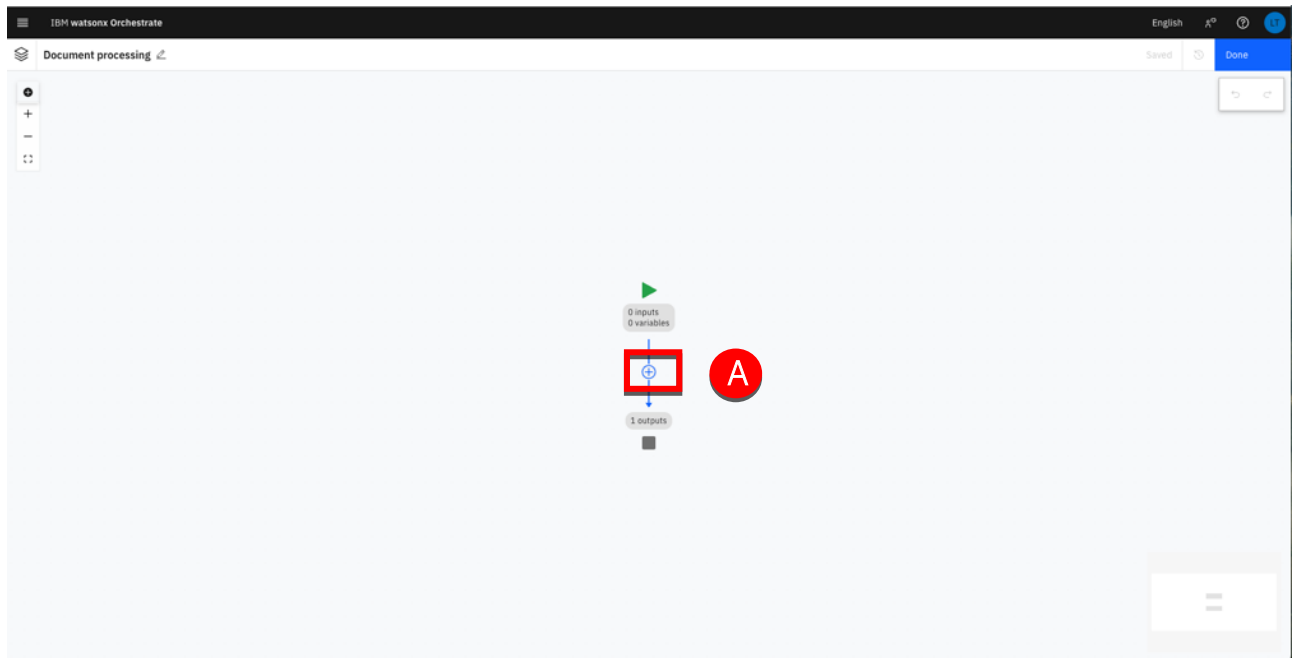
3. Type “Class_name” in the **Name** field (A). Click **Add** (B). Then click **Save** (C).



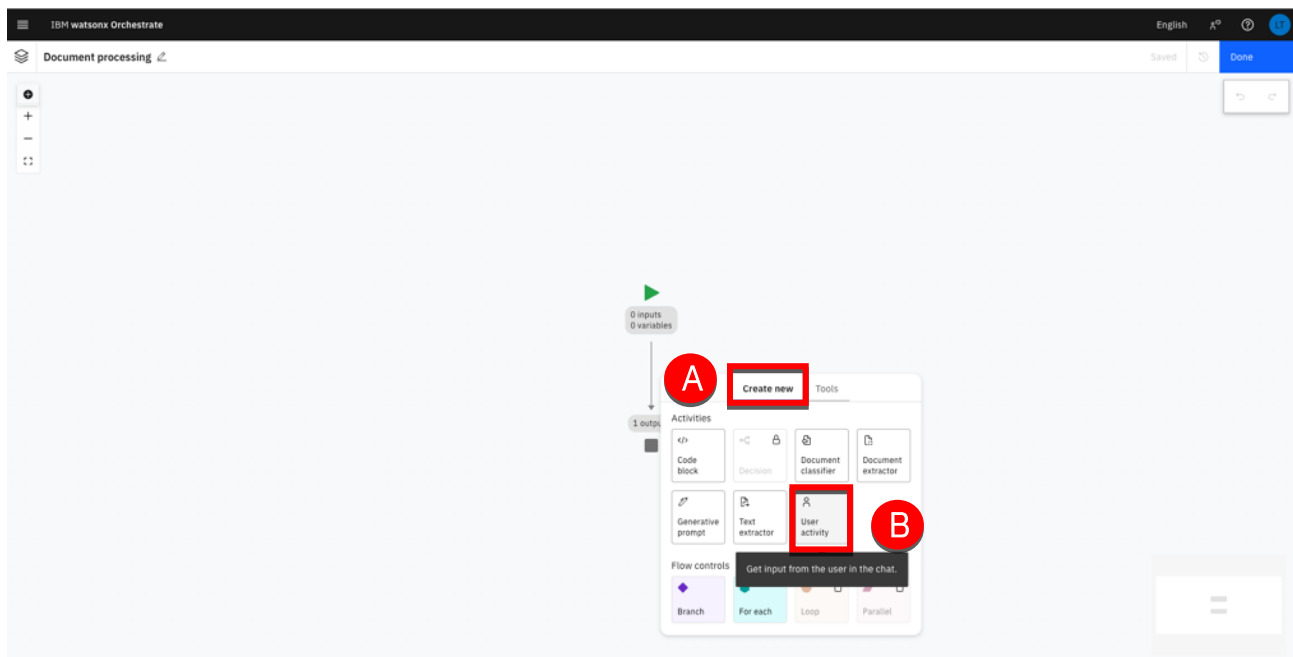
Design the workflow diagram

In this section, you will add the different workflow activities including the document classifier and the two extractors.

1. Move the cursor over the link until the + icon appears. Click the + to add an activity (A). The first activity of the process consists in uploading the document to process. It will be implemented as a user activity.

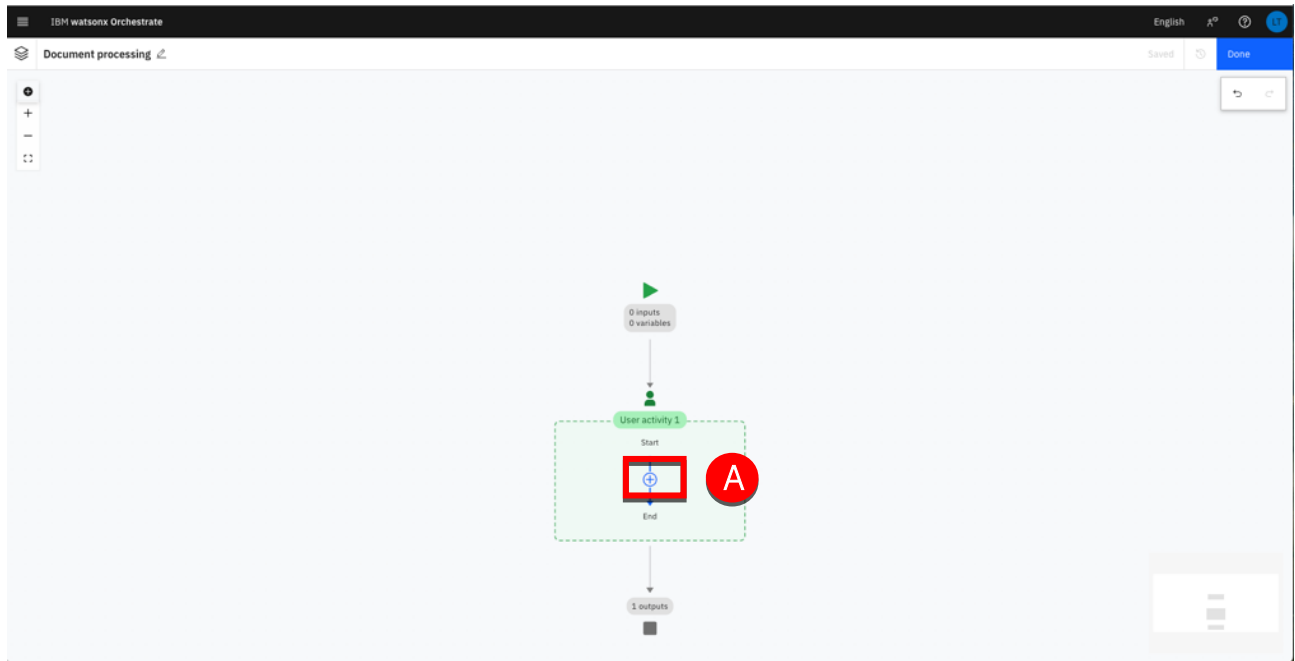


2. Select **Create new** (A) and then select **User activity** (B).

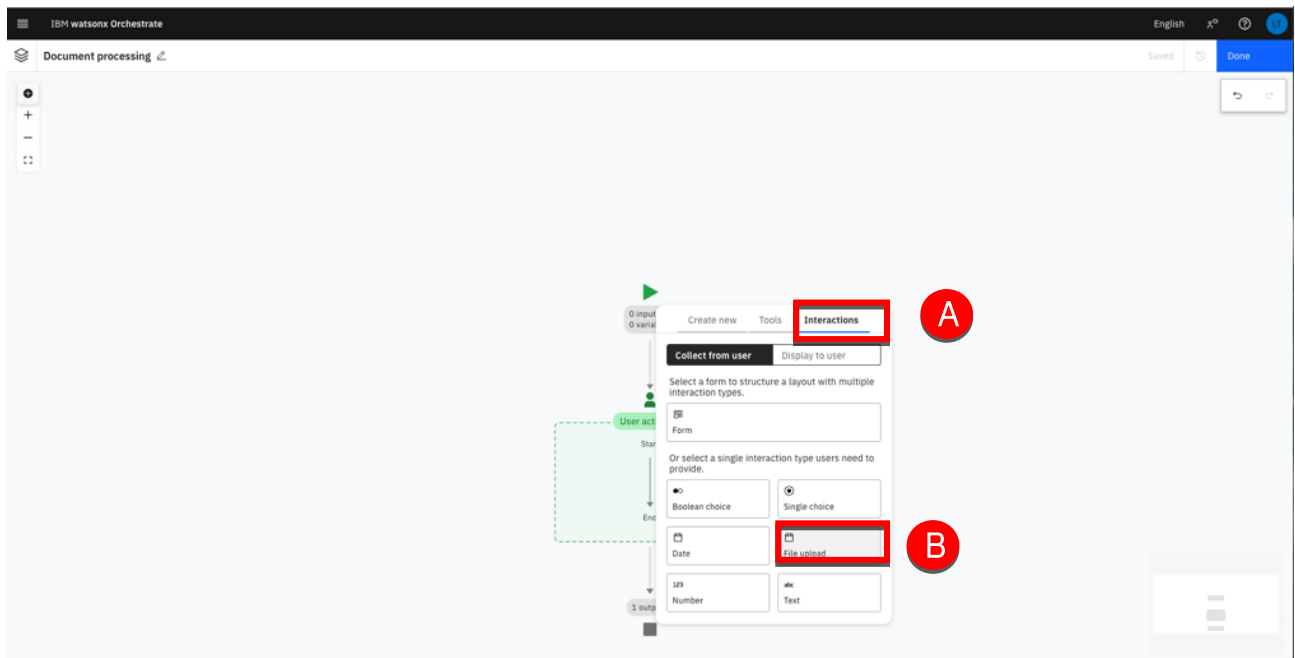


Note: A green box appears, indicating the user activity. This will contain the series of activities that the user must perform whilst interacting with the agent.

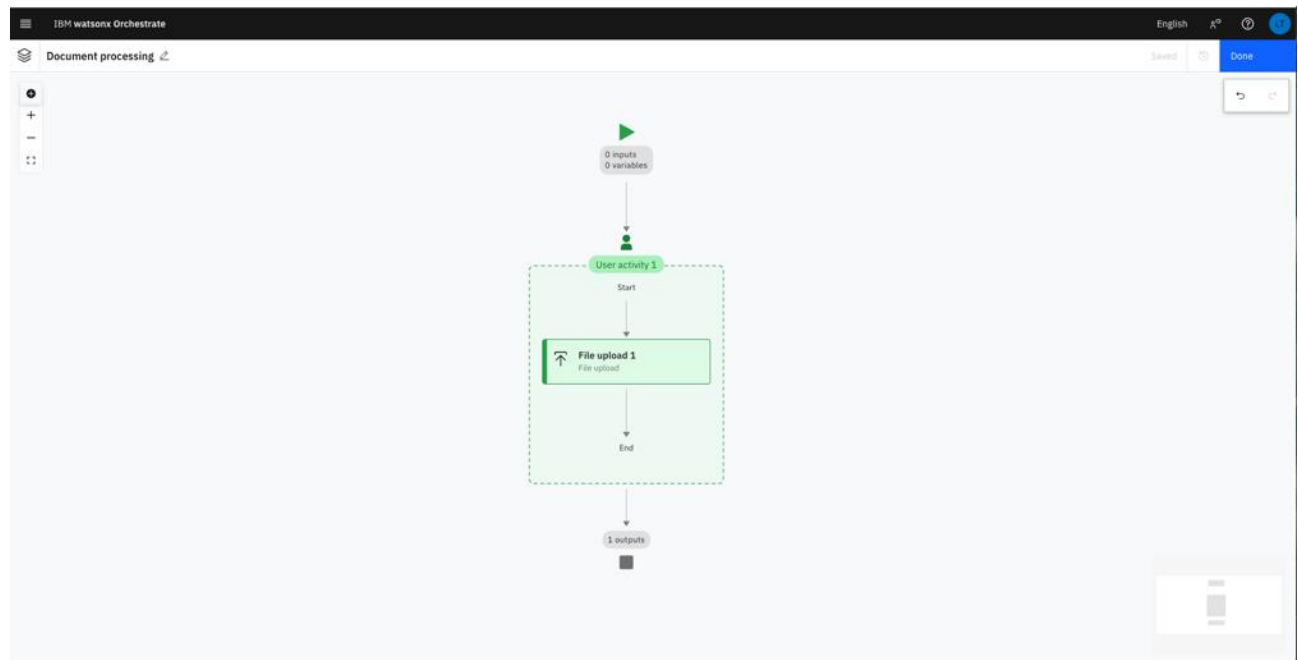
3. Move the cursor into the user activity and click + (A) to add the activities the user will have to perform.



4. Select the **Interactions** tab (B) then click **File upload** (C).



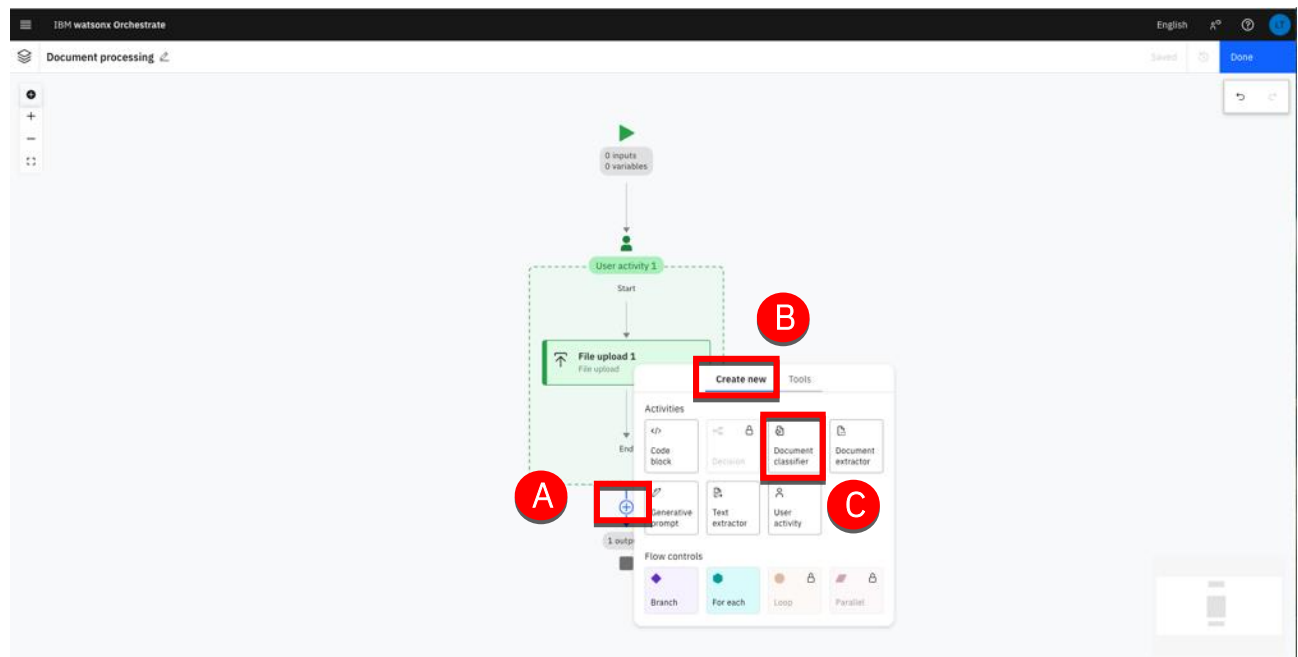
The first activity of the process is now defined. The next step is to classify the document the user will upload. To do this, you will add a watsonx Orchestrate Document Classifier.



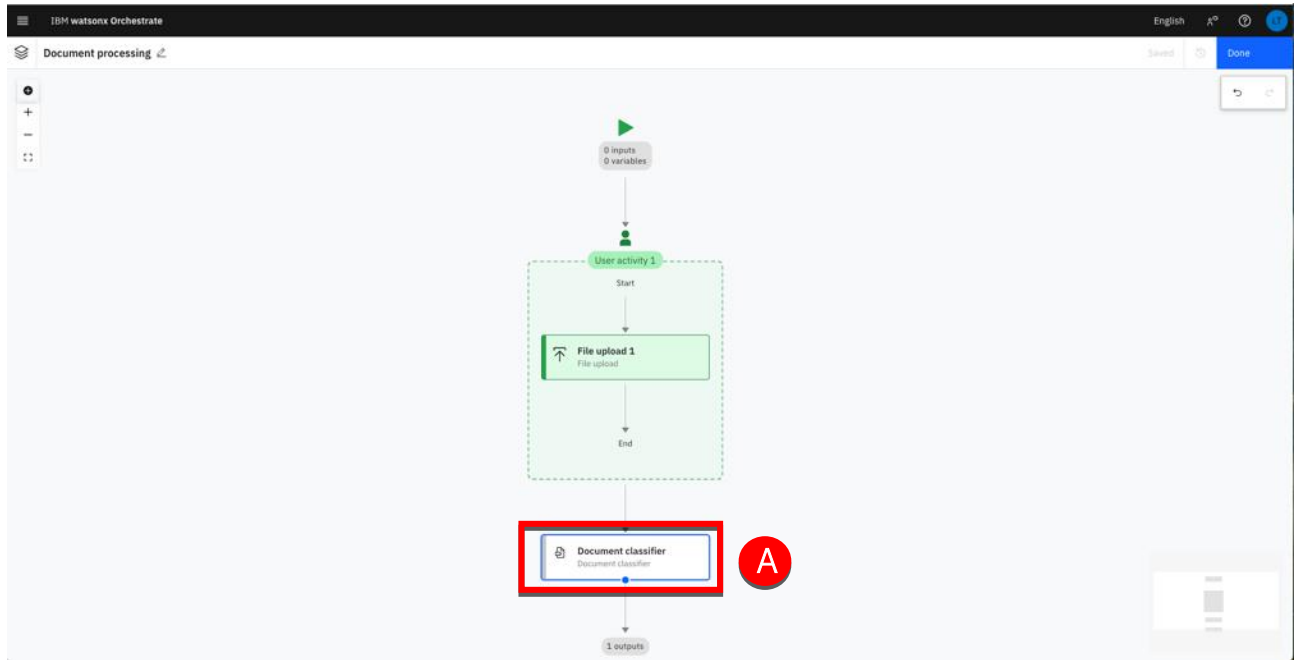
Add a document classifier

In this section, you will add a document classifier to distinguish Invoice documents from the Contract ones.

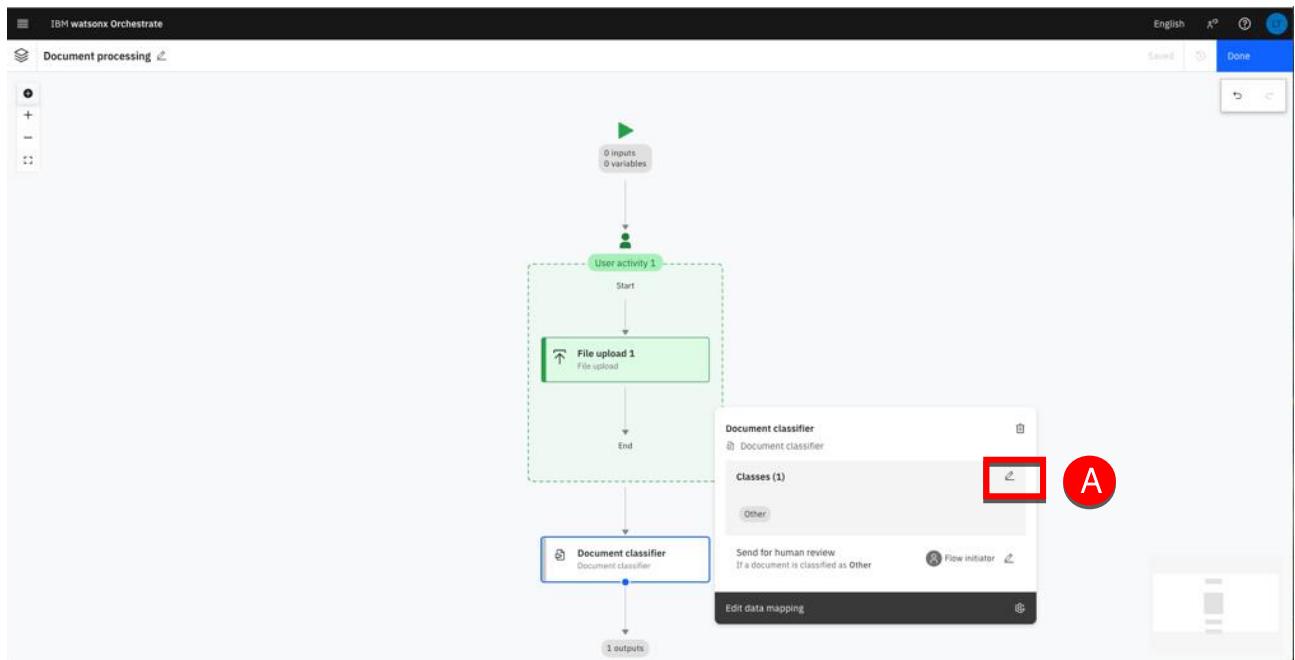
1. Move the cursor over the last link of the process and click + (A). Select **Create new** (B) then click **Document classifier** (C).



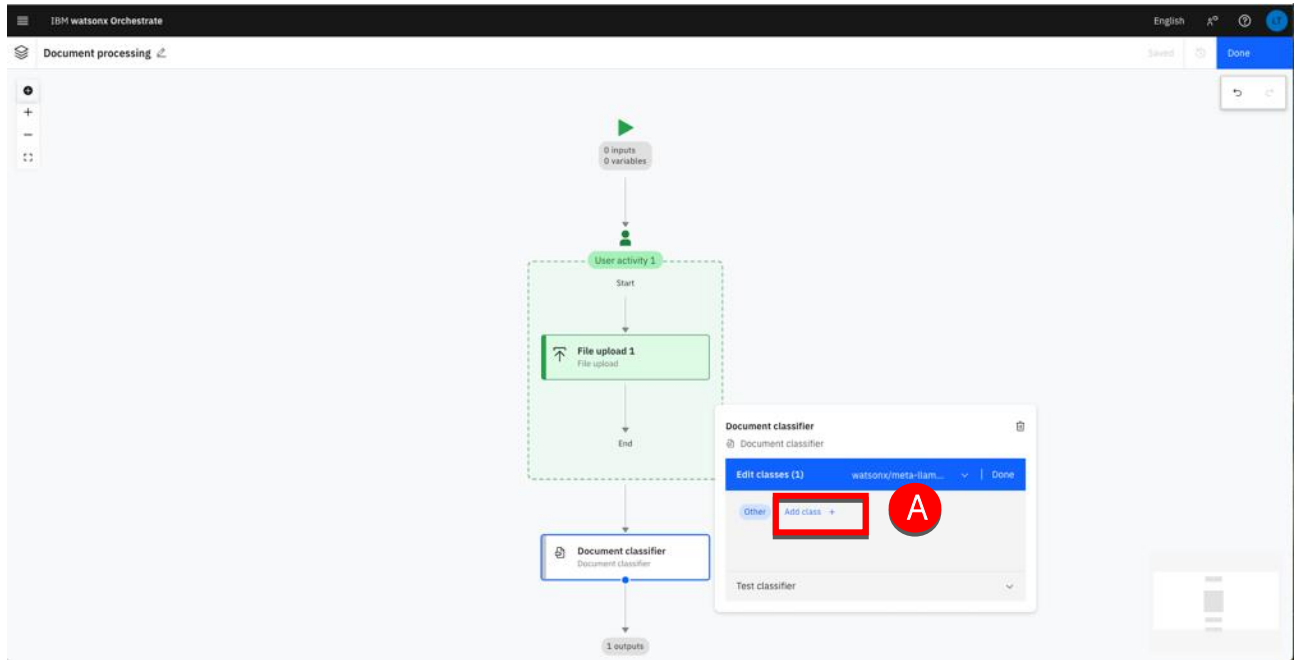
2. Click the **Document classifier** node (A).



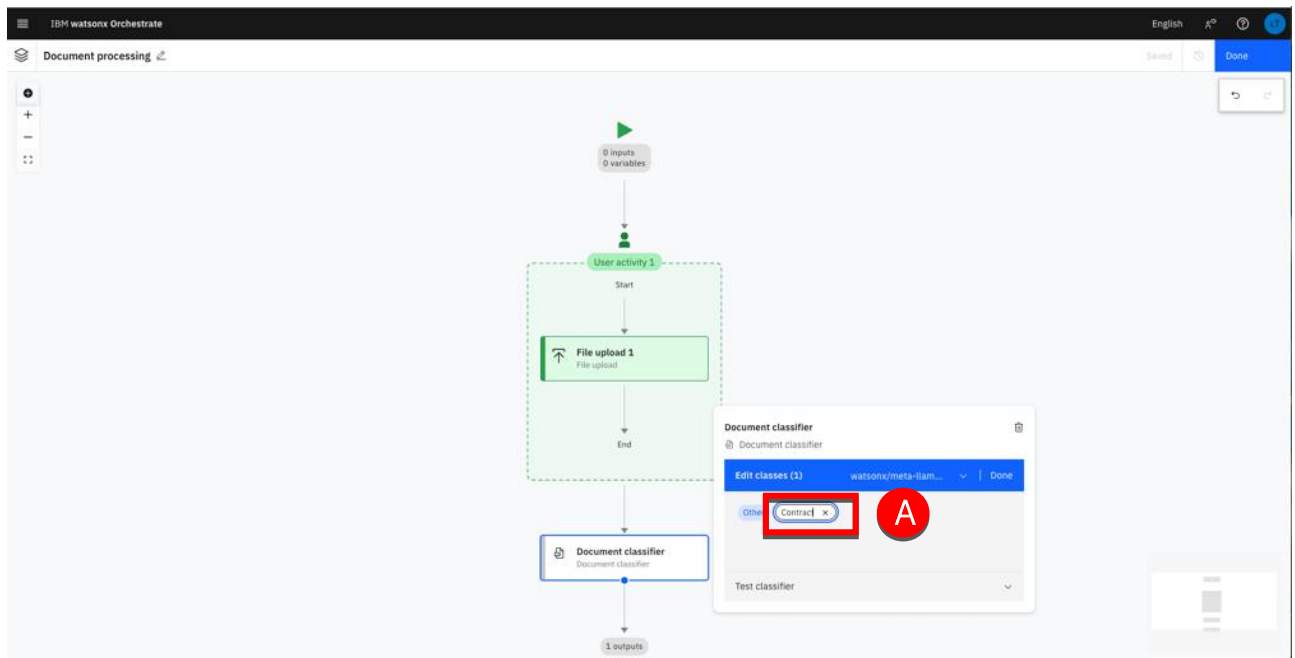
3. Click the edit icon (A).



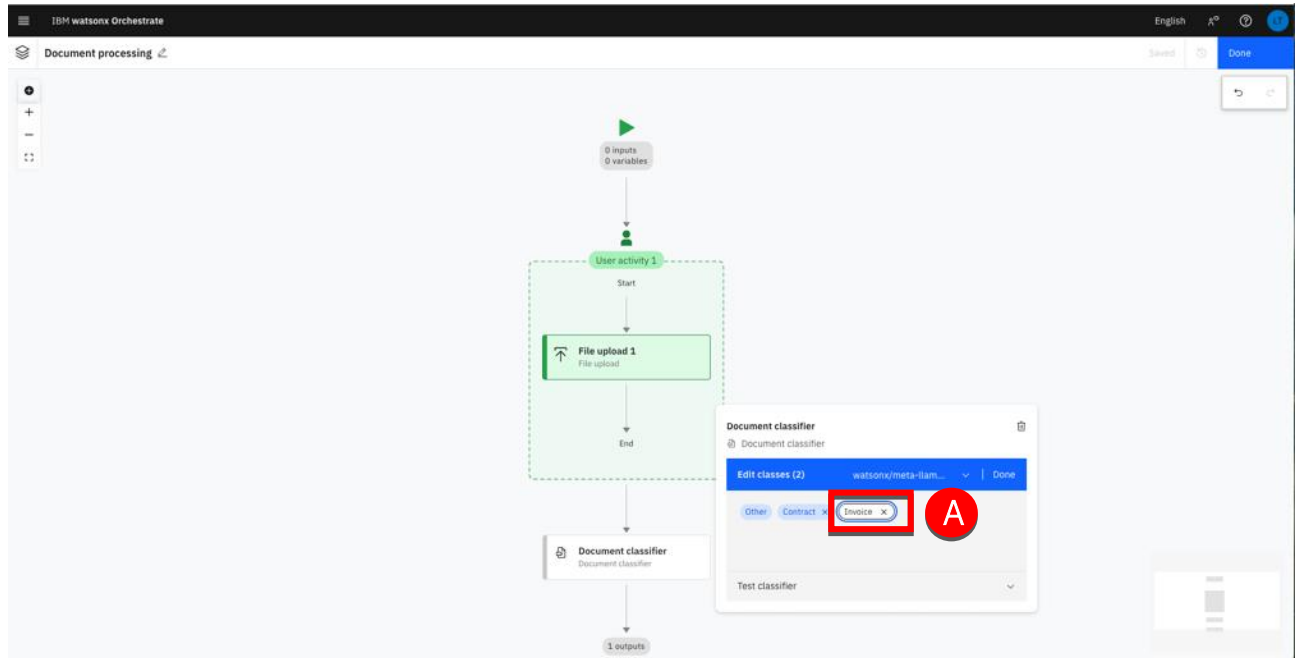
- Click Add class (A)



- Type 'Contract' (A), then Enter.

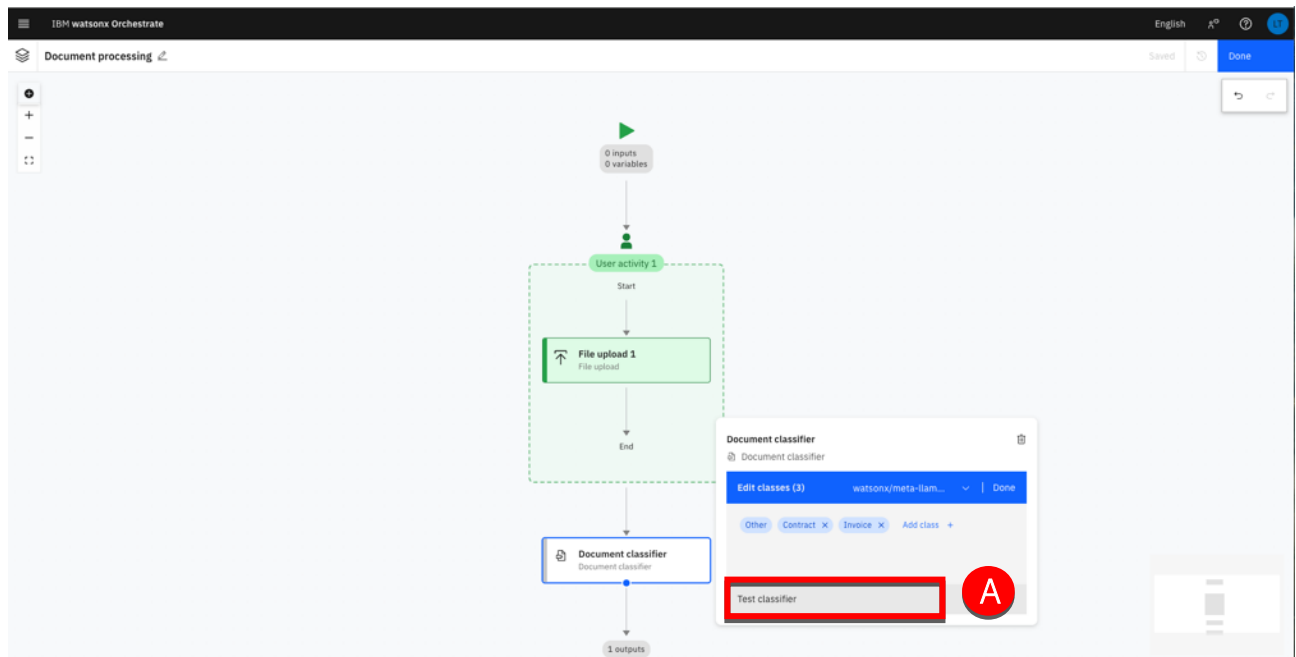


6. Type 'Invoice' into the new field (A), then TAB out of the field.

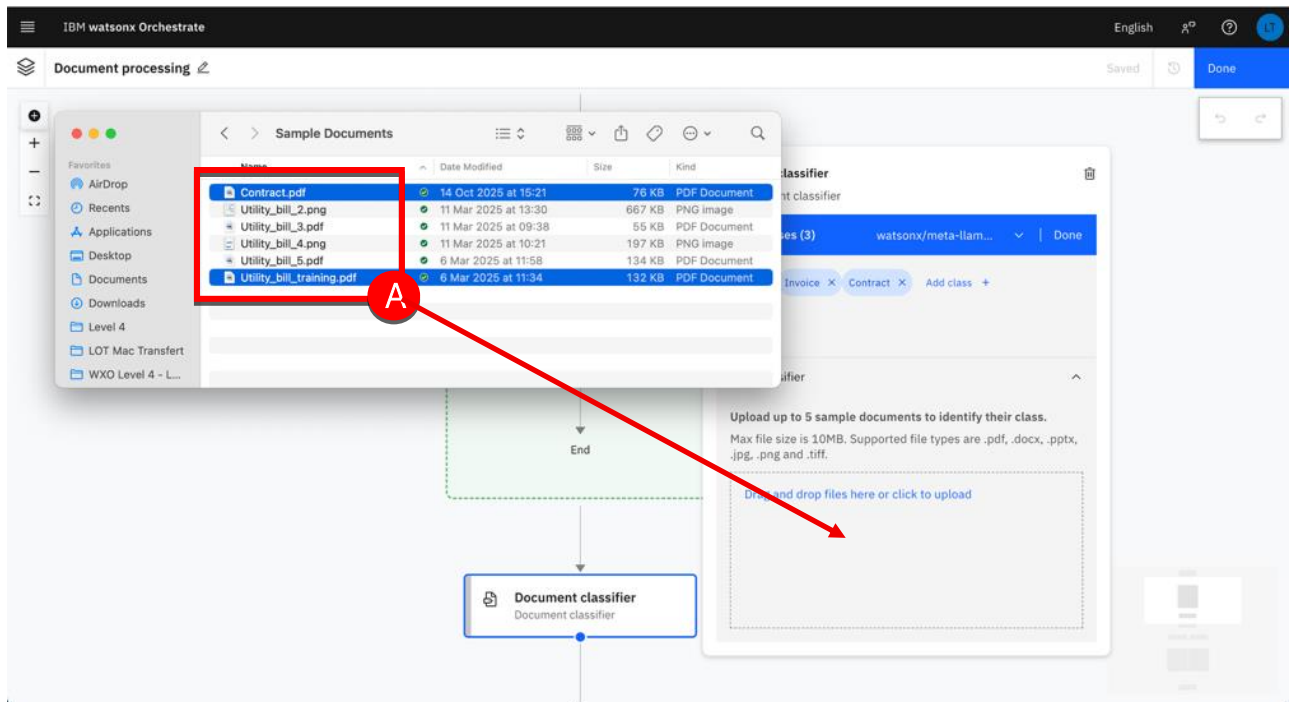


You are now ready to test your document classifier.

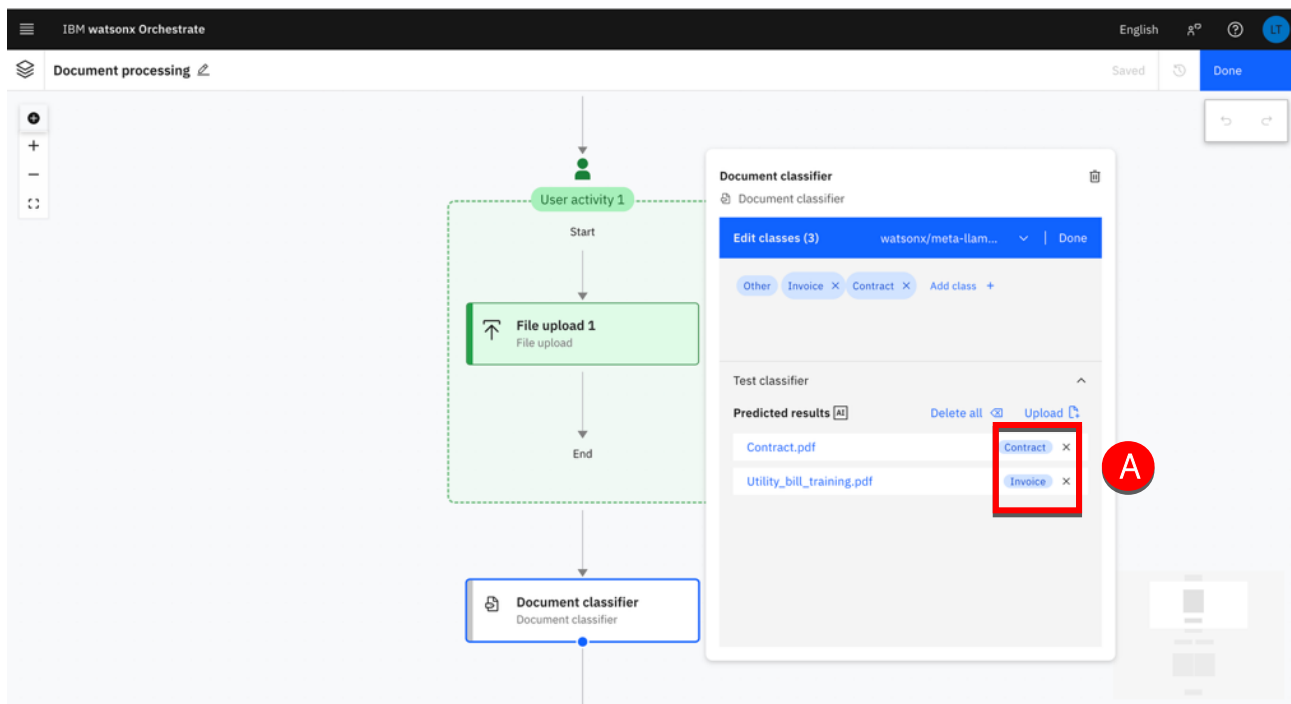
7. Click **Test classifier** (A).



8. Drag and drop the two sample files, **Utility_bill_training.pdf** and **Contract.pdf** from your local machine location (A). Note: ensure you select both files before dropping in the upload area, if you upload a single document you will need to click the **Upload** icon to upload the second document.



9. Wait for the documents to be analyzed and classified (A).



Note. The two documents have been successfully classified by the LLM based on their content. Contract.pdf has been classified as a Contract and Invoice.pdf has been classified as an Invoice. If your documents last too long to be classified, you may have to delete them and import them back again.

10. Click **Done** (A).

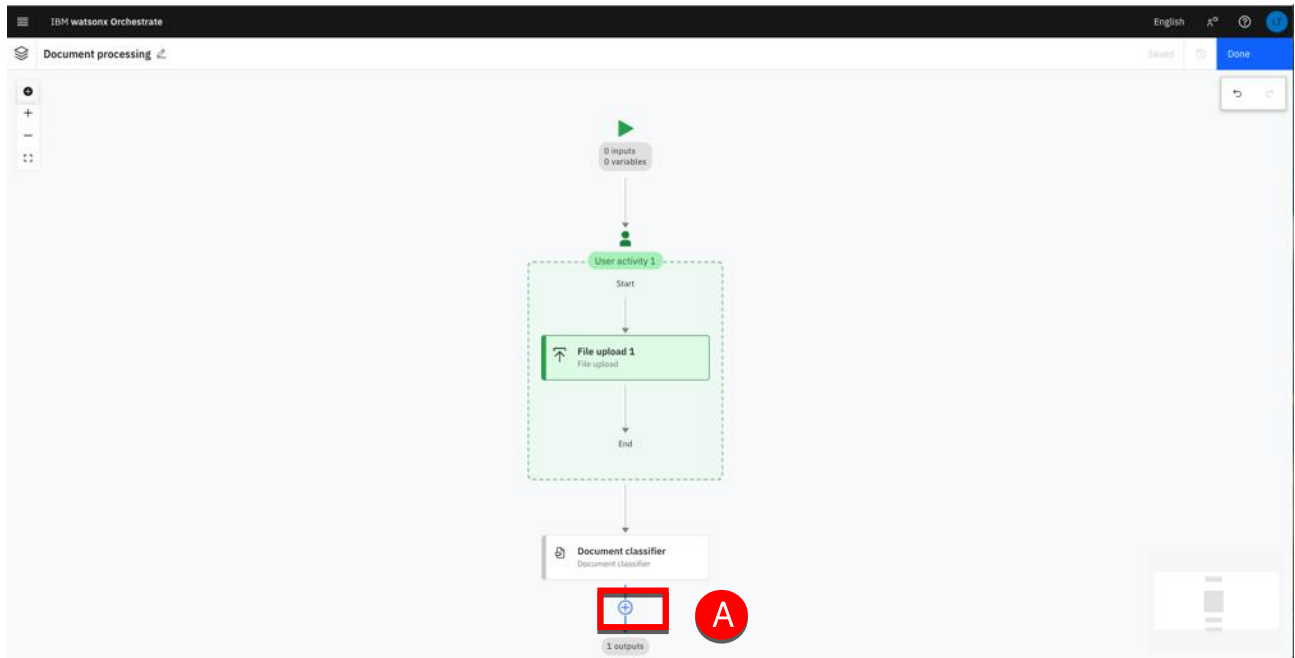
The screenshot displays the IBM watsonx Orchestrate interface. On the left, a flow diagram shows a process starting with 'User activity 1', followed by a 'File upload 1' node, and then a 'Document classifier' node. On the right, a configuration panel for the 'Document classifier' is open. It shows 'Edit classes (3)' with 'Contract' and 'Invoice' selected, and a 'Done' button highlighted with a red box and a red circle labeled 'A'. Below this, the 'Test classifier' section shows 'Predicted results' for 'Contract.pdf' (classified as 'Contract') and 'Utility_bill_training.pdf' (classified as 'Invoice').

Note: You can click the 'Upload' icon seen above to test out the classifier with more document types and verify its accuracy.

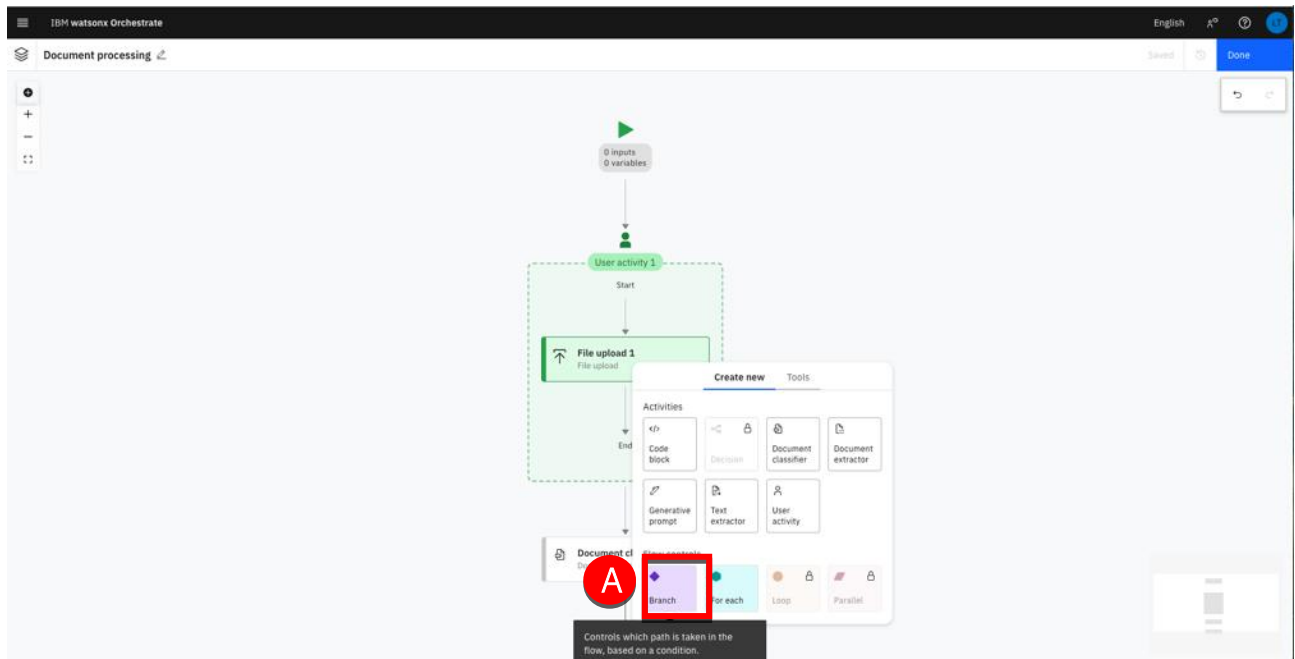
The next step is to route the process on the right extractor path depending on the document type. To do this, you will add a branch node responsible for the triage.

Add a Branch node

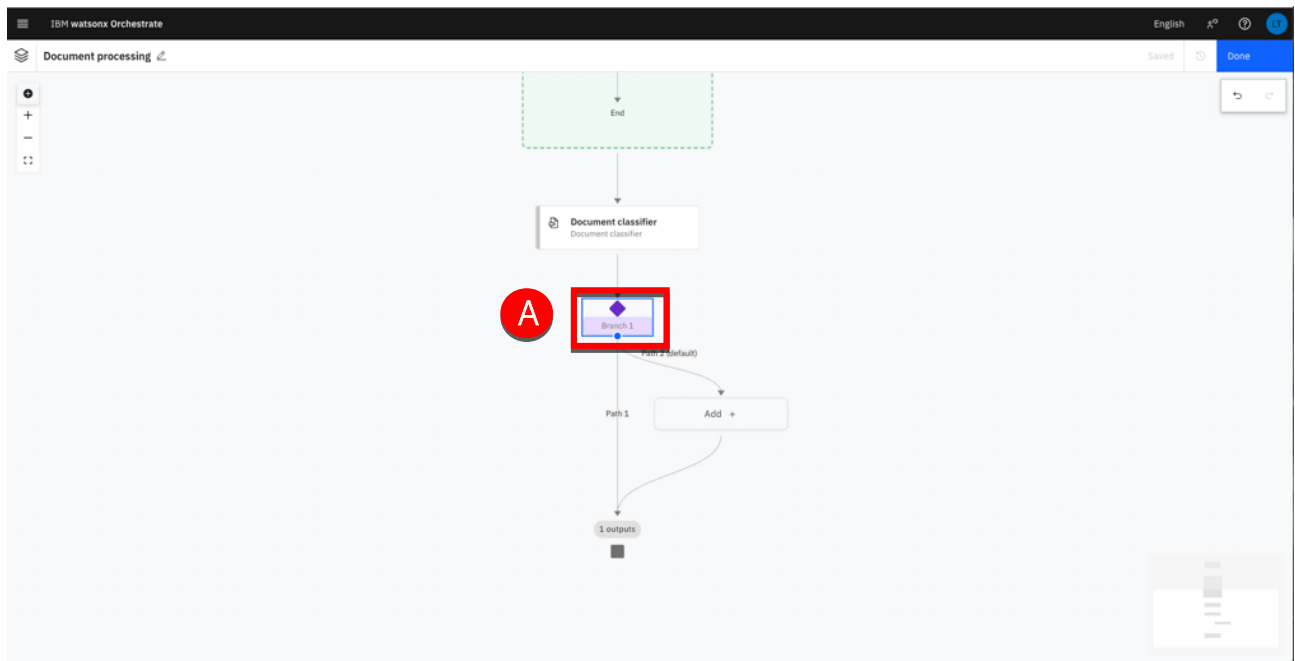
1. Move the cursor over the last link of the diagram then click + (A).



2. Click **Branch** (A).

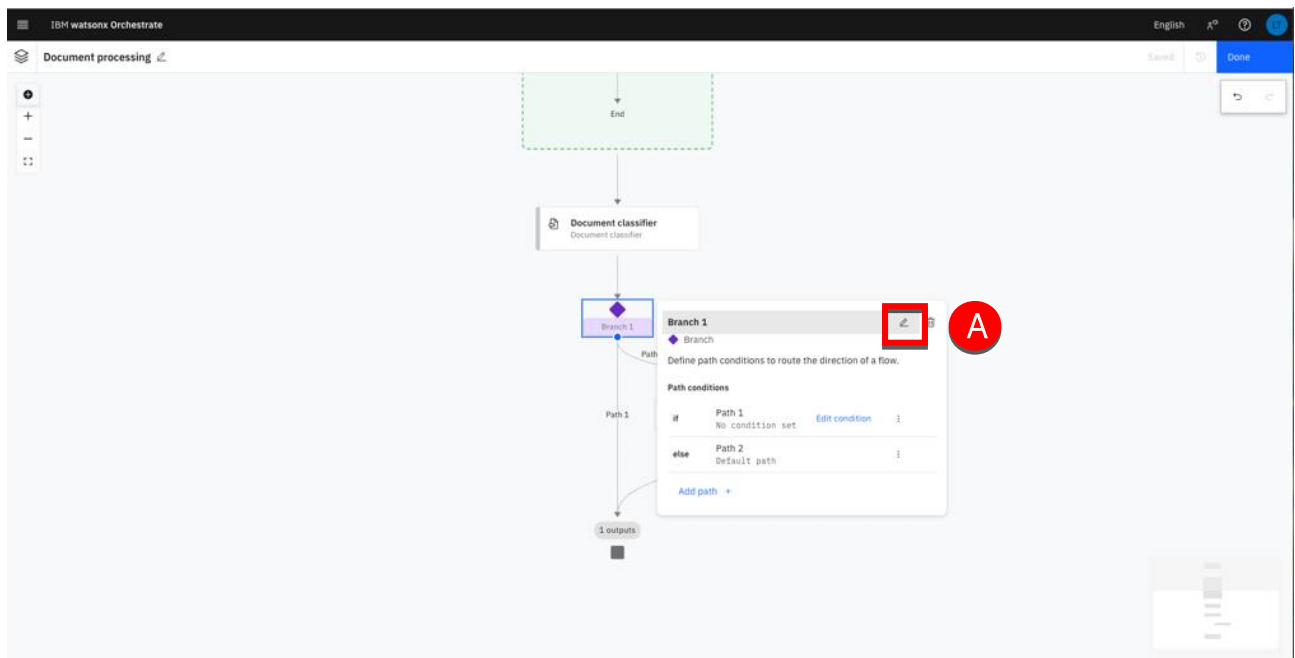


- Click the **Branch 1** node (A).

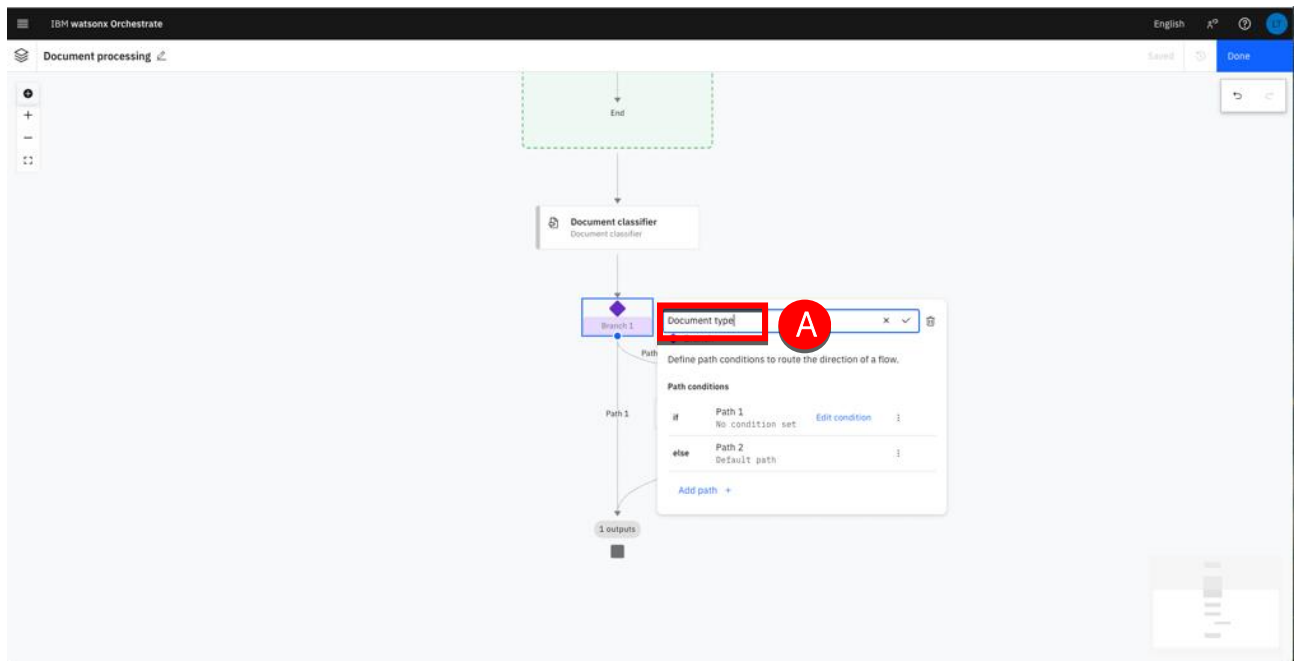


Note: When creating a branch from scratch, two paths are generated by default, but additional paths can be added as needed.

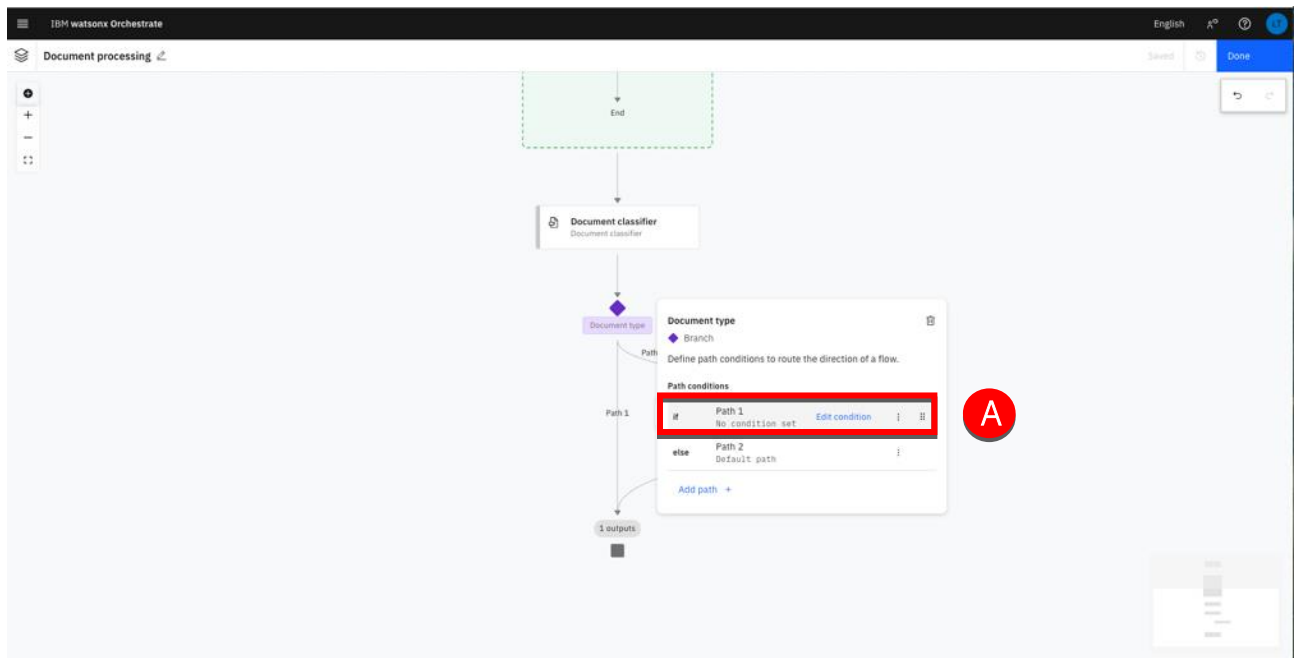
- Move the cursor over the **Branch 1** name then click the edit icon (A)



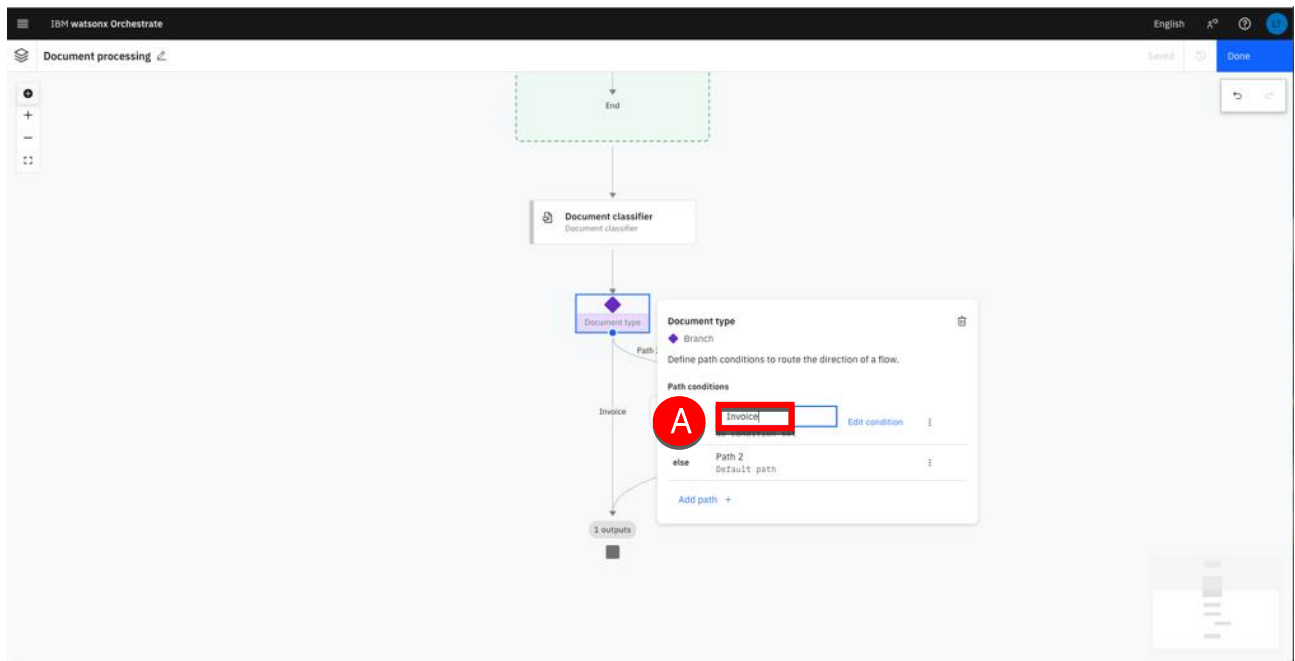
5. Type 'Document type' in the node name field (A) then hit Enter.



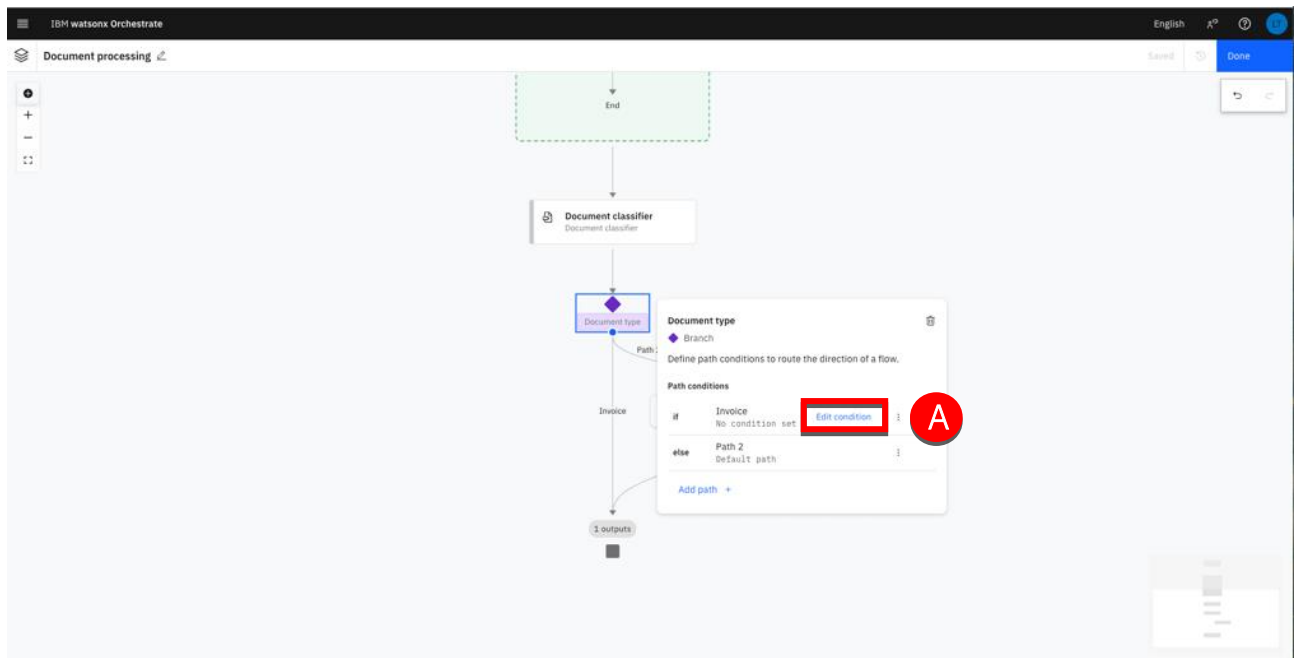
6. Click the Path 1 row (A)



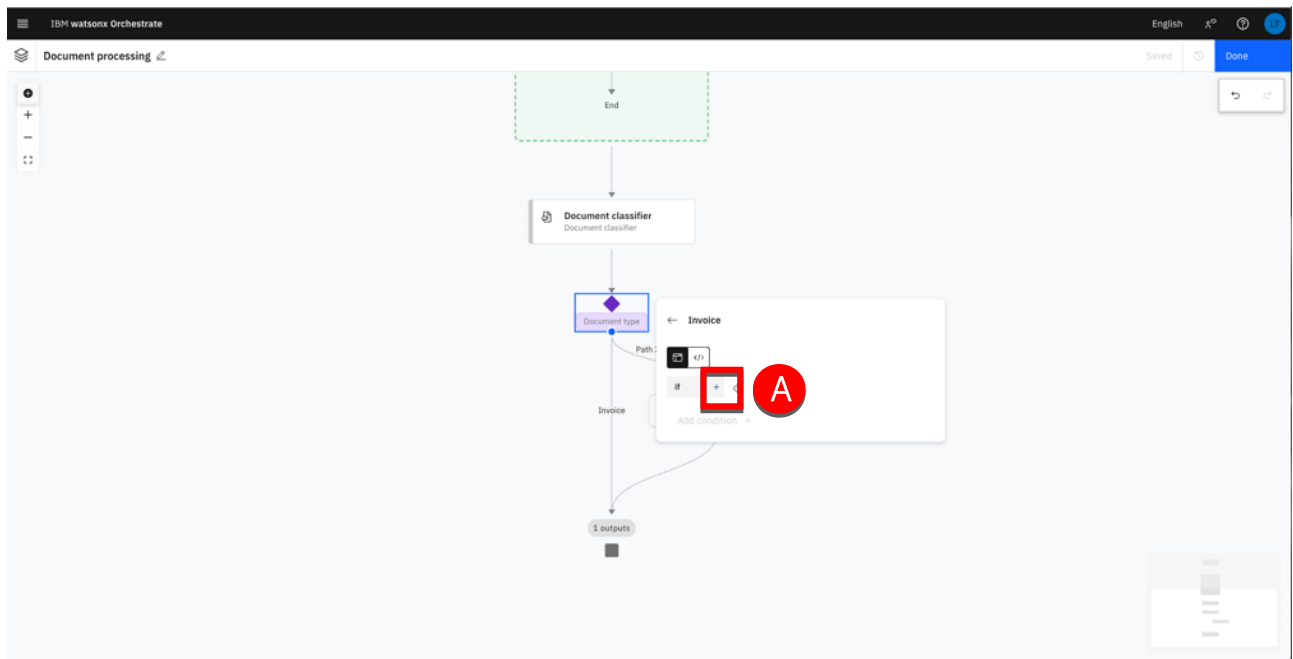
7. Type 'Invoice' (A).



8. Click **Edit condition** (A) to edit the condition and select invoice documents.

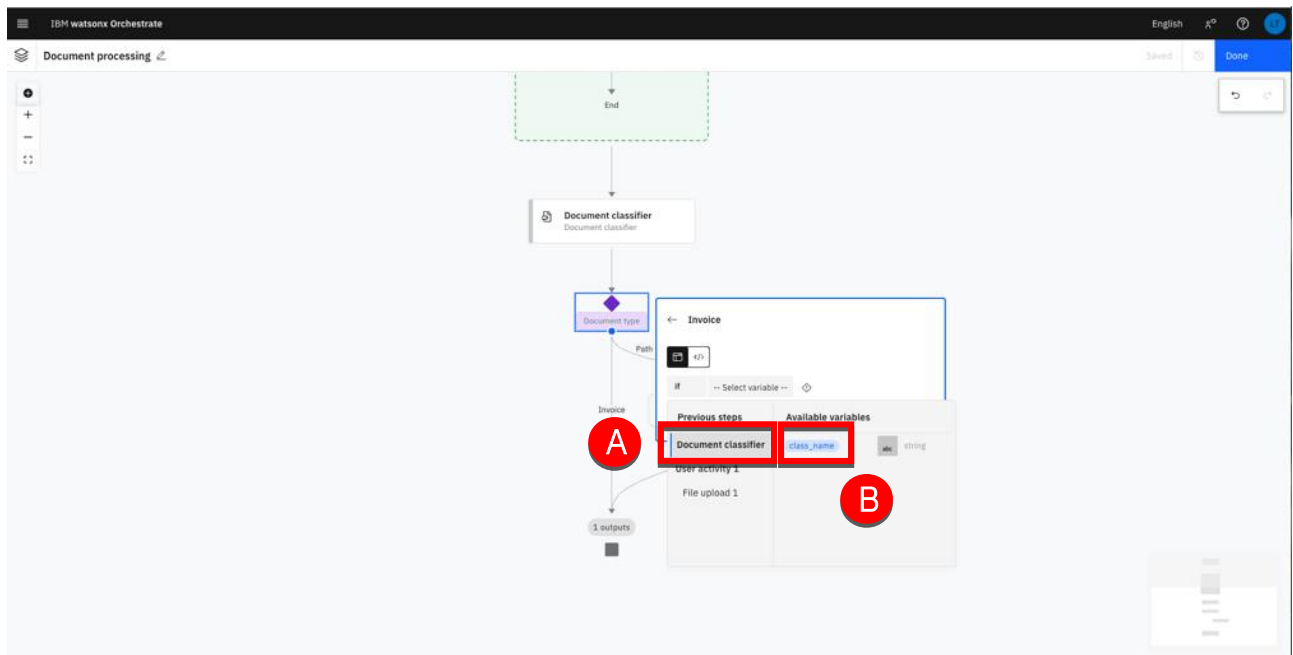


9. Click the + icon (A) to add conditions.

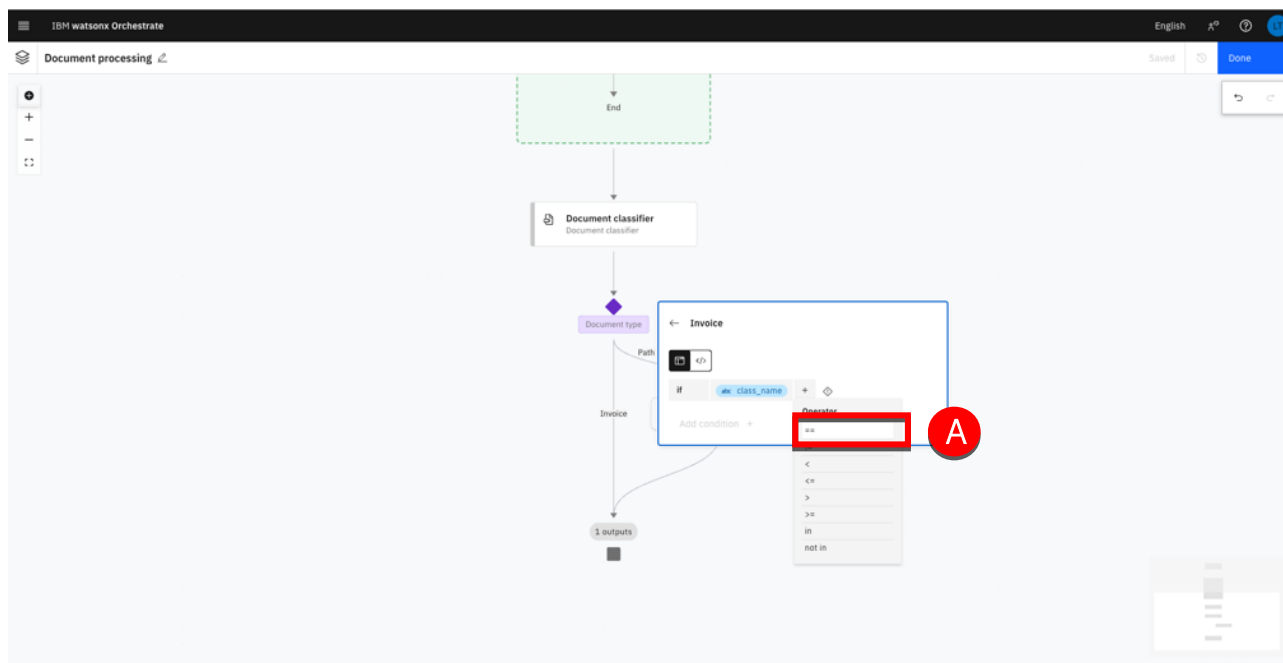


You will use the result of the Document classifier step as the variable to evaluate for the routing.

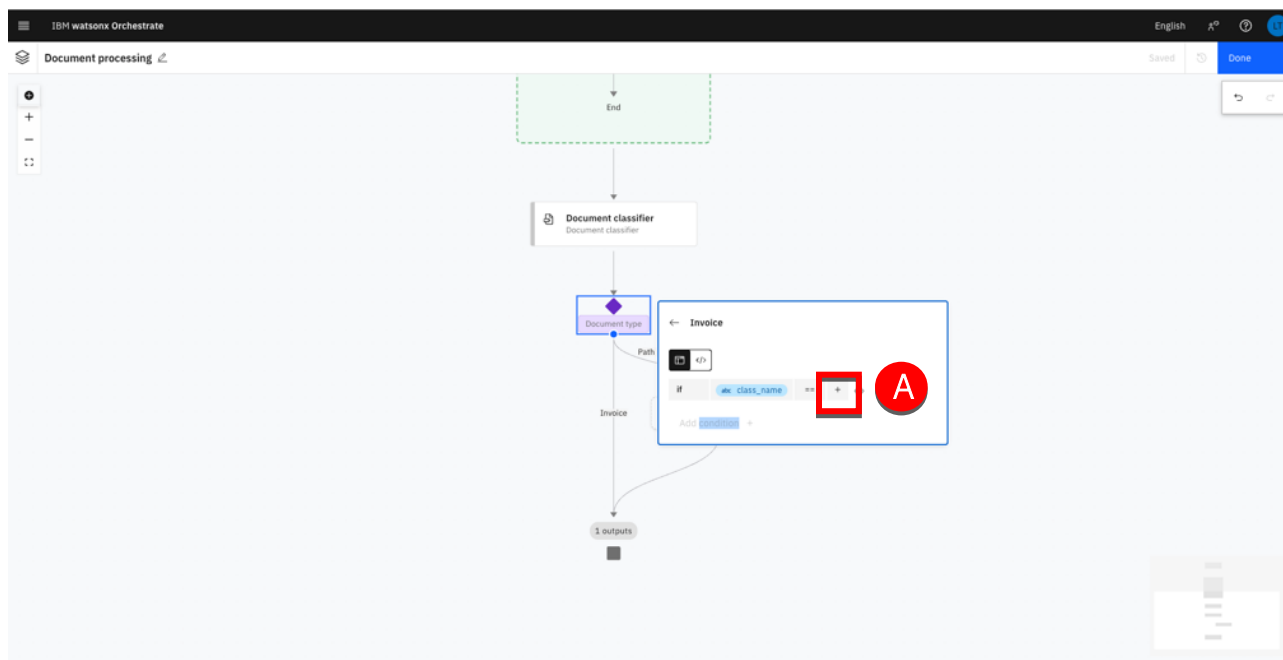
10. Click **Document classifier** (A) and then click **class_name** (B).



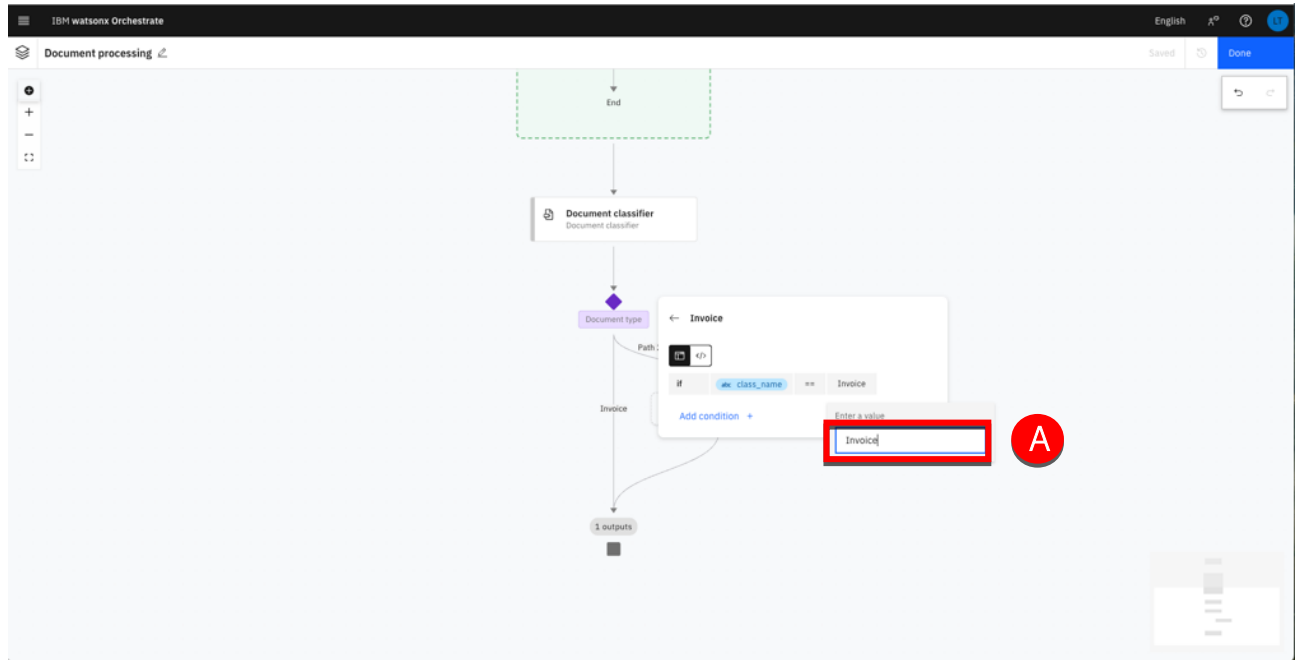
11. Select the '==' operator (A).



12. Click + to specify the value to check (A).



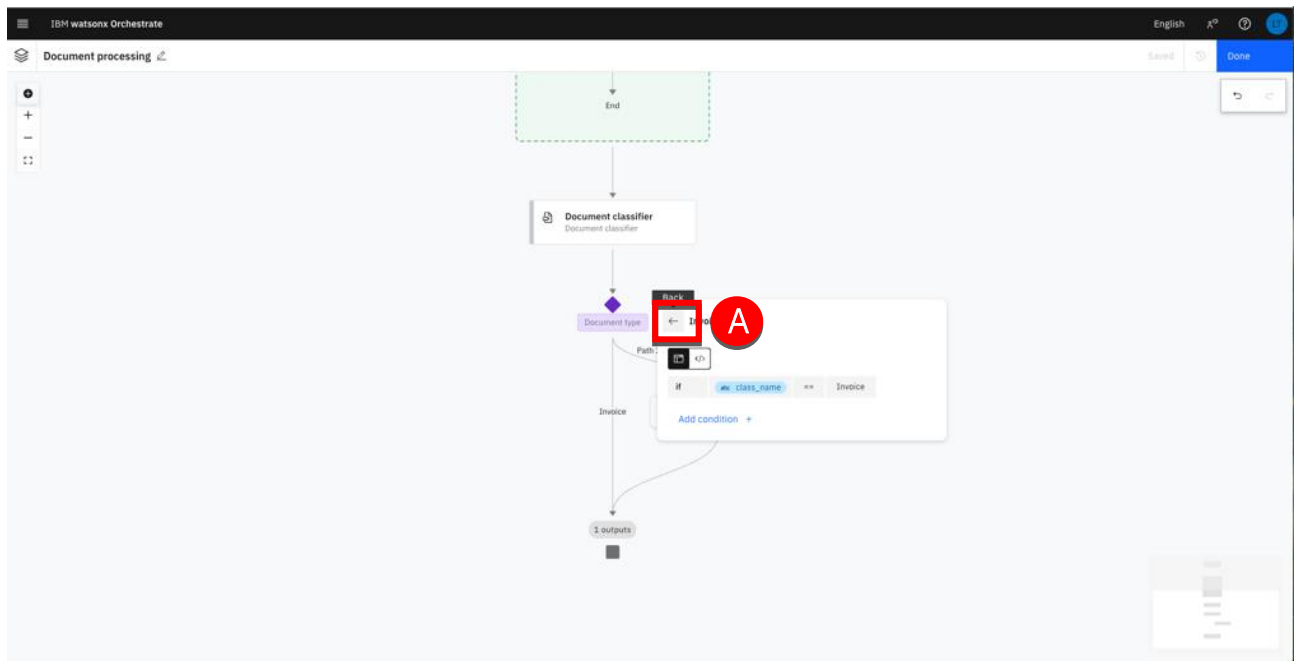
13. Type 'Invoice' (A) then hit Enter.



The process will be routed in this branch if the document type is Invoice.

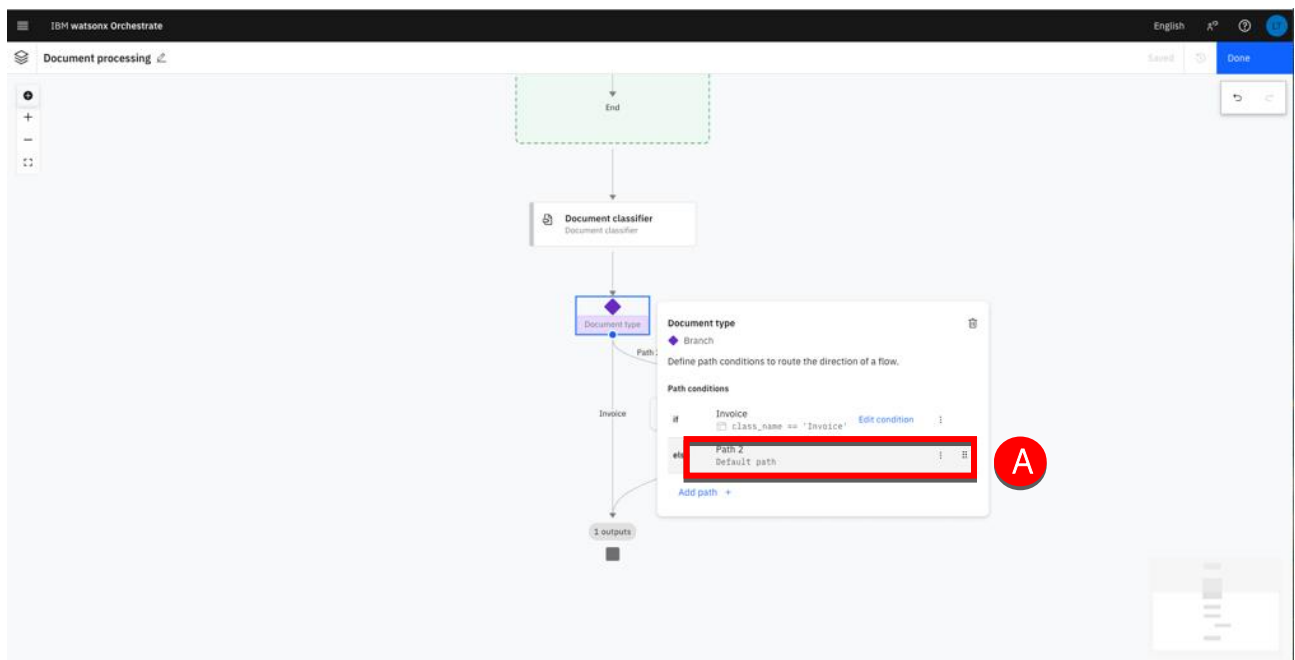
Note: Conditions can also be edited with the expression editor (</> icon). The equivalent expression to enter using the expression editor will then be: `flow["Document classifier"].output.class_name == "Contract"`

14. Click the **Back** icon (A).

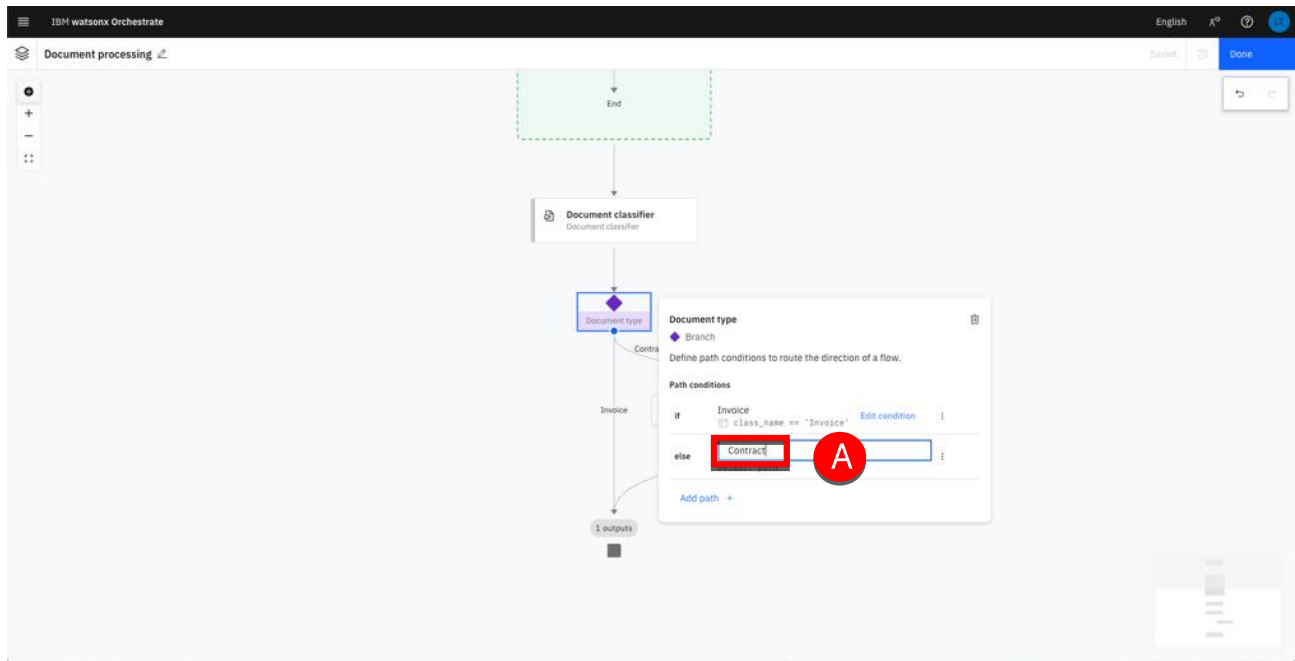


Let's just rename the second path Contract. As you will just have 2 types of documents, any document that is not an invoice (i.e. contracts) will be routed to this branch.

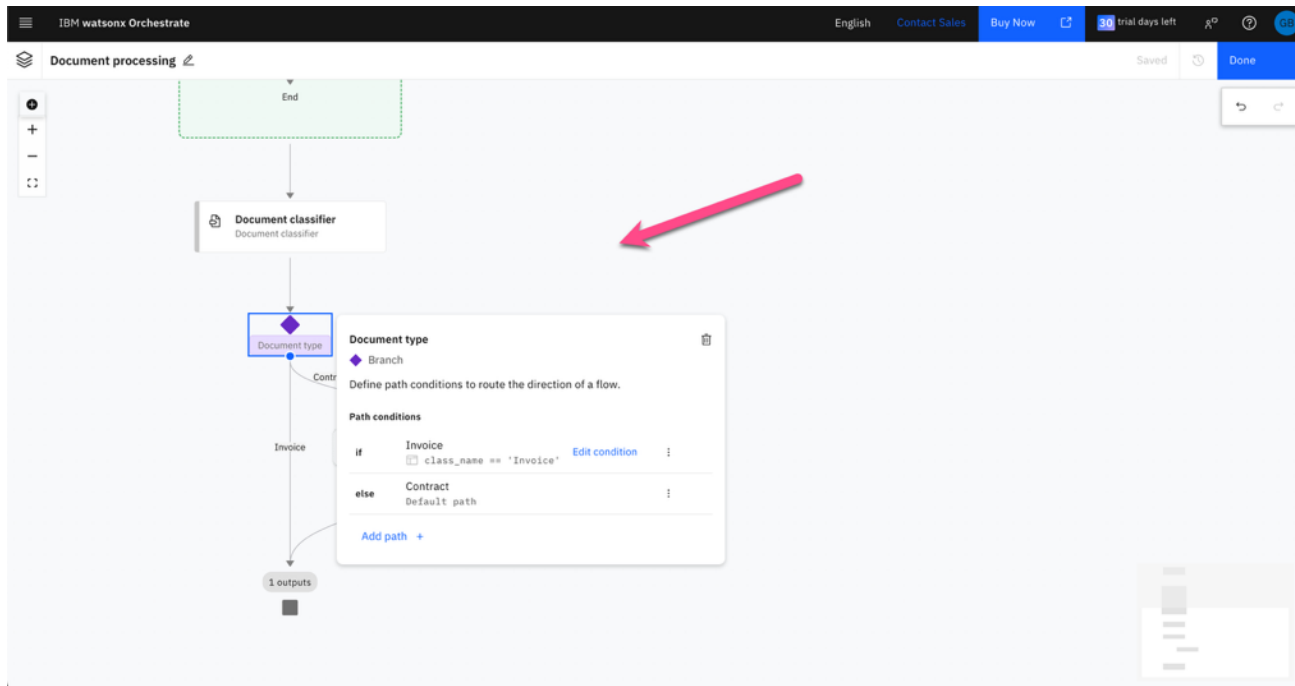
15. Click the **Path 2** row (A).



16. Type 'Contract' (A) then hit Enter.



17. Click anywhere on the canvas to close the **Document type** branch panel.



You are now ready to add the 2 document extractors, one for each document type.

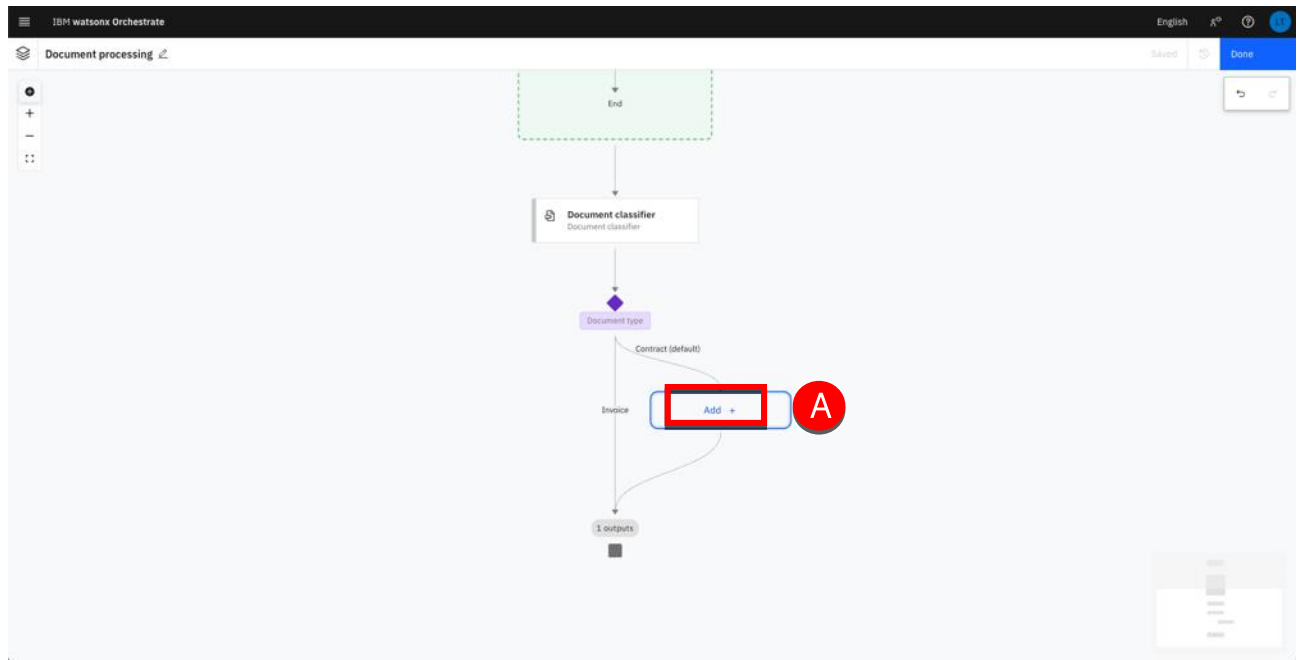
Add a document extractor

In this section, you will create two document extractors. Each document extractor will be responsible for extracting specific data from each document class. You will start with the Contract type of documents for which you will extract the following fields:

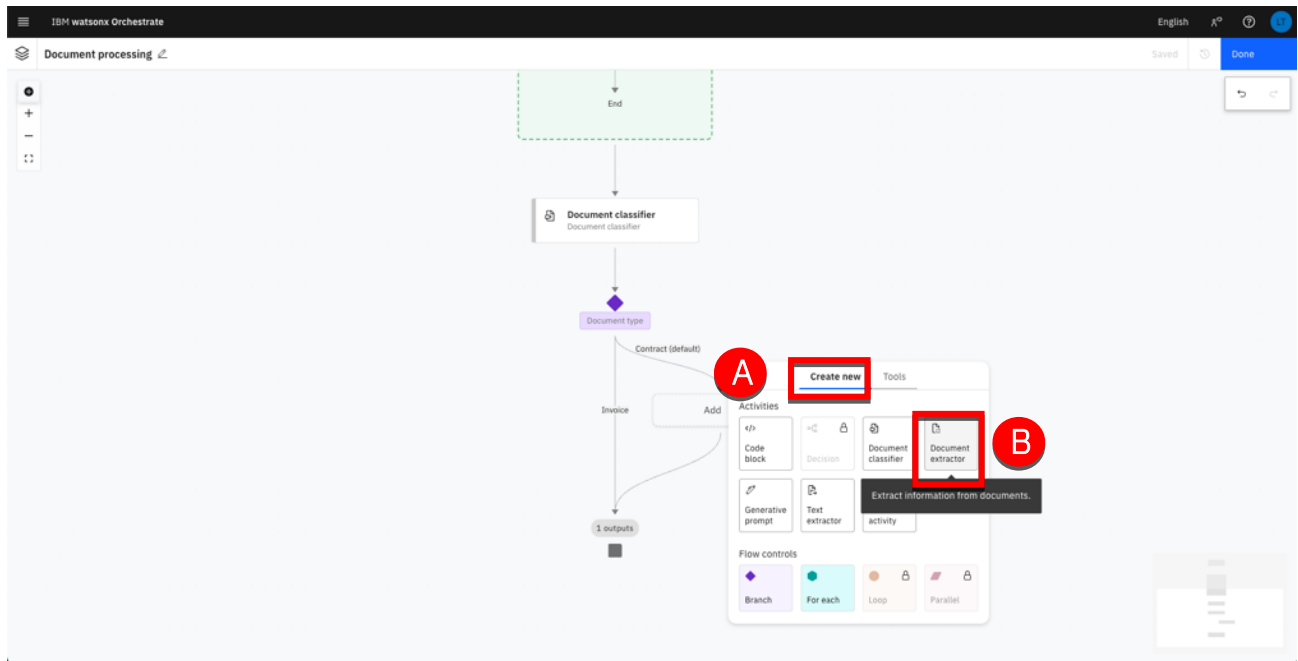
- Buyer
- Supplier
- Effective date

Note: During this section, extractors may return different values from the one shown in this guide. LLM are evolving and may produce slightly different results. In many situations, you will have to refresh the interface manually closing/re-opening or clicking on the interface to see the extracted values updated after a few seconds (minutes depending on your LLM shared usage).

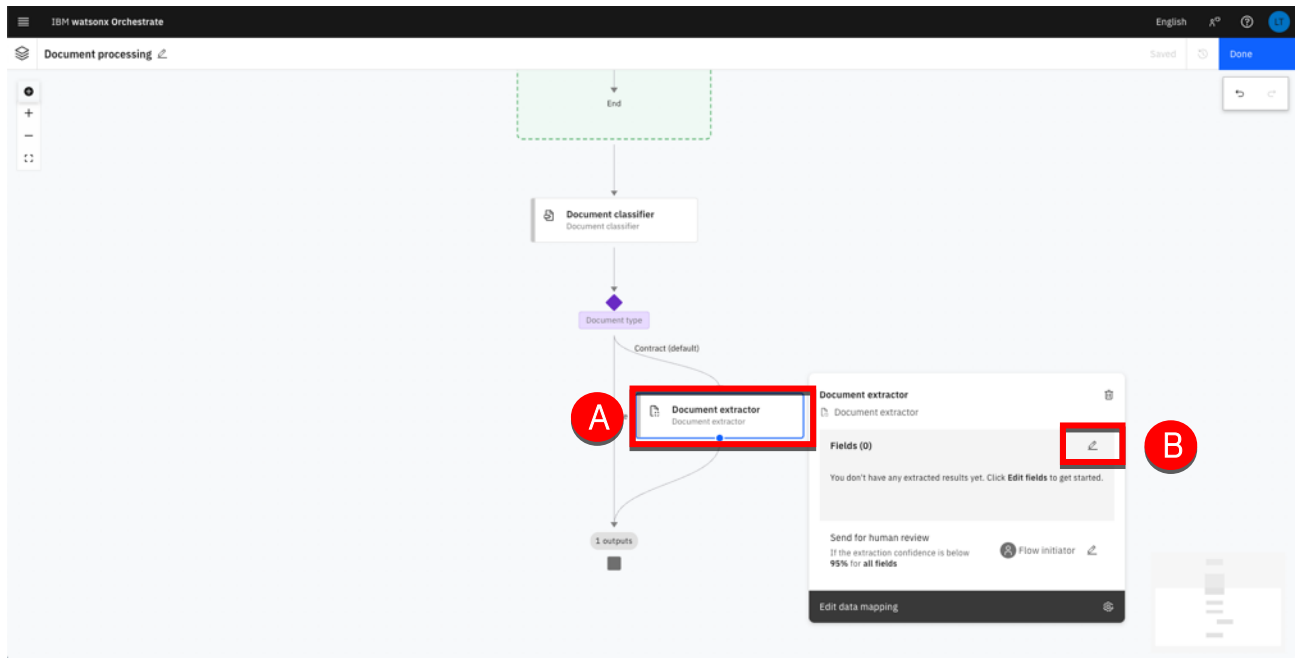
1. Click **Add + (A)**.



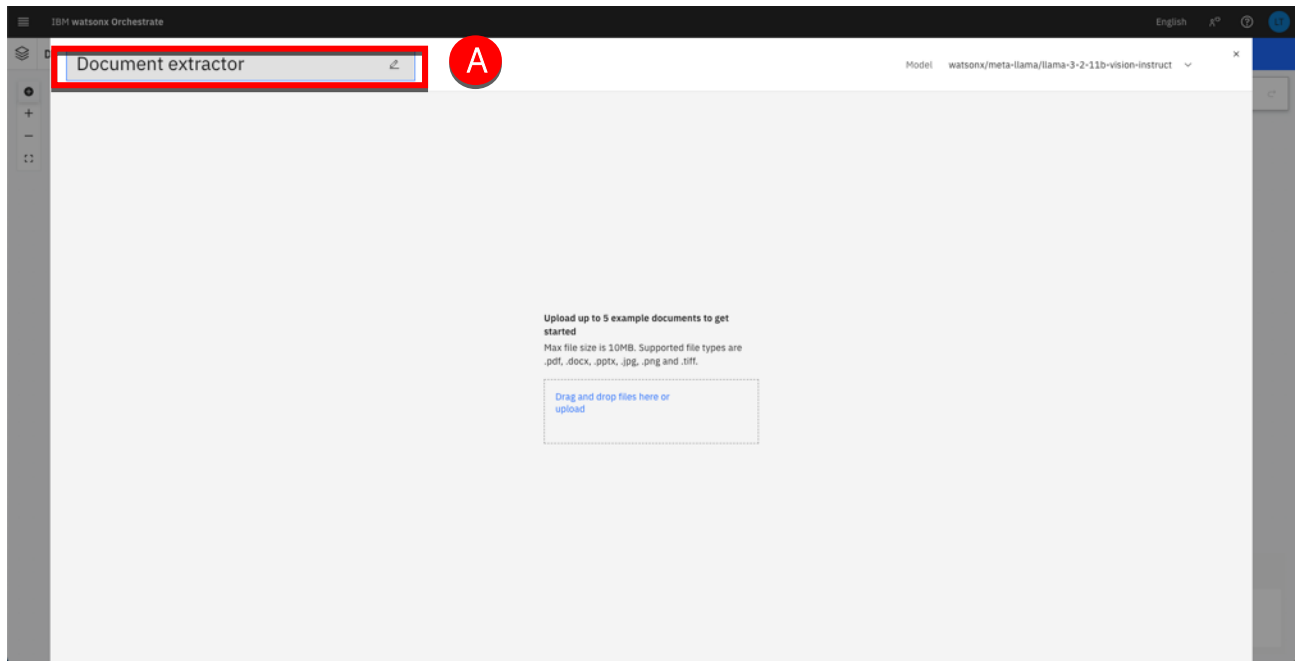
2. Select **Create new** (A) then click **Document extractor** (B).



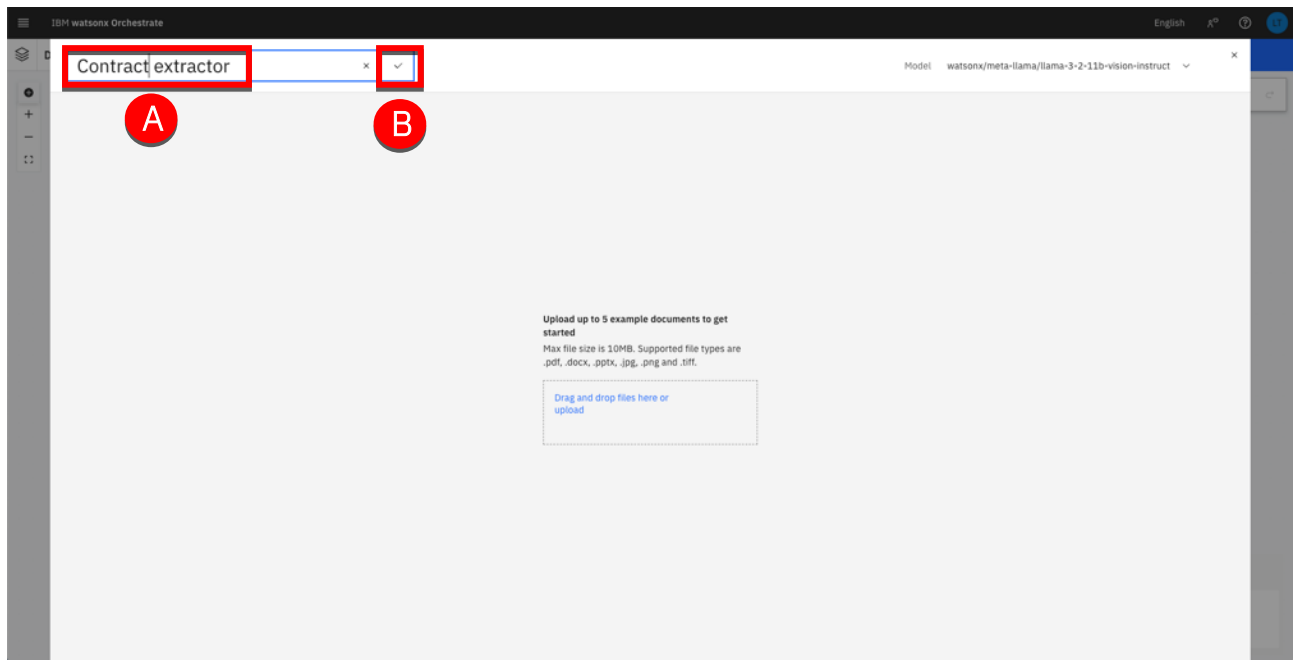
3. Click the **Document extractor** node (A) then click the edit icon (B).



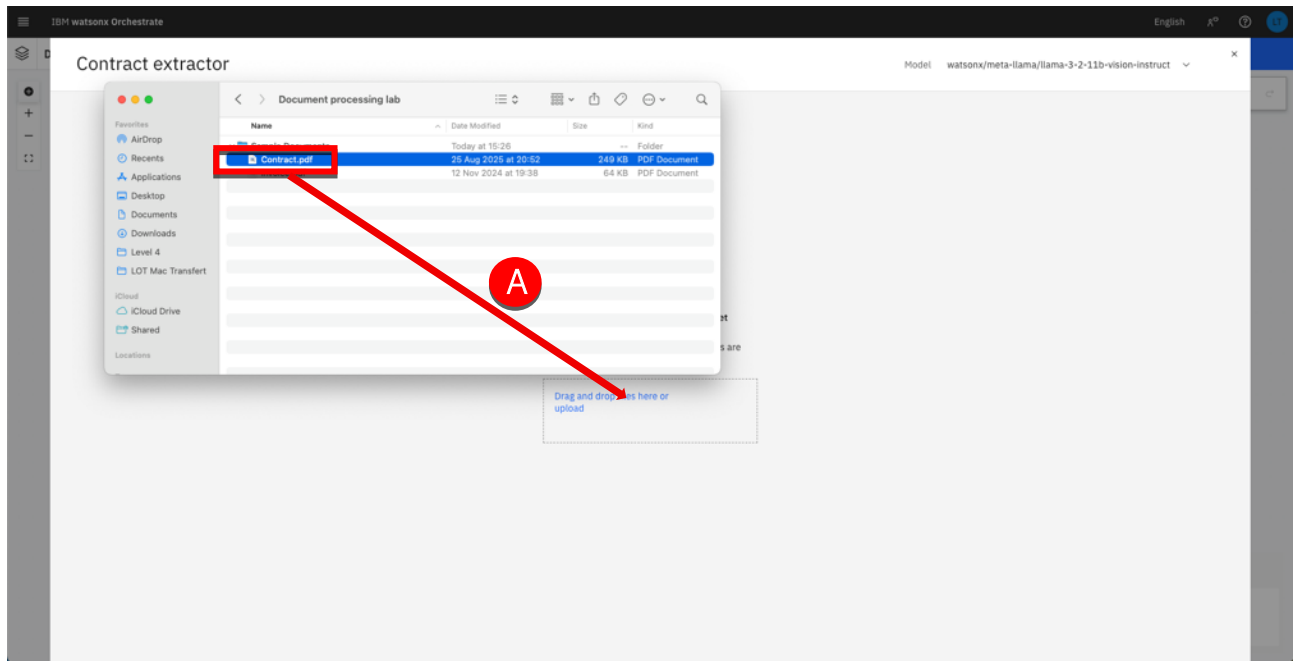
- Click the **Document extractor** name (A) to edit it.



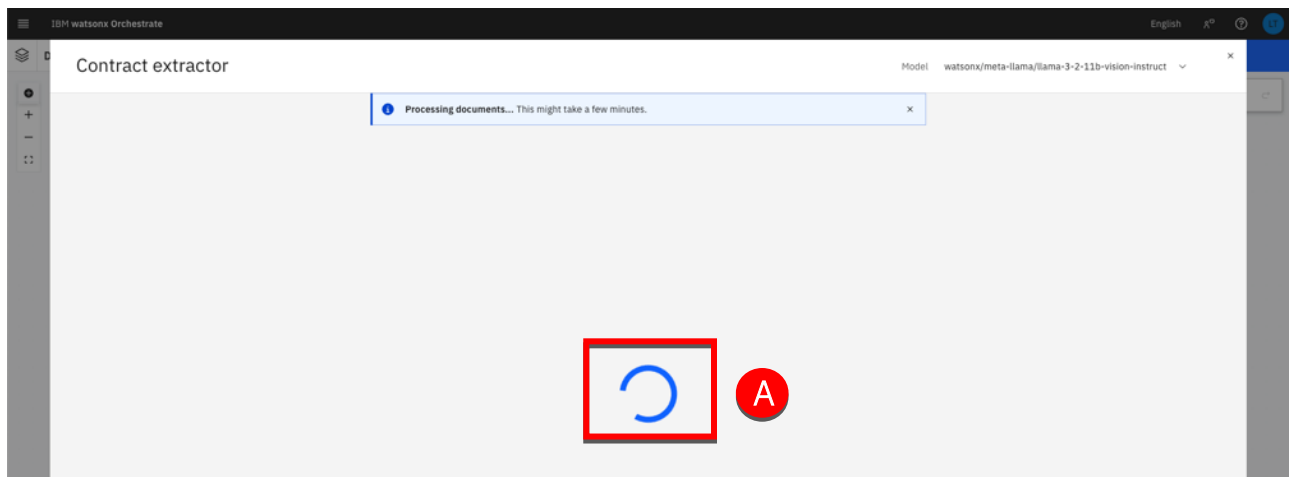
- Type 'Contract extractor' (A) as a new name then click the save icon (B).



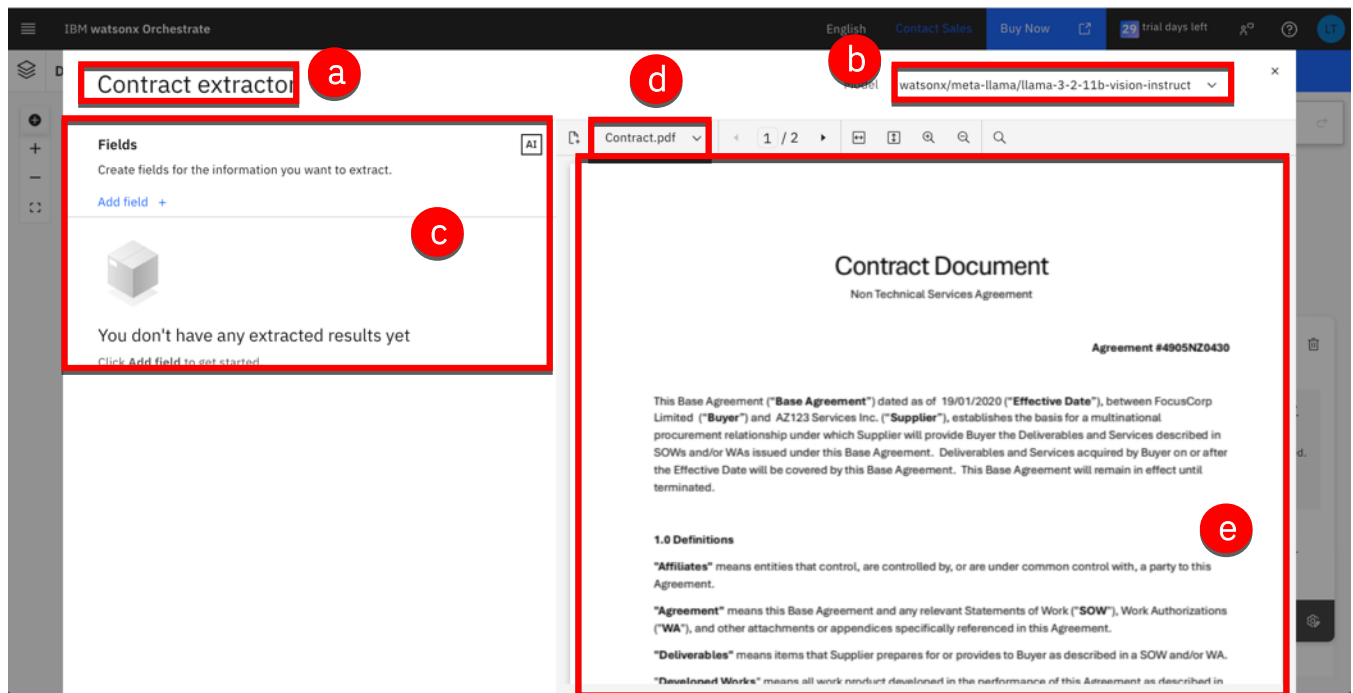
6. Drag and drop the **Contract.pdf** file (A) from your computer to the dropping area.



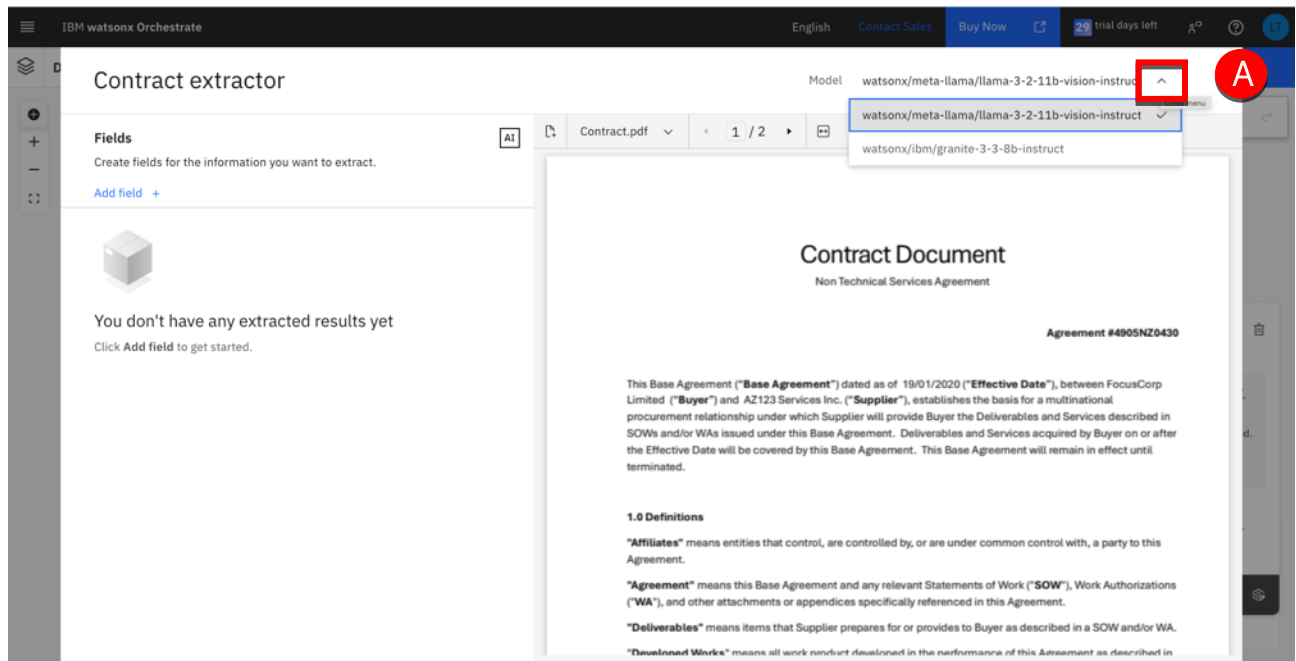
7. Wait for the document to be processed.



8. Once uploaded, the sample document is displayed, and the page is divided into the following key segments:
- a. Activity Name – Displays the name of the document extractor activity. Rename it to reflect the sample document you've uploaded.
 - b. LLM Model Selection – Choose your preferred LLM model from the drop-down menu. You can switch models at any time while prompting them for document extraction.
 - c. Field Definitions – Define the list of fields you want to extract from the sample document. The selected LLM will retrieve the corresponding values.
 - d. Document Navigation & Upload – Use the drop-down to navigate between uploaded documents. You can upload up to five additional documents for prompting and testing.
 - e. Document Viewer: Shows the uploaded document with the specified fields for extraction highlighted.



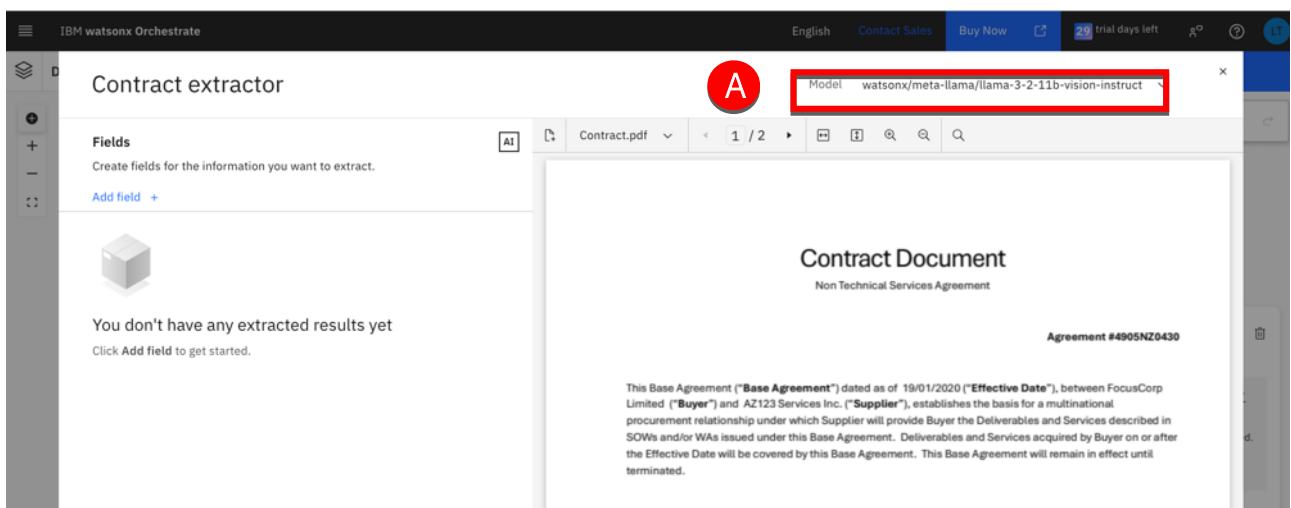
9. Click the model expand icon (A)



Note: You can change the LLM model by clicking on the 'Model' drop-down. Here you'll see all the available models that you can select. You can identify the current model that is being used, as you'll see a tick icon.

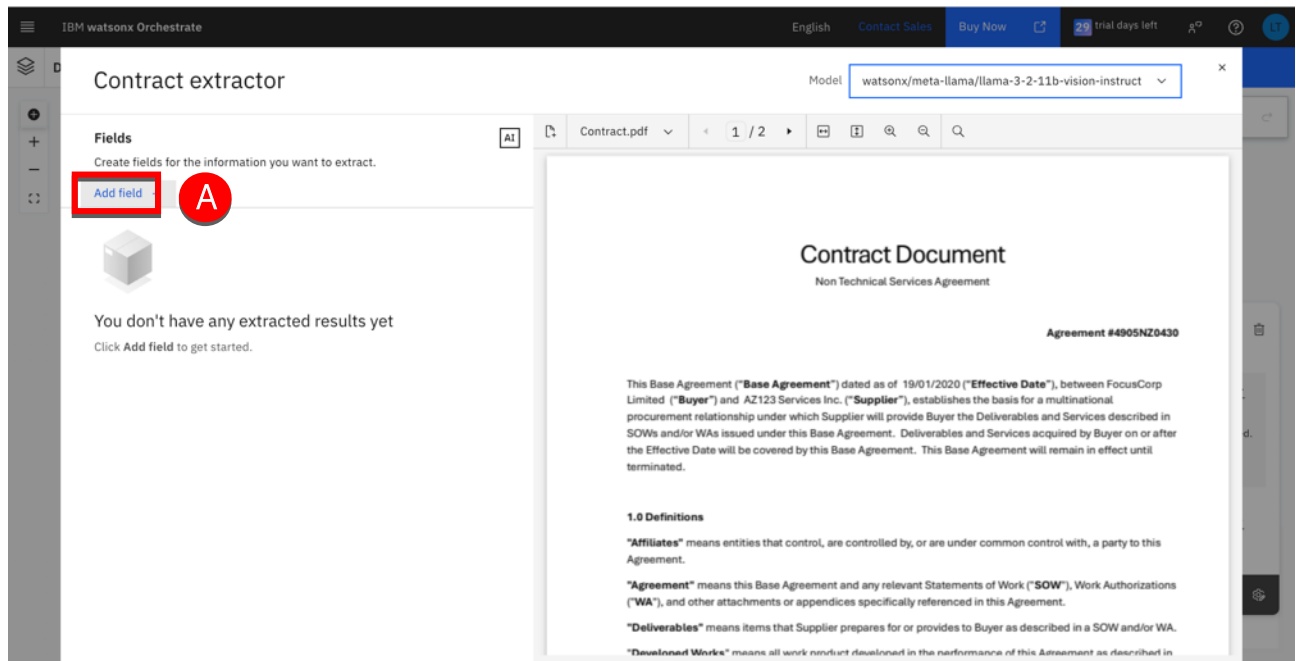
You can change the model at any time while you test to see which one is most accurate at extracting fields

10. Keep the meta-lama model (A).

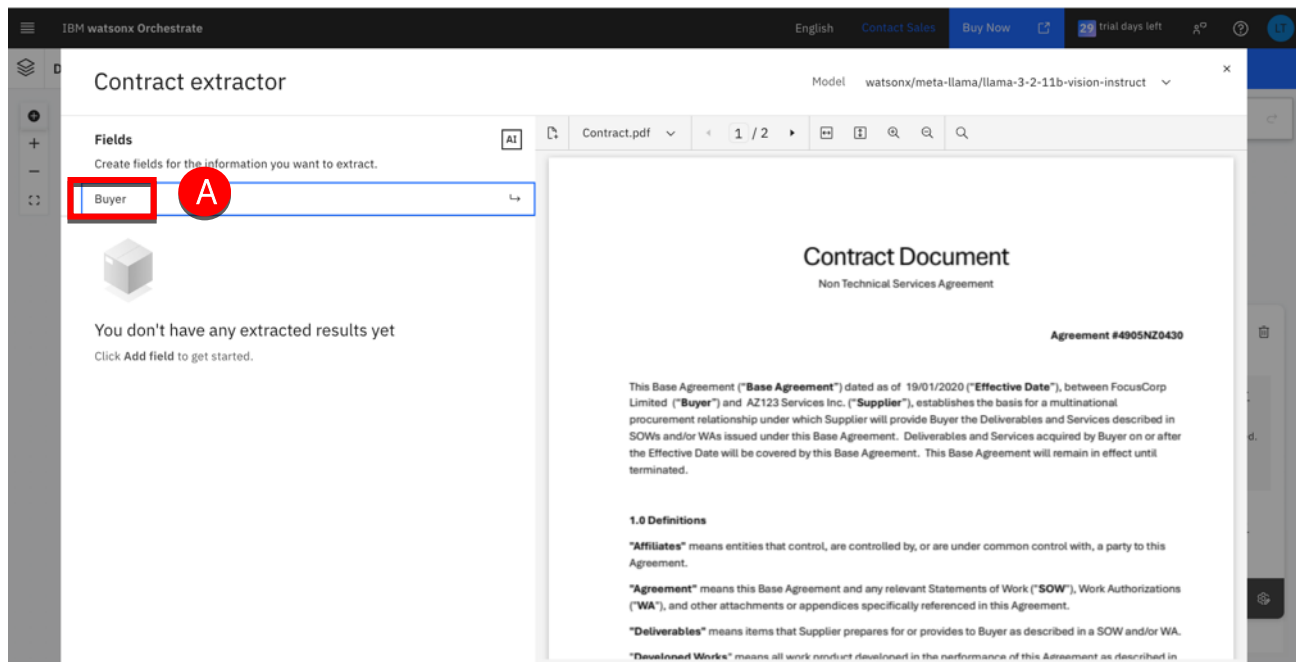


Let's now define the fields to extract.

11. Click Add field (A).



12. Type Buyer (A) and hit Enter.

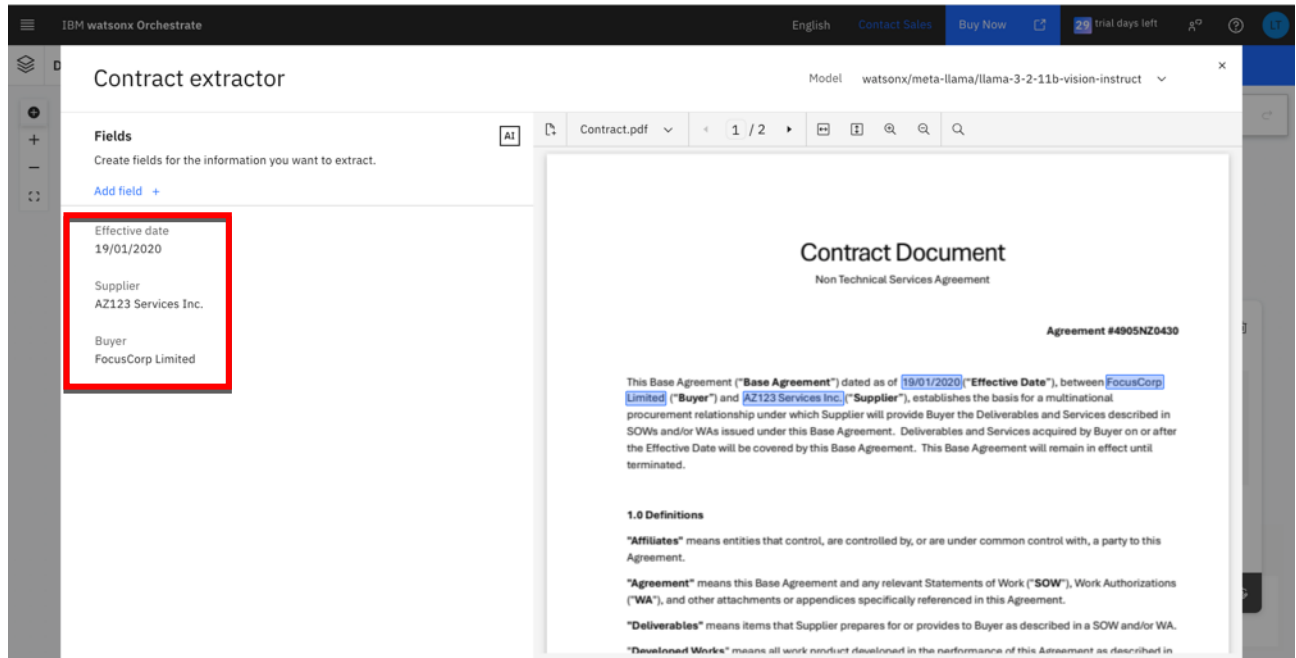


Note: The LLM will retrieve a value associated with the key you define here, which will auto-highlight in the document preview.

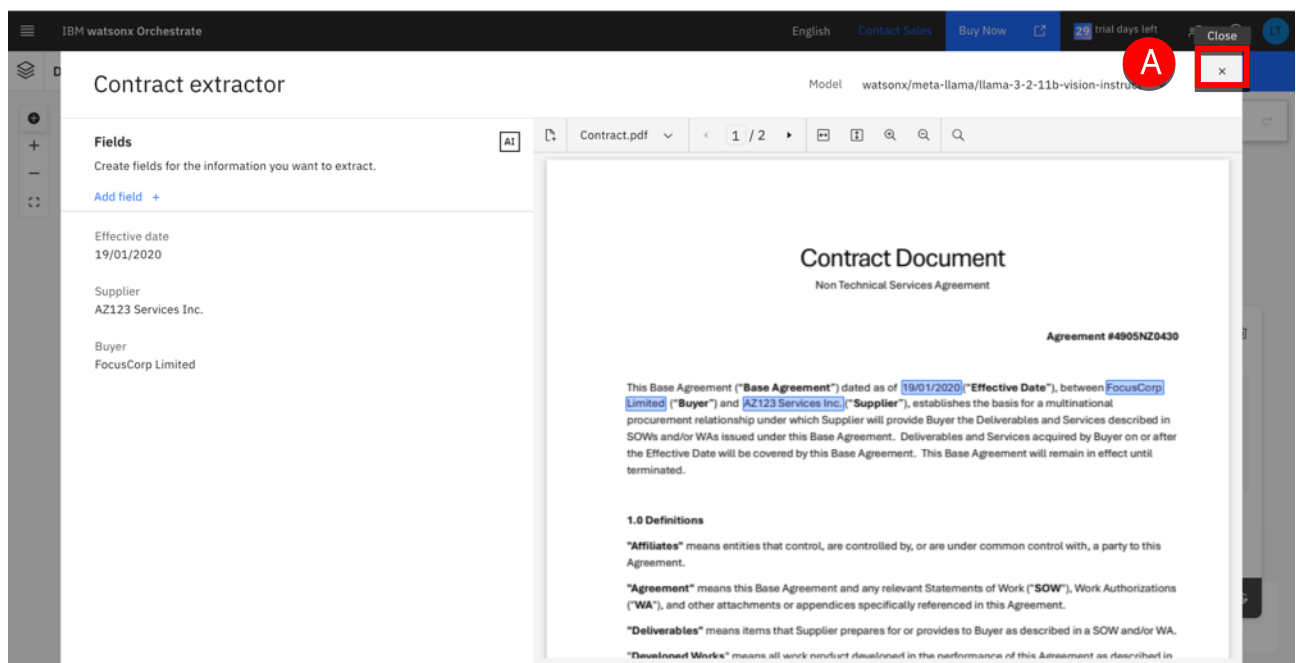
13. In the same way, add the 2 following fields:

- Supplier
- Effective date

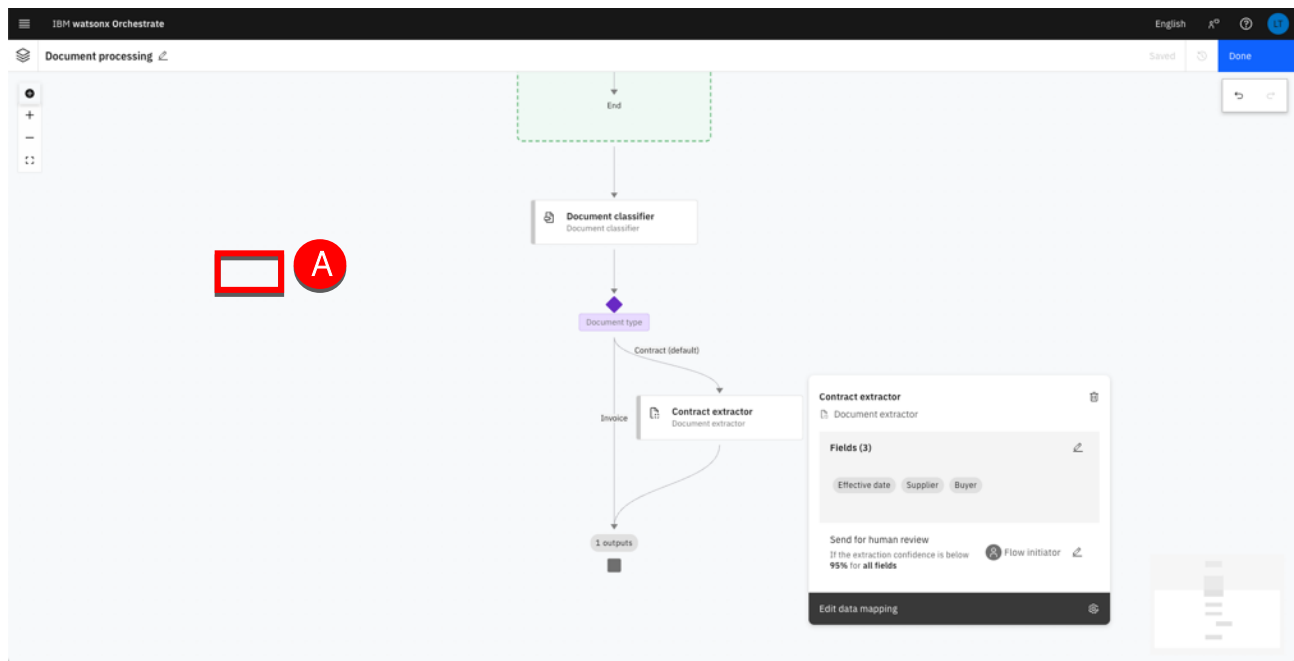
Your screen should look like:



14. Click x (A) to close the extractor.



15. Click in the background to close the **Contract extractor** property view (A).

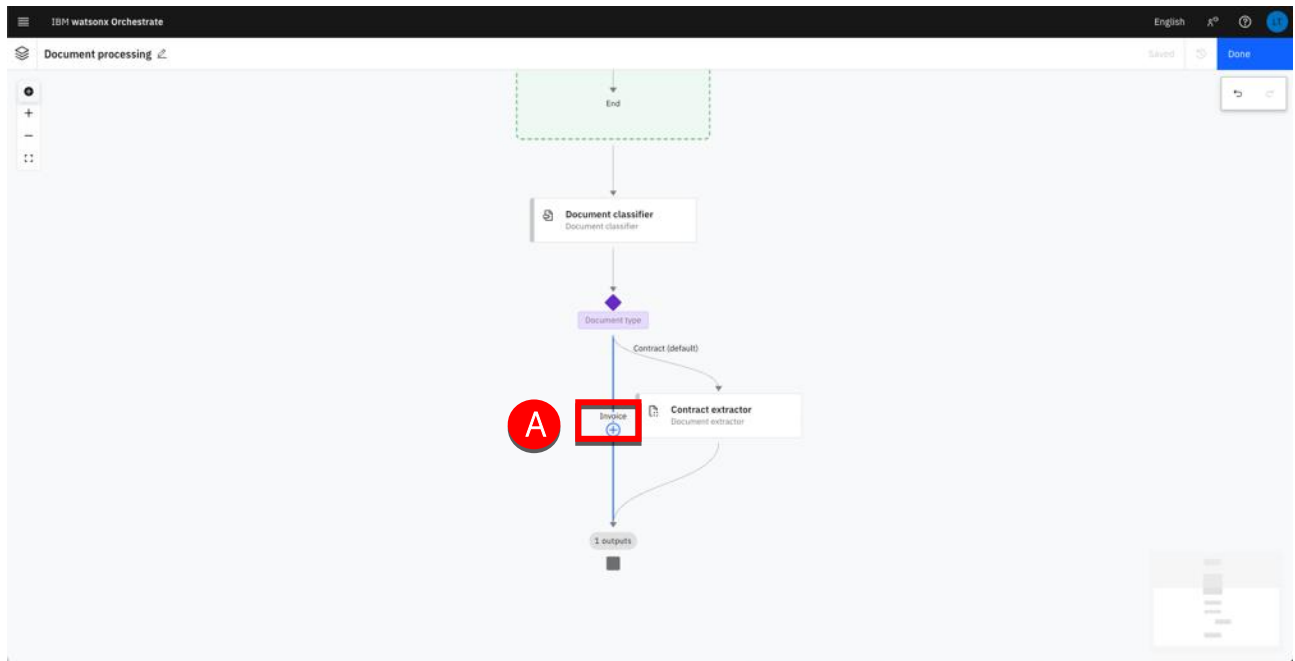


Next, you will create the Invoice extractor in the corresponding branch of the process. This extractor will be more complex and harder to implement. You will learn how to train the model to extract values from different kind of documents (PDF files or images).

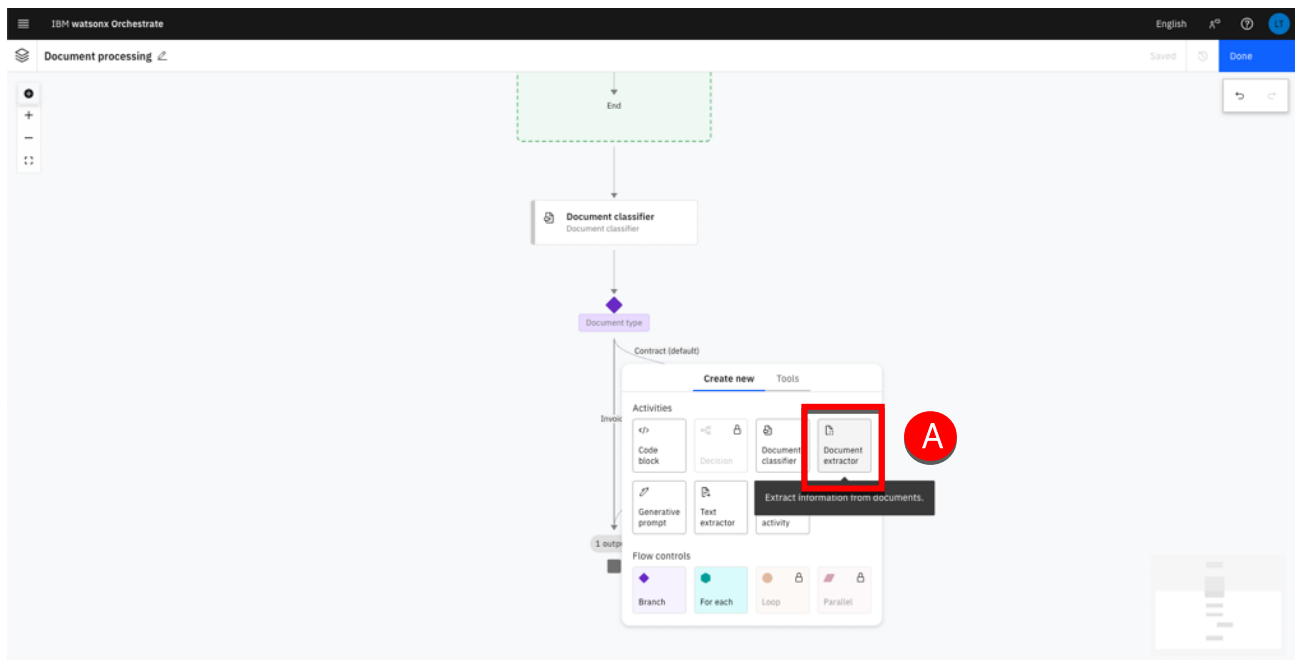
For the invoice documents, you will extract the following fields:

- Customer name
- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage

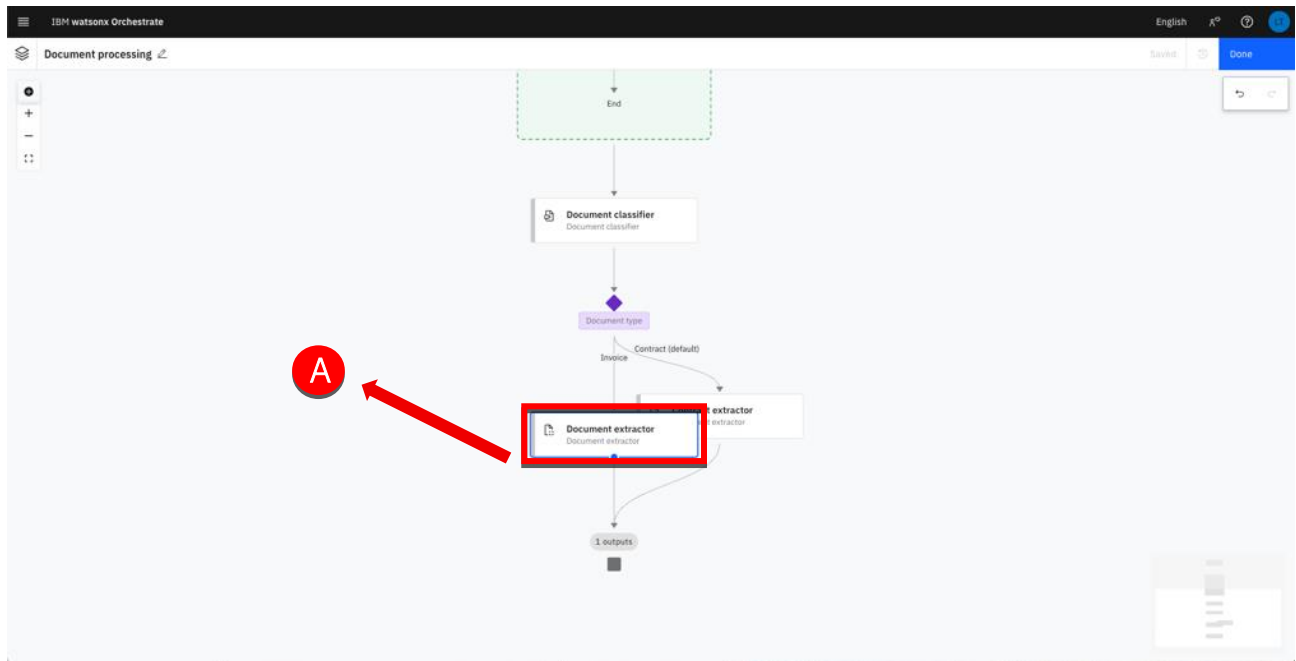
16. Move the mouse over the **Invoice** branch and click + (A)



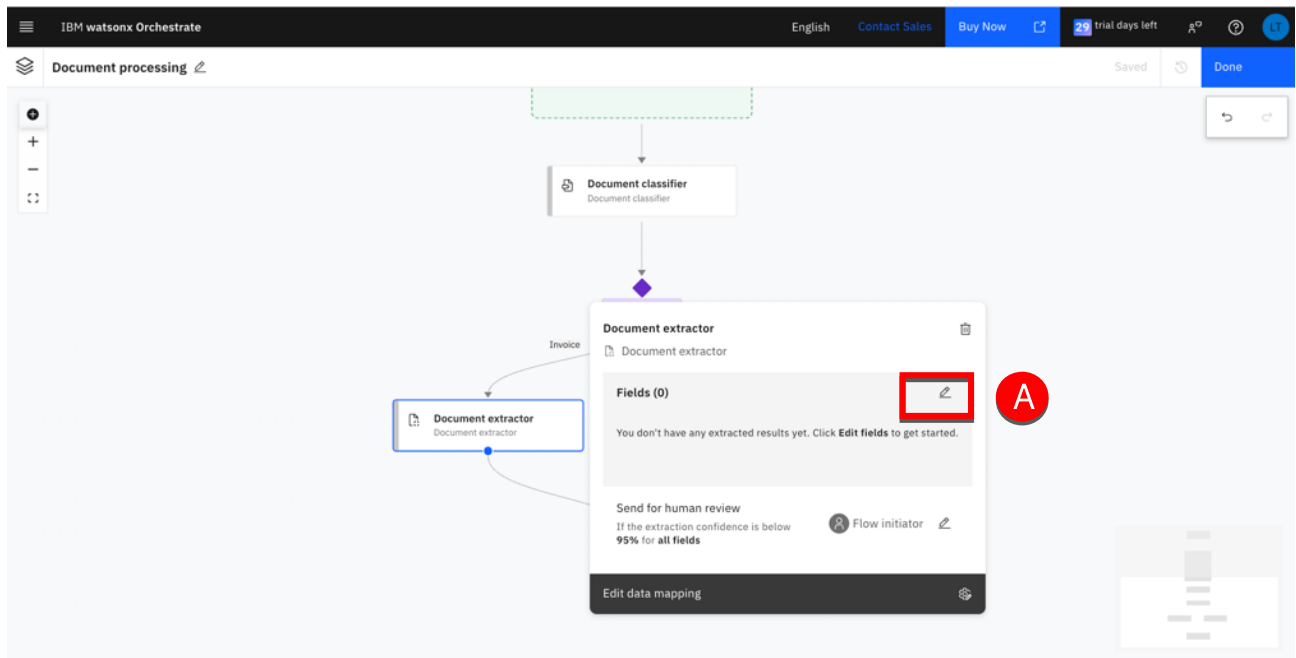
17. Select **Document extractor** (A).



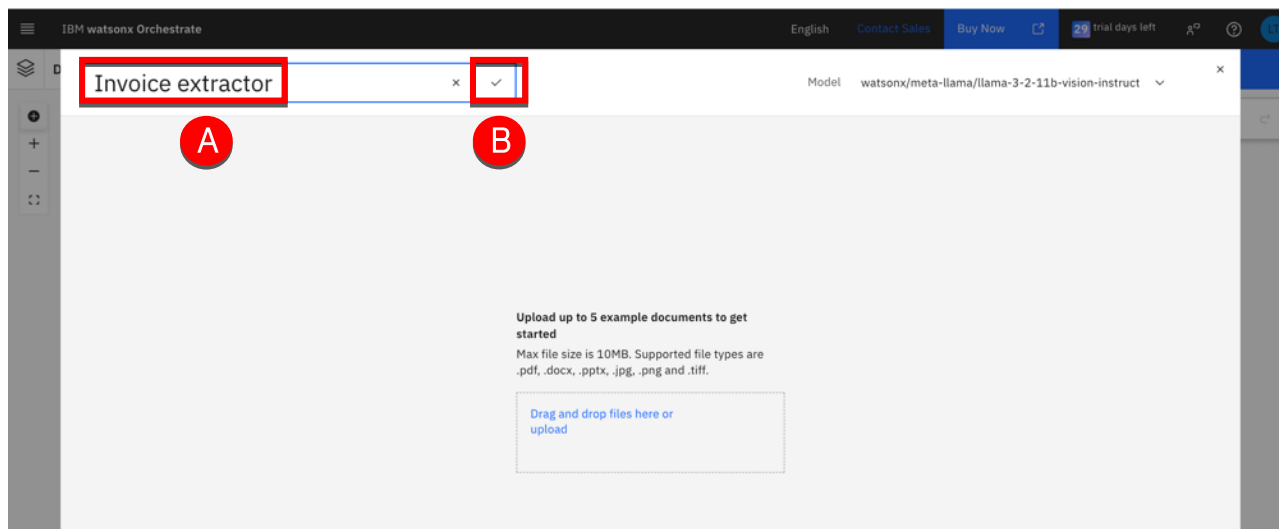
18. Click the **Document extractor** node and move it for a clearer layout (A).



19. Click the edit icon (A)

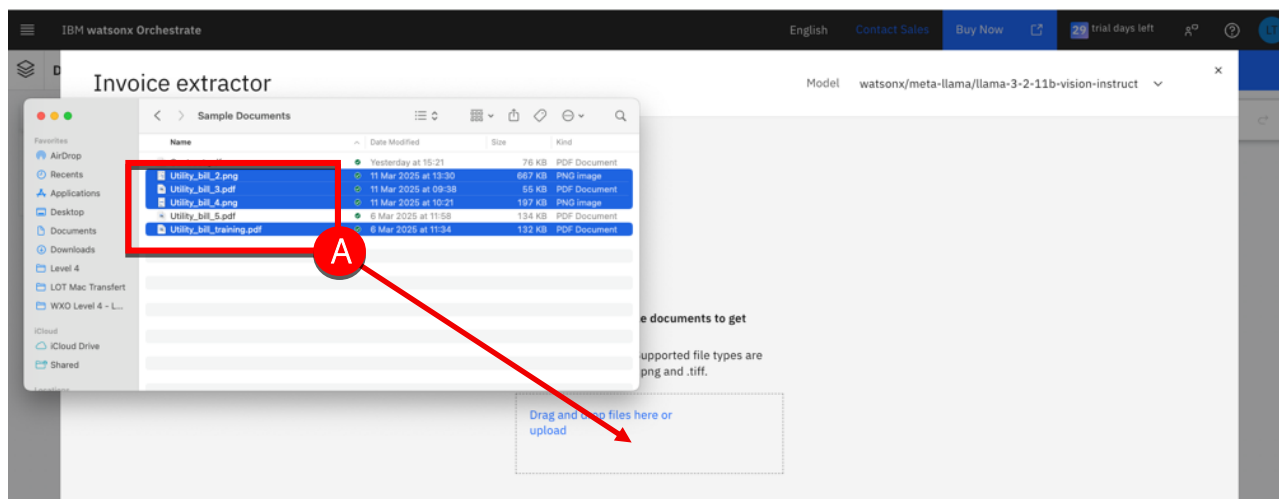


20. Type 'Invoice extractor' in the name field (A). Click the save icon (B).

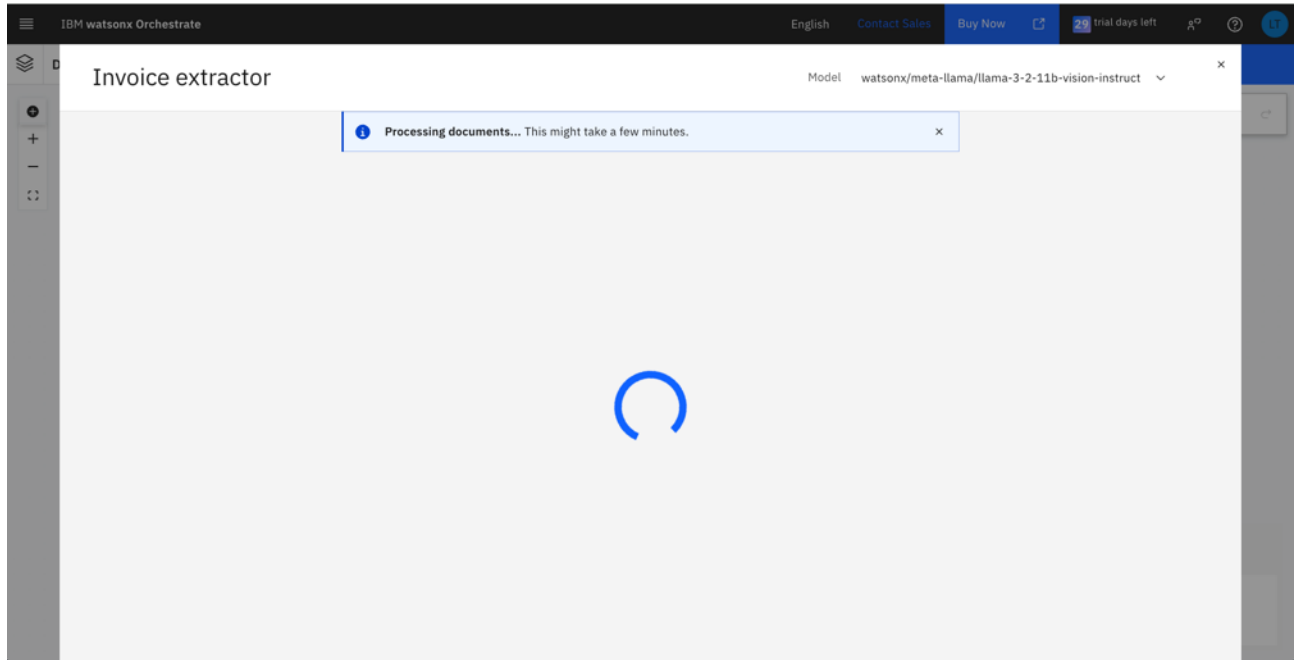


21. Drag and drop the four following files in the upload area (A)

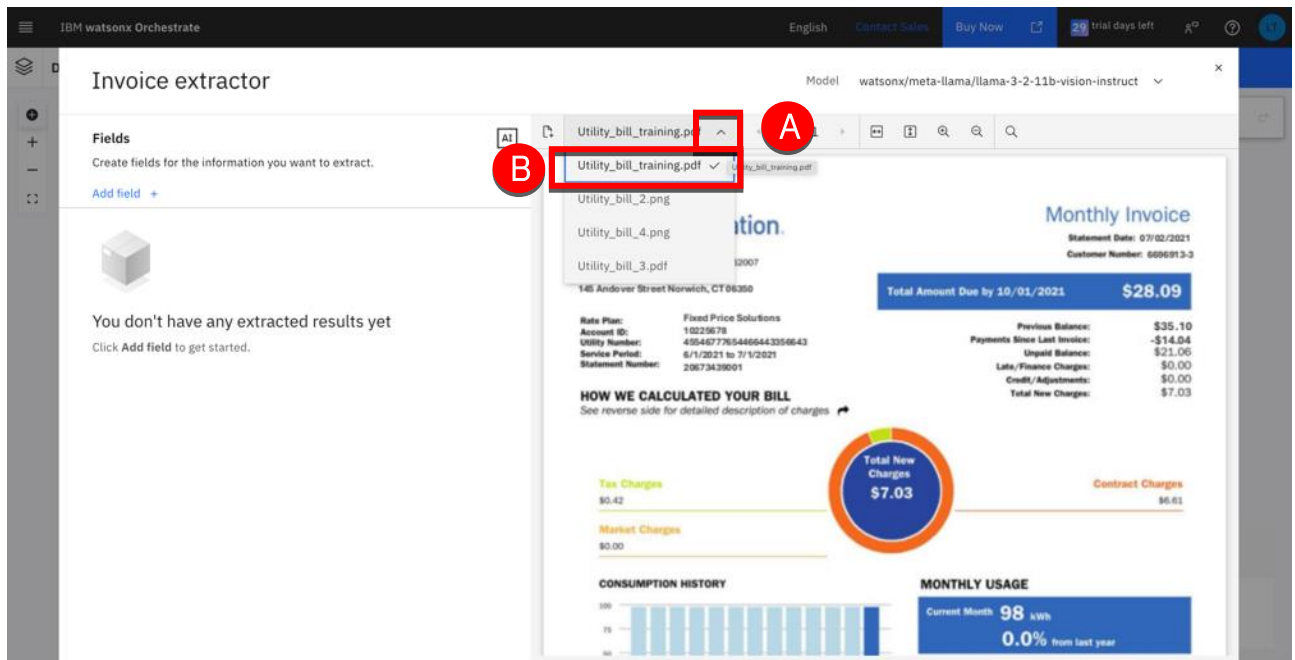
- Utility_bill_training.pdf
- Utility_bill_2.png
- Utility_bill_3.pdf
- Utility_bill_4.png



22. Wait for the 4 files to be processed.



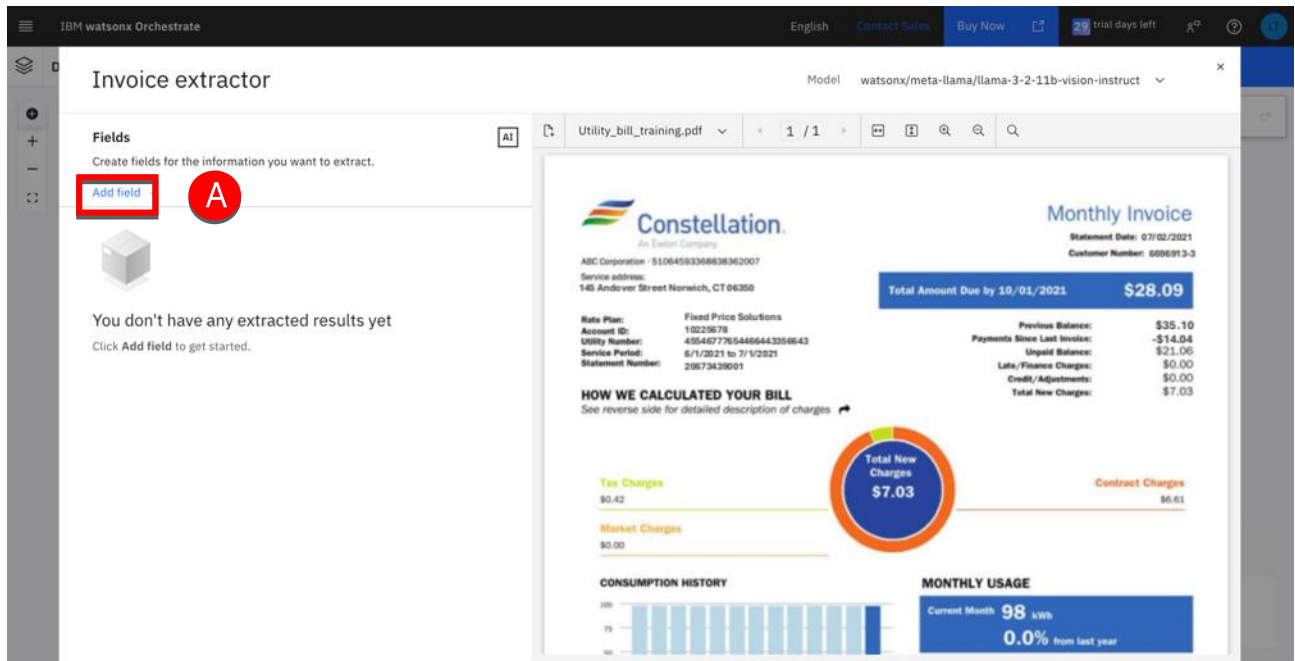
23. Expand the file selector (A) and select the `Utility_bill_training.pdf` file (B).



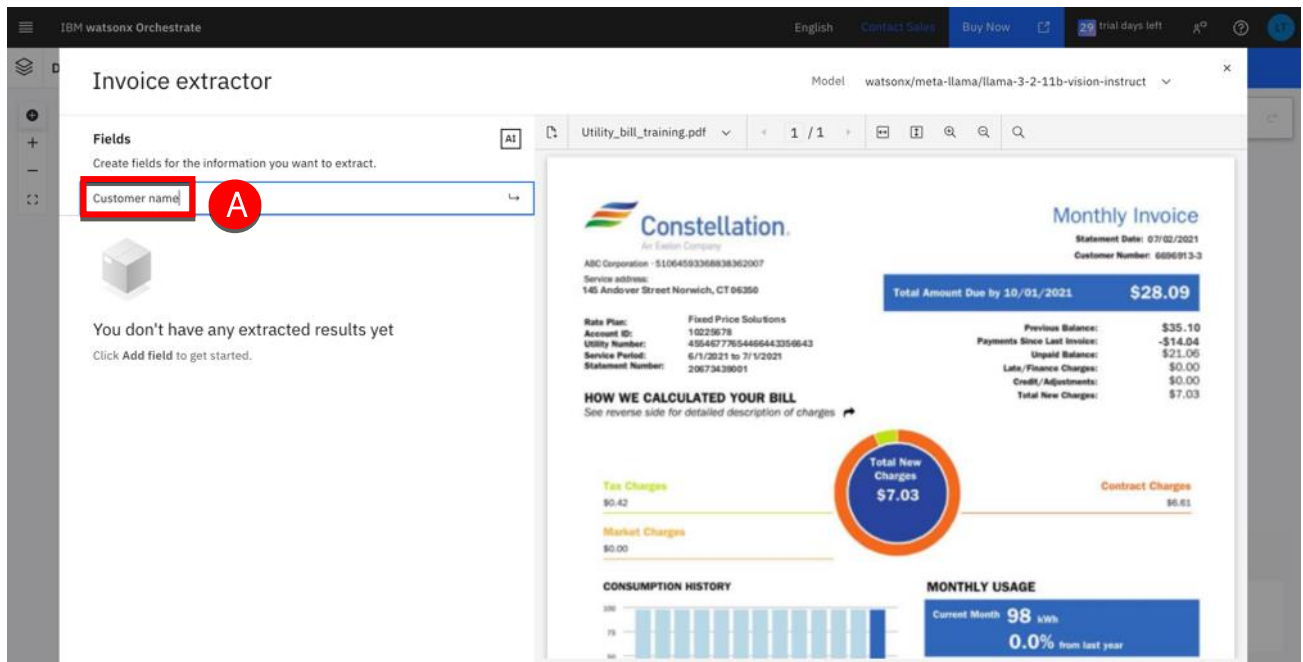
Define the fields to extract

You are now ready to specify the fields to be extracted.

1. Click Add field (A).



2. Type 'Customer name' into the field name (A) then press Enter.



- Click **Add field** (A).

The screenshot shows the 'Invoice extractor' interface in IBM watsonx Orchestrate. On the left, under the 'Fields' section, there is a button labeled 'Add field' which is highlighted with a red rectangle and a red circle labeled 'A'. Below this button, the 'Customer name' field is populated with 'ABC Corporation'. The main area of the interface displays a sample 'Monthly Invoice' from Constellation Energy. The invoice includes details such as the statement date (07/02/2021), customer number (6096913-3), and a total amount due of \$28.09. It also features a breakdown of charges (Tax, Market, Total New, Contract) and a monthly usage summary showing 98 kWh.

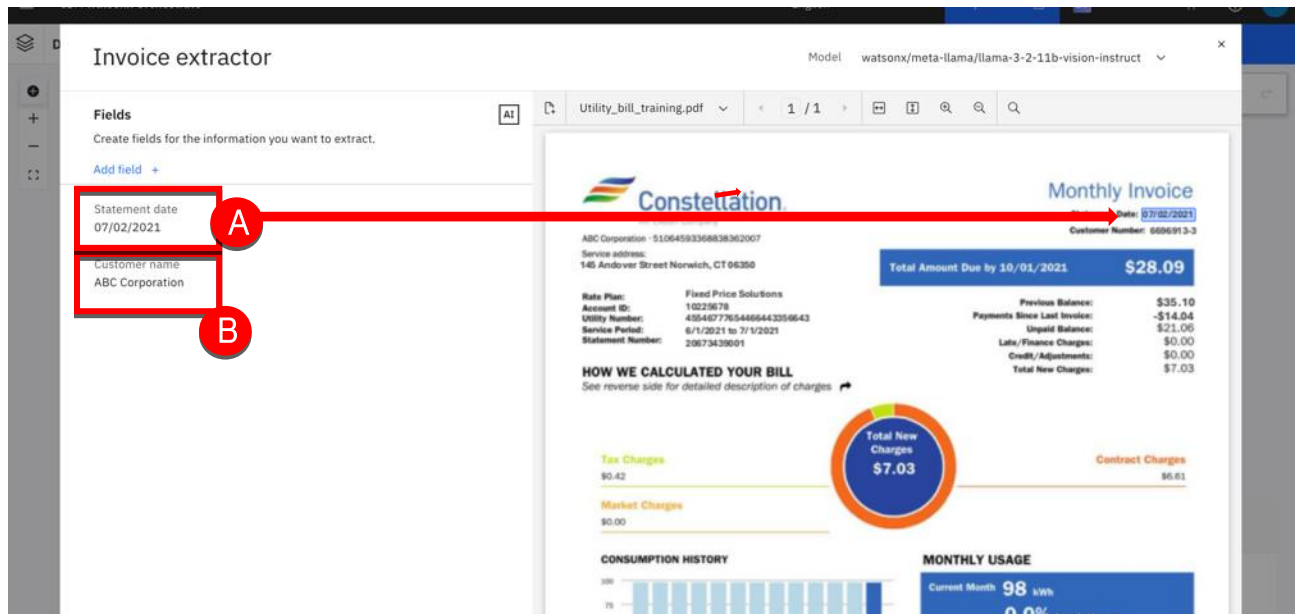
Note: The extractor may not recognize the field directly. We will see in the next step how to train the extractor.

- Enter 'Statement date' into the field name (A) then press Enter.

The screenshot shows the 'Invoice extractor' interface after the 'Statement date' field has been added. In the 'Fields' section on the left, the 'Statement date' field is now listed and highlighted with a red rectangle and a red circle labeled 'A'. The main area of the interface remains the same, displaying the sample 'Monthly Invoice' from Constellation Energy.

5. Review the extractor results:

- The **Statement date** was recognized (A).
- The **Customer name** was recognized (B).



Note: The recognition results you see may vary from these screenshots depending on the maturity of the LLM model and its ability to recognize utility bill fields as LLM models are evolving rapidly.

You will add additional fields described in the next section before training the model to extract the missing fields.

6. Repeat the steps above to add the following fields (A):

- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage

The screenshot shows the IBM watsonx Orchestrate interface. On the left, under the 'Fields' section, a list of fields to be extracted is shown, enclosed in a red box and labeled 'A':

- Year to year usage
98 kWh
- Monthly usage
98 kWh
- Due date
10/01/2021
- Amount due
\$28.09
- Customer address
145 Andover Street Norwich, CT 06350
- Customer number
6696913-3

The main area displays a sample utility bill from Constellation. The bill includes the following information:

- Constellation** (An Exelon Company)
- ABC Corporation - 510645933688362007
- Service address: 145 Andover Street Norwich, CT 06350
- Rate Plan: Fixed Price Solutions
- Account ID: 10225678
- Utility Number: 40546777654466443350643
- Service Period: 6/1/2021 to 7/1/2021
- Statement Number: 20673439001
- Statement Date: 07/02/2021
- Customer Number: 6696913-3
- Total Amount Due by 10/01/2021: \$28.09
- Previous Balance: \$35.10
- Payments Since Last Invoice: -\$14.04
- Unpaid Balance: \$21.06
- Late/Fines/Charges: \$0.00
- Credit/Adjustments: \$0.00
- Total New Charges: \$7.03
- HOW WE CALCULATED YOUR BILL: See reverse side for detailed description of charges.
- Tax Charges: \$0.42
- Market Charges: \$0.00
- CONSUMPTION HISTORY: A bar chart showing usage over time.
- MONTHLY USAGE: Current Month 98 kWh, 0.0% from last year.

Note: In this example, the extractor was able to correctly recognize all the fields. This may not be true in your situation. We will see in the next chapter how to improve the model.

Test on an image file

In the previous step, the document extractor has been set-up using a pdf document. Scanned images can also be used in the same way. Let's use an image of the same document populated with different data.

7. Click the manage document icon (A).

IBM watsonx Orchestrate

Invoice extractor

Model: watsonx/meta-llama/llama-3-2-11b-vision-instruct

Utility_bill_training.pdf

Fields

Create fields for the information you want to extract.

Add field +

Year to year usage
98 kWh

Monthly usage
98 kWh

Due date
10/01/2021

Amount due
\$28.09

Customer address
145 Andover Street Norwich, CT 06350

Customer number
6696913-3

Statement date
07/02/2021

Customer name
ABC Corporation

Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Rate Plan: Fixed Price Solutions
Account ID: 10229678
Utility Number: 49546777654466443356643
Service Period: 6/1/2021 to 7/1/2021
Statement Number: 20673439001

Previous Balance: \$35.10
Payments Since Last Invoice: -\$14.04
Unpaid Balance: \$21.06
Late/Finance Charges: \$0.00
Credit/Adjustments: \$0.00
Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL
See reverse side for detailed description of charges

Tax Charges: \$0.42
Market Charges: \$0.00
Total New Charges: \$7.03
Contract Charges: \$6.61

CONSUMPTION HISTORY

MONTHLY USAGE
Current Month: 98 kWh
0.0% from last year

8. Select Utility bill 2.png (A).

IBM watsonx Orchestrate

Invoice extractor

Model: watsonx/meta-llama/llama-3-2-11b-vision-instruct

Utility_bill_training.pdf

Utility_bill_training.pdf ✓

Utility_bill_2.png

Utility_bill_4.png

Utility_bill_3.png

Fields

Create fields for the information you want to extract.

Add field +

Year to year usage
98 kWh

Monthly usage
98 kWh

Due date
10/01/2021

Amount due
\$28.09

Customer address
145 Andover Street Norwich, CT 06350

Customer number
6696913-3

Statement date
07/02/2021

Customer name
ABC Corporation

Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Rate Plan: Fixed Price Solutions
Account ID: 10229678
Utility Number: 49546777654466443356643
Service Period: 6/1/2021 to 7/1/2021
Statement Number: 20673439001

Previous Balance: \$35.10
Payments Since Last Invoice: -\$14.04
Unpaid Balance: \$21.06
Late/Finance Charges: \$0.00
Credit/Adjustments: \$0.00
Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL
See reverse side for detailed description of charges

Tax Charges: \$0.42
Market Charges: \$0.00
Total New Charges: \$7.03
Contract Charges: \$6.61

CONSUMPTION HISTORY

MONTHLY USAGE
Current Month: 98 kWh
0.0% from last year

9. Note that some fields may not have been recognized with this image version of the document (A).

The screenshot displays the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' agent is configured with a list of fields to extract from a document. A red box highlights the 'Year to year usage' and 'Customer name' fields, both of which show 'No results found. Try editing the field details.' The right panel shows the extracted data from a Constellation utility bill. The bill includes account details, charges, and usage information. A red circle with the letter 'A' is placed over the 'Year to year usage' field in the left panel.

Fields to Extract:

- Year to year usage
No results found. Try editing the field details.
- Monthly usage
98 kWh
- Due date
10/01/2021
- Amount due
\$13.07
- Customer address
145 Andover Street Norwich, CT 06350
- Customer number
6696913-3
- Statement date
07/02/2021
- Customer name
No results found. Try editing the field details.

Extracted Data (Constellation Monthly Invoice):

- Statement Date: 07/02/2021
- Customer Number: 6696913-3
- Total Amount Due by 10/01/2021: \$13.07
- Rate Plan: Fixed Price Solutions
- Account ID: 10225678
- Utility Number: 45546777654466443356643
- Service Period: 6/1/2021 to 7/1/2021
- Statement Number: 20673439001
- Previous Balance: \$35.10
- Payments Since Last Invoice: -\$14.04
- Unpaid Balance: \$21.06
- Late/Finance Charges: \$0.00
- Credit/Adjustments: \$0.00
- Total New Charges: \$7.03
- Contract Charges: \$6.61
- Current Month: 98 kWh
- 0.0% from last year

Note: It may take a few seconds before the screen is refreshed with the extracted values. If all the fields are showing “No result found”, refresh the view by switching to another document, then back again.

Note: We will work to improve the LLM model in the next chapter.

Test on a new document model

Not all invoices will have the same structure. In this section, you will test your document extractor on a utility invoice from a different energy company.

10. Click the expand icon to expand the document list (A) then select **Utility bill 4.png** (B).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Fields' section lists extracted data: Year to year usage (+24%), Monthly usage (124 kWh), Due date (May 15, 2025), Amount due (\$45.99), Customer address (Gran plaza D 3rd Floor-Appt 345 234 Main Street CA 1234, Los Angeles), Customer number (FC-45678), Statement date (May 5th, 2025), and Customer name (Mr. Smith bill). The main view displays a 'Monthly Invoice' from 'FCE' with various fields extracted, including 'Year to year usage', 'Monthly usage', 'Due date', 'Amount due', 'Customer address', 'Customer number', 'Statement date', and 'Customer name'. The invoice also shows a 'Total Amount Due' of \$13.07 and a 'MONTHLY USAGE' of 98 kWh.

11. Note that some fields may not have been correctly extracted with this new version of the document (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Fields' section lists extracted data: Year to year usage (+24%), Monthly usage (124 kWh), Due date (May 15, 2025), Amount due (\$45.99), Customer address (Gran plaza D 3rd Floor-Appt 345 234 Main Street CA 1234, Los Angeles), Customer number (FC-45678), Statement date (May 5th, 2025), and Customer name (Mr. Smith bill). The main view displays a 'Monthly Invoice' from 'FCE' with various fields extracted, including 'Year to year usage', 'Monthly usage', 'Due date', 'Amount due', 'Customer address', 'Customer number', 'Statement date', and 'Customer name'. The invoice also shows a 'Total Amount Due' of \$13.07 and a 'MONTHLY USAGE' of 98 kWh.

The document extractor may have been unable to recognize and extract all the fields with this new version of the document, and some errors may also be present. You will improve the results from the extractor in the next section by giving better descriptions of the fields and providing examples.

Note: Based on your LLM history and version, un-identified fields or errors may differ from the one presented in this lab. You may have to adapt the following script to correct errors encountered with your extractor.

Improve your extractor

In this section, you will apply techniques to improve your results, helping the model to better identify and extract relevant data from your unstructured documents.

Add field descriptions

In our example (that may differ from yours), the Statement date is extracted but requires a reformatting. We would like all the dates to be formatted the same way with the mm/dd/yyyy format. Let's add a precise description of the field that will be used by the LLM to better extract the information in the format we want.

12. Identify a field displaying an error (A) then click the **View field details** icon (B).

13. Enter 'The statement date is the date the invoice was issued. Return the date using in the MM/DD/YYYYY format. May 1st, 2025 will be displayed as 05/01/2025. ' into the **Description** field (A), then click **Show on document** (B).

The screenshot shows the IBM watsonx Orchestrate interface for the 'Invoice extractor' agent. The 'Edit 'Statement date'' panel is active, showing the 'Description' field with the prompt: 'The statement date is the date the invoice was issued. Return the date using in the MM/DD/YYYYY format. May 1st, 2025 will be displayed as 05/01/2025.' This field is highlighted with a red box and labeled 'A'. The 'Show on document' button at the bottom of the panel is also highlighted with a red box and labeled 'B'. The background shows a sample invoice document for 'FCE Energies' with a due date of 'May 15th, 2025' and an amount of '\$45.99'.

14. The field is now correctly identified and formatted the right way (A). Click the <- icon of the edit panel (B).

The screenshot shows the IBM watsonx Orchestrate interface for the 'Invoice extractor' agent. The 'Edit 'Statement date'' panel is active, showing the 'Results' field with the output: '05/05/2025'. This field is highlighted with a red box and labeled 'A'. The '<-' icon in the top left of the panel is highlighted with a red box and labeled 'B'. The background shows the same sample invoice document as in the previous screenshot.

Note: If your extractor does not show the correctly formatted date, click the <- icon, then return back to the field details view to refresh the values.

15. Repeat the instruction definition (adapt it to the due date) so that the Due date is also displayed in the MM/DD/YYYY format (A).

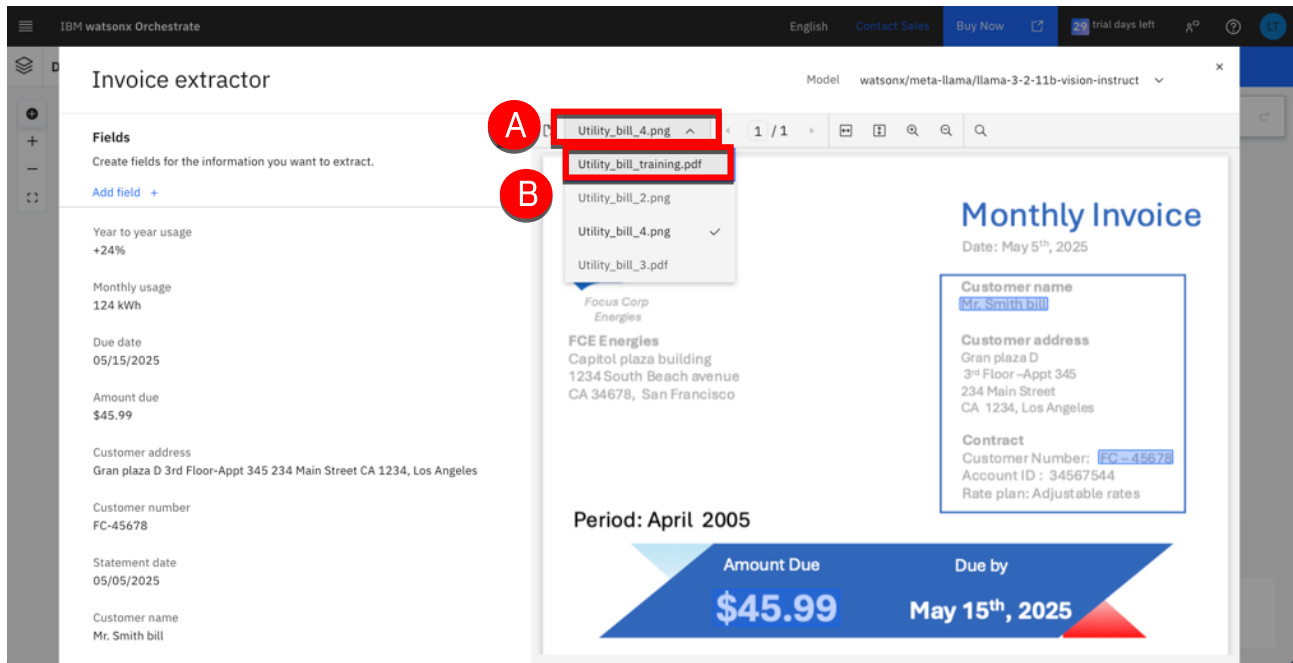
The screenshot displays the IBM watsonx Orchestrate interface for configuring an 'Invoice extractor'. On the left, the 'Edit 'Due date'' field is shown with a description: 'Return the date using in the MM/DD/YYYY format. May 1st, 2025 will be displayed as 05/01/2025.' This description is highlighted with a red box, and a red circle with the letter 'A' is placed next to it. The data type is set to 'String'. On the right, a sample utility bill from FCE Energies is displayed, showing a due date of May 15th, 2025, and an amount due of \$45.99.

Note: At the time of writing, there is a minor issue with the position of the bounding box showing the extracted values position on the document (as you can see on the previous screen shot). This may be corrected by the time the lab is published so your output may be slightly different.

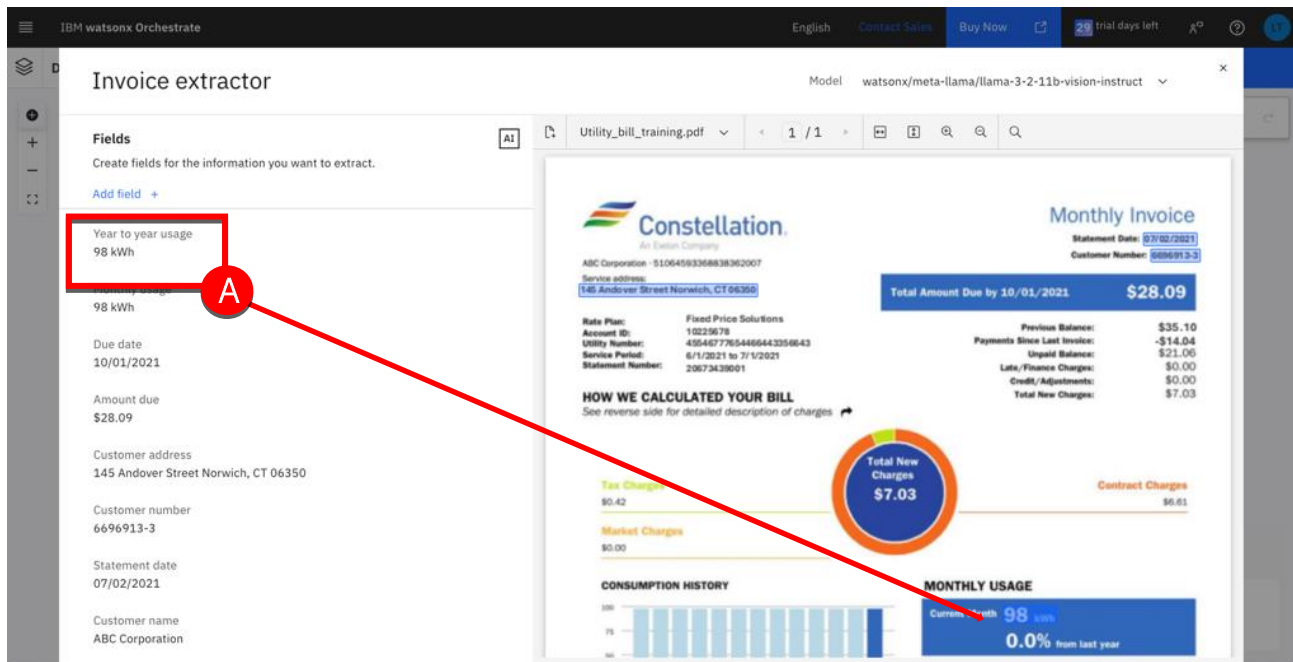
Adding examples

In the previous section, we have seen the benefit of using field descriptions to guide the model. When descriptions do not provide the expected result, you can add examples to improve your extraction results.

1. Expand the file selector (A). Select **Utility_bill_training.pdf** (B).



2. The **Year to year usage** is being extracted from the wrong element in the document (A), you will now add examples to help the model to extract the Amount due accurately.



- Click the **View field details** icon of the Year to year usage field (A).

IBM watsonx Orchestrate

Invoice extractor

Model: watsonx/meta-llama/llama-3-2-11b-vision-instruct

Fields

Create fields for the information you want to extract.

Add field +

Year to year usage
98 kWh

Monthly usage
98 kWh

Due date
10/01/2021

Amount due
\$28.09

Customer address
145 Andover Street Norwich, CT 06350

Customer number
6696913-3

Statement date
07/02/2021

Customer name
ABC Corporation

Utility_bill_training.pdf

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Constellation Energy

Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Rate Plan: Fixed Price Solutions
Account ID: 10225678
Utility Number: 4354577854465443354643
Service Period: 6/1/2021 to 7/1/2021
Statement Number: 20673439001

Previous Balance: \$35.10
Payments Since Last Invoice: -\$14.04
Unpaid Balance: \$21.06
Late/Finesse Charges: \$0.00
Credit/Adjustments: \$0.00
Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL
See reverse side for detailed description of charges.

Tax Charges: \$0.42
Market Charges: \$0.00
Contract Charges: \$6.61

CONSUMPTION HISTORY

MONTHLY USAGE
Current Month: 98 kWh
0.0% from last year

- Click **Enter example** (A).

IBM watsonx Orchestrate

Invoice extractor

Model: watsonx/meta-llama/llama-3-2-11b-vision-instruct

Edit 'Year to year usage'

Enter the field details used to prompt the LLM for the expected result.

Name
Year to year usage

Description
The term that is used to identify this field.

Data type
String

Examples (Optional)
Add 3-5 examples to help the LLM understand what output you're expecting.

Enter example +

Delete field

Utility_bill_training.pdf

1 / 1

Constellation Energy

Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Rate Plan: Fixed Price Solutions
Account ID: 10225678
Utility Number: 4354577854465443354643
Service Period: 6/1/2021 to 7/1/2021
Statement Number: 20673439001

Previous Balance: \$35.10
Payments Since Last Invoice: -\$14.04
Unpaid Balance: \$21.06
Late/Finesse Charges: \$0.00
Credit/Adjustments: \$0.00
Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL
See reverse side for detailed description of charges.

Tax Charges: \$0.42
Market Charges: \$0.00
Contract Charges: \$6.61

CONSUMPTION HISTORY

MONTHLY USAGE
Current Month: 98 kWh
0.0% from last year

Last Month: 98 kWh
Last Year: 98 kWh

- Click inside the example **Input** cell (A), then draw a rectangle around the area containing the **Amount due** value in the document (B). This will automatically capture the text contained in your selected area in the Input field.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' agent configuration is visible. The 'Name' field is 'Year to year usage'. The 'Description' field is 'Provide a definition of the information'. The 'Data type' is 'String'. The 'Examples (Optional)' section shows an input field (A) with a red box around it, containing the text 'e.g. the "CREDIT LINE" for the borrower on the amount of Two Hundred Thousand Dollars (\$200,000.00)'. The 'Output' field shows '\$200,000.00'. On the right, a sample utility bill document is displayed. The document includes a 'Total Amount Due by 10/01/2021' of '\$28.09'. It also shows 'Total New Charges' of '\$7.03' and 'Contract Charges' of '\$6.61'. The 'MONTHLY USAGE' section shows 'Current Month: 98 kWh' and '0.0% from last year' (B), which is highlighted with a red box. The 'CONSUMPTION HISTORY' section shows a bar chart of usage over the last 12 months.

- Let's now specify how to extract the output value from that selection. Click the Output cell (A) then just select the percentage value to extract (B).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' agent configuration is visible. The 'Name' field is 'Year to year usage'. The 'Description' field is 'Provide a definition of the information'. The 'Data type' is 'String'. The 'Examples (Optional)' section shows an input field (A) with a red box around it, containing the text 'e.g. \$200,000.00'. The 'Output' field shows '0.0%'. On the right, the same sample utility bill document is displayed. The 'MONTHLY USAGE' section shows 'Current Month: 98 kWh' and '0.0% from last year' (B), which is highlighted with a red box. The 'CONSUMPTION HISTORY' section shows a bar chart of usage over the last 12 months.

7. Click **Show on document** (A).

The screenshot shows the 'Invoice extractor' interface in IBM watsonx Orchestrate. On the left, the configuration for the 'Year to year usage' field is visible. The 'Show on document' button at the bottom of this configuration is highlighted with a red rectangle and labeled with a red circle 'A'. The main area displays a sample utility bill from ABC Corporation for \$28.09, including sections for charges, consumption history, and monthly usage.

Note: As soon as text under the selected area is recognized, it is added to the input field example. If no text is dynamically selected when surrounding the area, this means the document extractor will not recognize this area as containing text.

8. If the value (A) is not correctly identified (A), a description may be added as shown in the next step.

The screenshot shows the 'Invoice extractor' interface in IBM watsonx Orchestrate. The 'Results' section at the bottom left shows the 'Input' field containing '0.0% from last year', which is highlighted with a red rectangle and labeled with a red circle 'A'. The 'Output' field is empty. The main area displays the same utility bill sample as in the previous screenshot.

9. If you still have an incorrect output (ie, more text is displayed than expected) provide a description. Type 'The year to year usage is a percentage number. Always display the percentage value followed by the % sign. Add no more text to the result.' in the **Description** (A). Click Show on document (B) then wait a few seconds for the result to be updated.

The screenshot displays the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' configuration panel is visible. It includes a 'Description' field (A) containing the text: 'The year to year usage is a percentage number. Always display the percentage value followed by the % sign. Add no more text to the result.' Below this is a 'Data type' dropdown set to 'String'. An 'Examples (Optional)' section shows an input-output example where '0.0% from last year' is input and '0.0%' is output. At the bottom of the configuration panel is a 'Show on document' button (B). On the right, a sample utility bill document is displayed. The bill includes customer information, a total amount due of \$28.09, a breakdown of charges (Tax: \$0.42, Market: \$0.00, Total New: \$7.03, Contract: \$6.61), and a consumption history bar chart. The 'MONTHLY USAGE' section shows 'Current Month 98 kWh' and '0.0% from last year'. A red circle with the letter 'A' is placed over the 'Description' field, and a blue circle with the letter 'B' is placed over the 'Show on document' button.

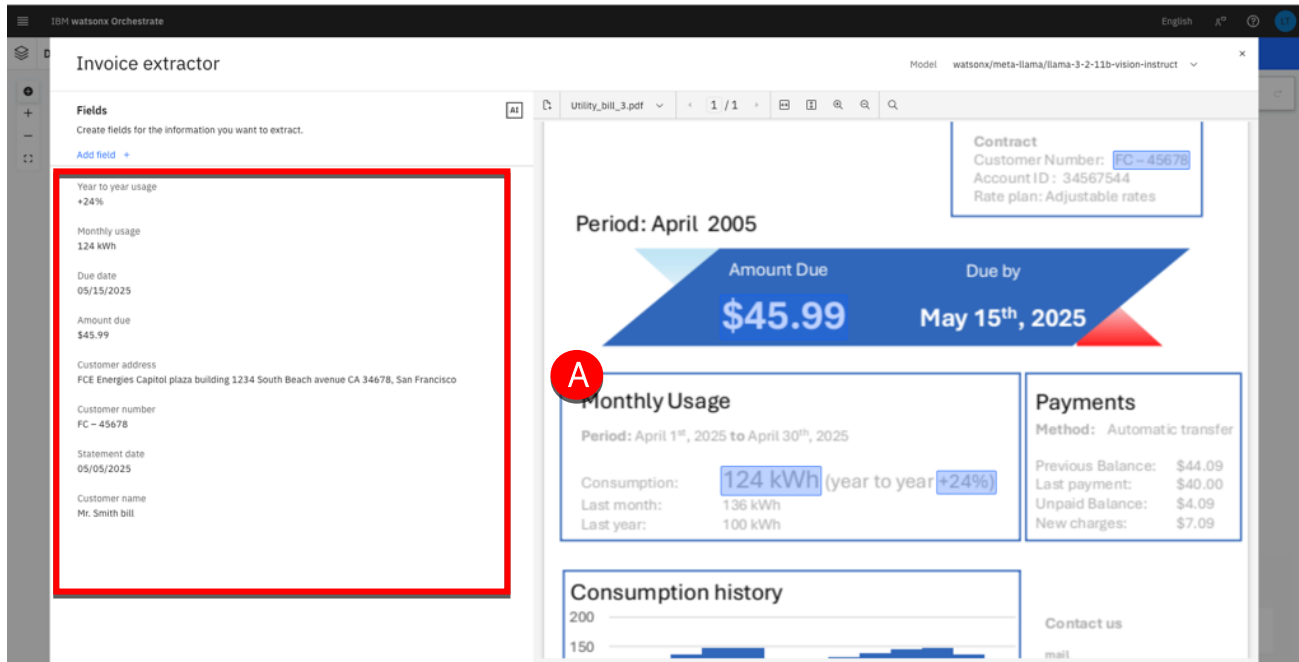
Note: Depending on your environment, it may take some time before the result is correctly displayed. Don't hesitate to add more examples until the correct result is displayed should you model fail to display the expected value. Click anywhere in the interface to refresh the screen and see the updated values if required.

10. Click the <- icon (A).

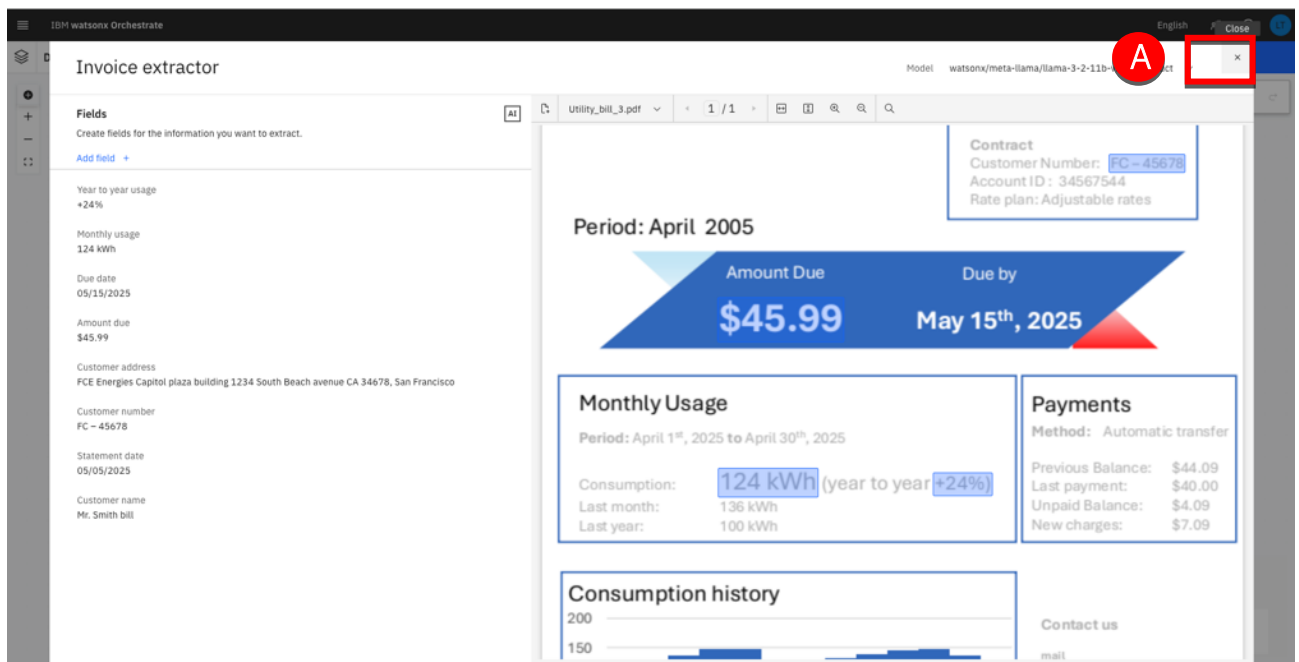
The screenshot shows the 'Invoice extractor' interface in IBM watsonx Orchestrate. On the left, the 'Fields' list is visible, with a red box highlighting the '<-' icon and a red circle 'A' next to it. The main area displays a sample utility bill from ABC Corporation. The bill includes a table of charges, a consumption history bar chart, and a monthly usage summary. The 'Total Amount Due by 10/01/2021' is \$28.09.

11. Repeat the two improvement methods (Description / Add example) for each field in each of the different document samples you have provided until all the fields are correctly extracted for each file (A).

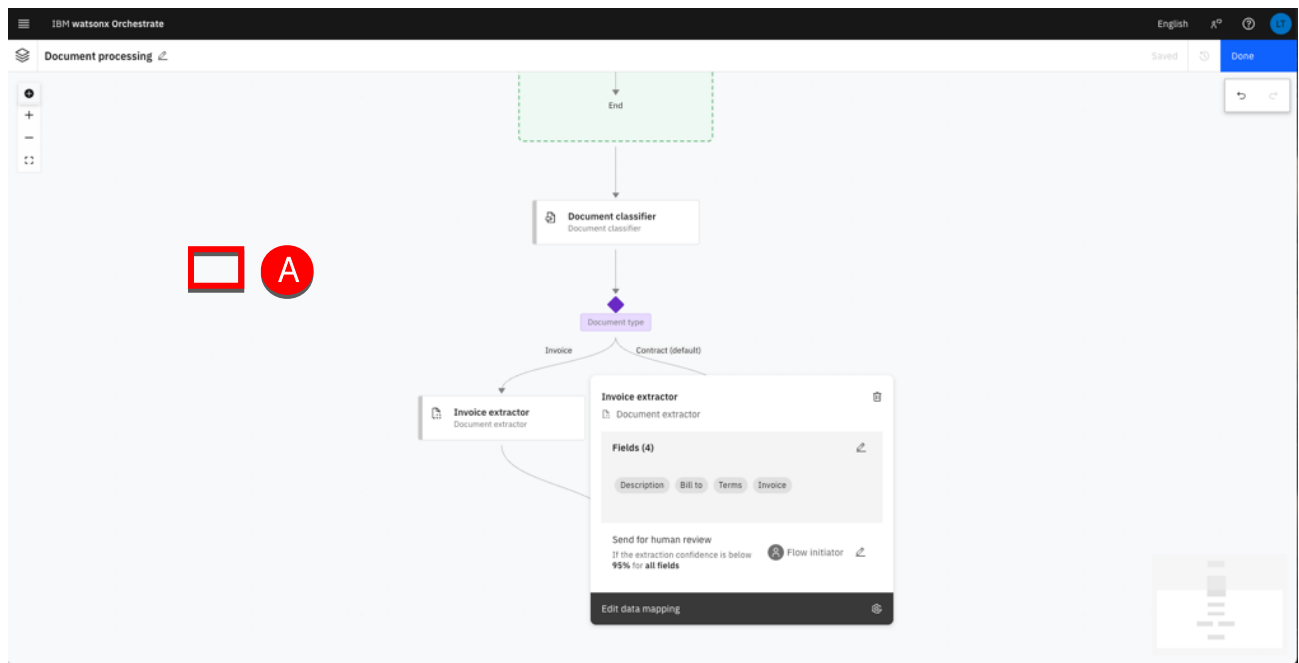
The screenshot shows the 'Invoice extractor' interface in IBM watsonx Orchestrate. On the left, the 'Fields' list is visible, containing extracted fields from the utility bill. A red box highlights the list, and a red circle 'A' is next to it. The main area shows the same utility bill sample as in the previous screenshot.



12. Click X to close the extractor editor (A).



13. Click on the diagram background to close the **Invoice extractor** property view (A).

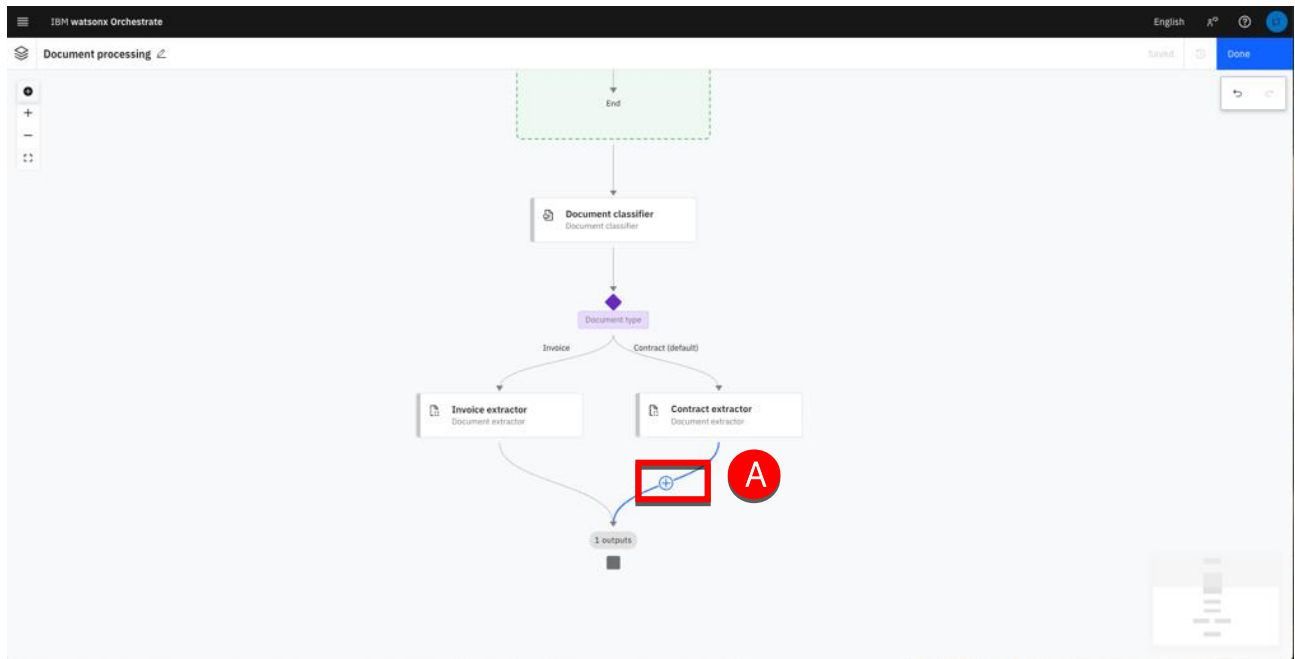


You will now add the user activities for each branch to display the extracted values.

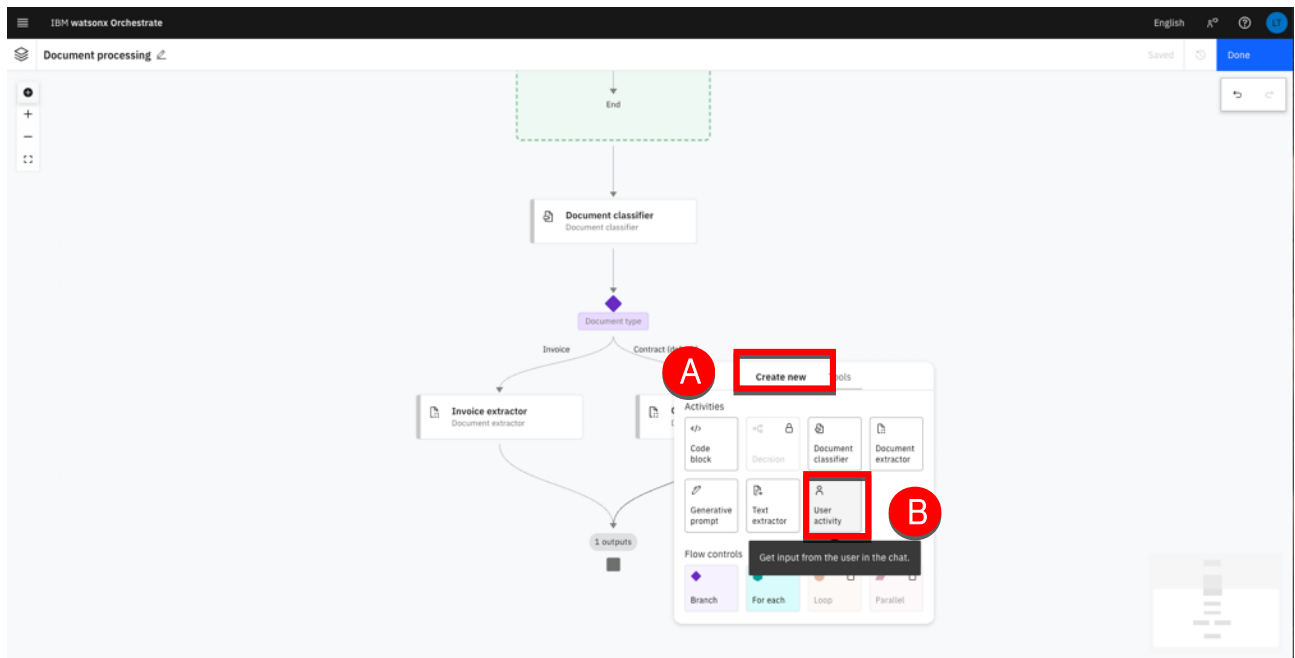
Add a Display to user activity

To complete the flow, we need to add a final activity that presents the extracted values to the user. This step allows us to format and display the results clearly, based on the core fields that we defined when we trained each extractor.

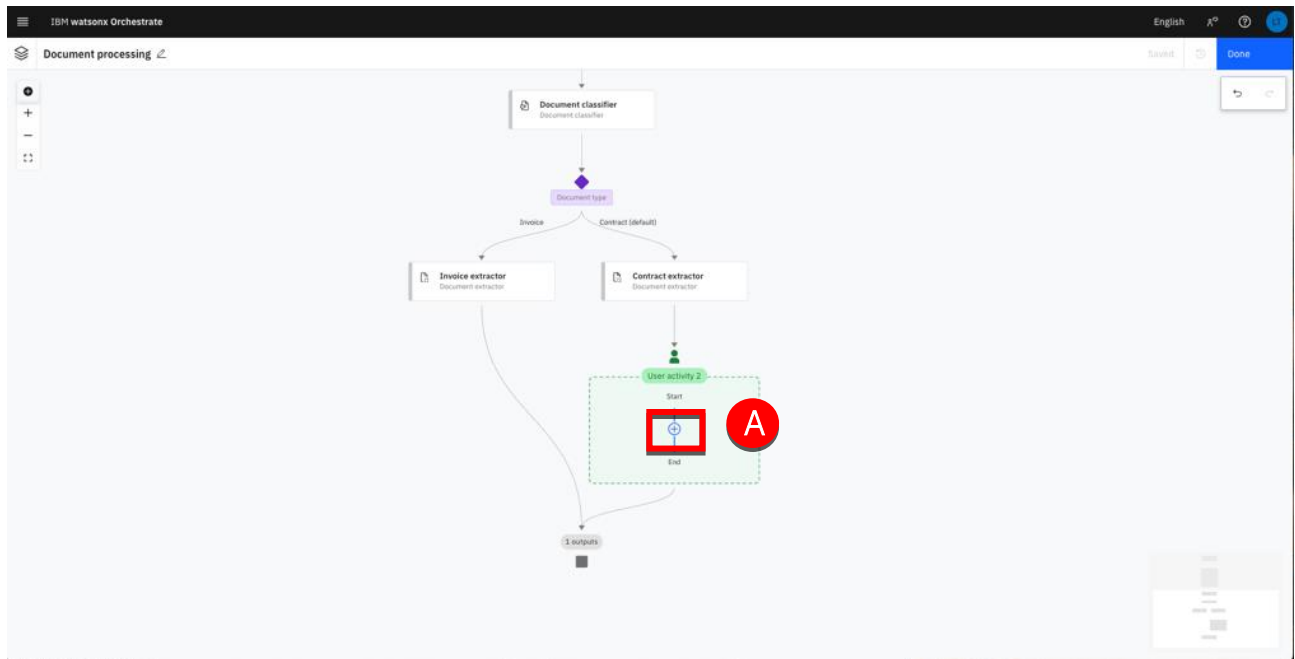
1. Hover the last link in the Contract extractor branch and click + (A).



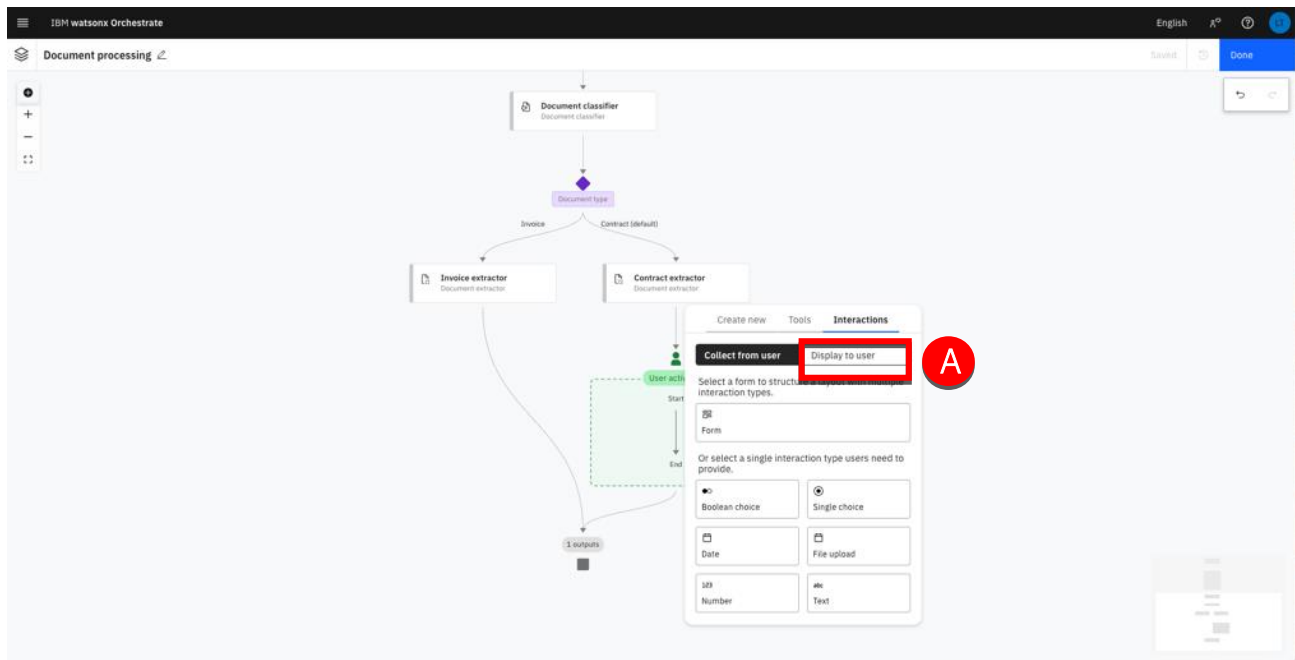
2. Select **Create new** (A) then click **User activity** (B).



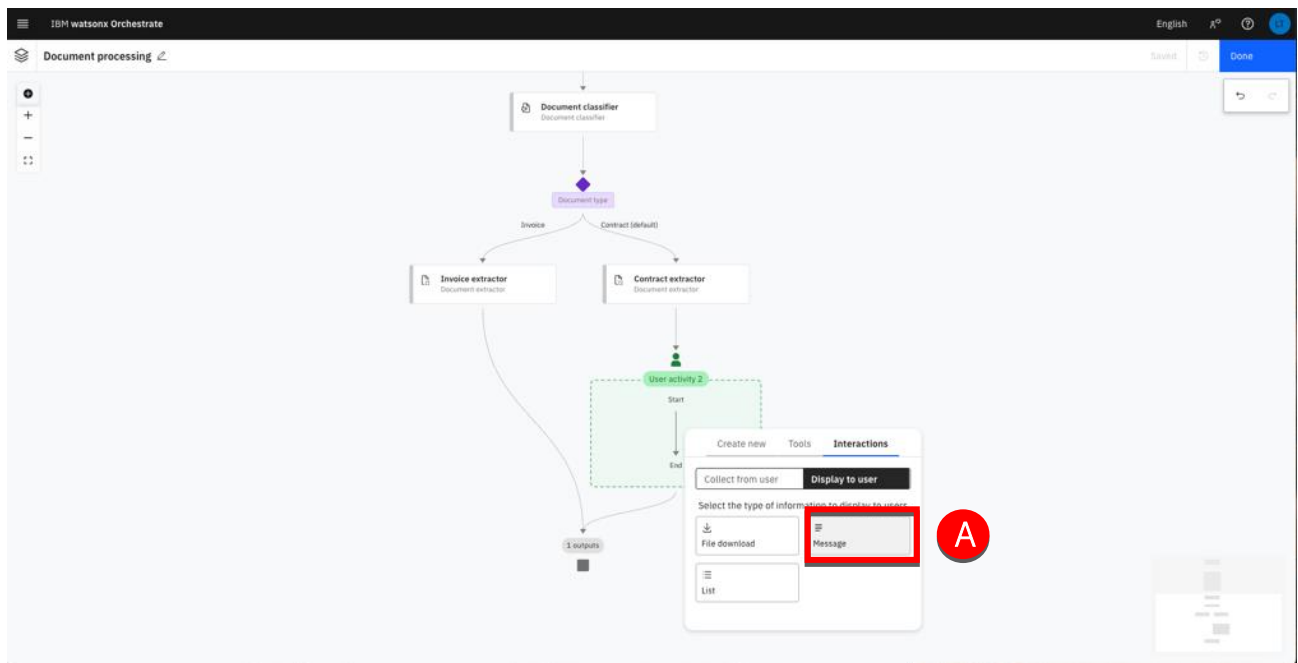
- Click + (A) to add a new action in the user activity node.



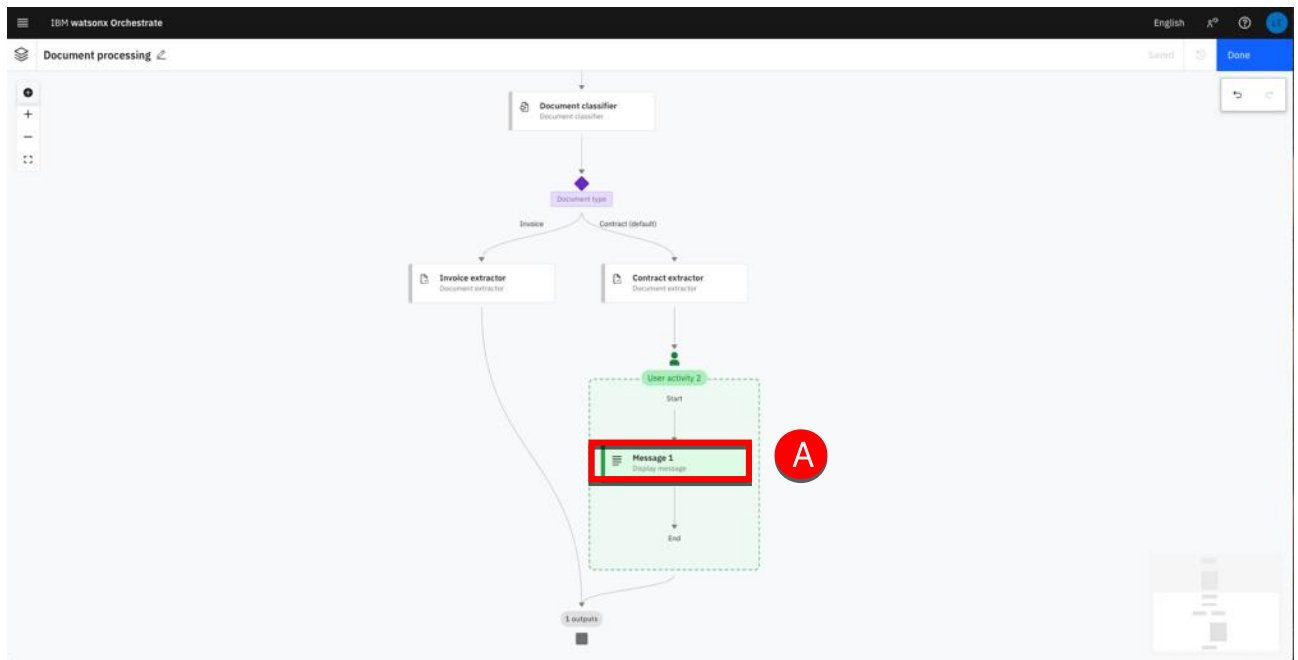
- Click **Display to user** (A).



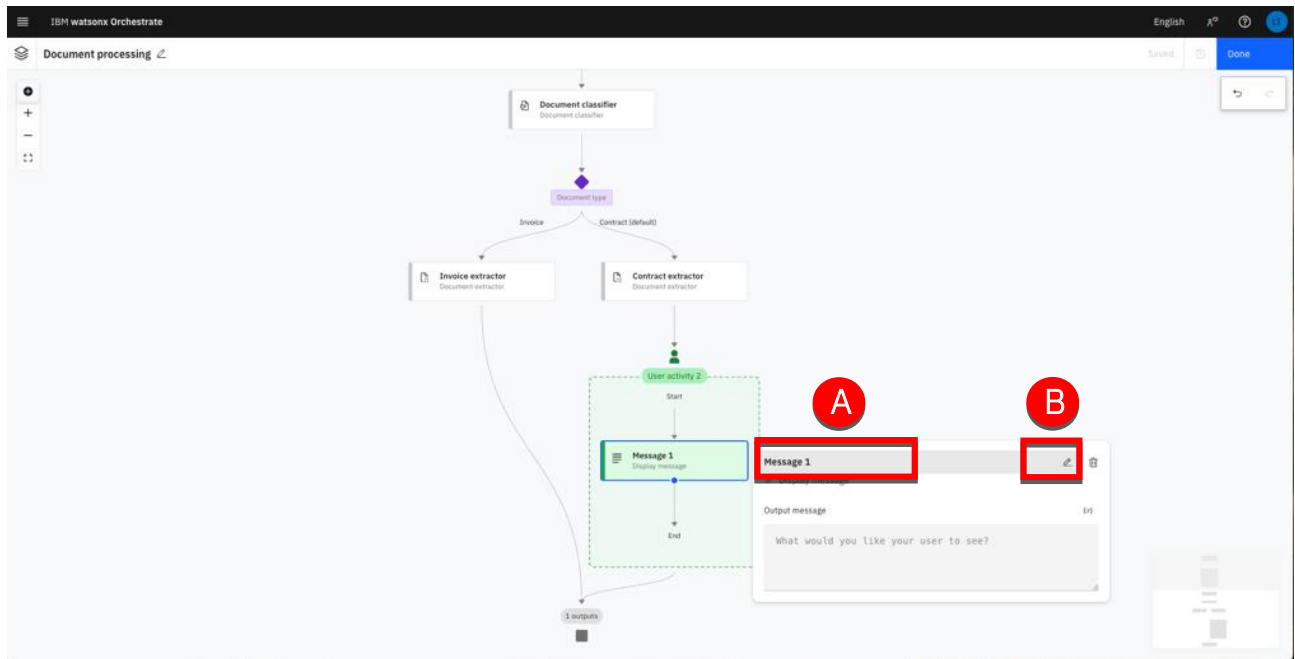
5. Select **Message** (A).



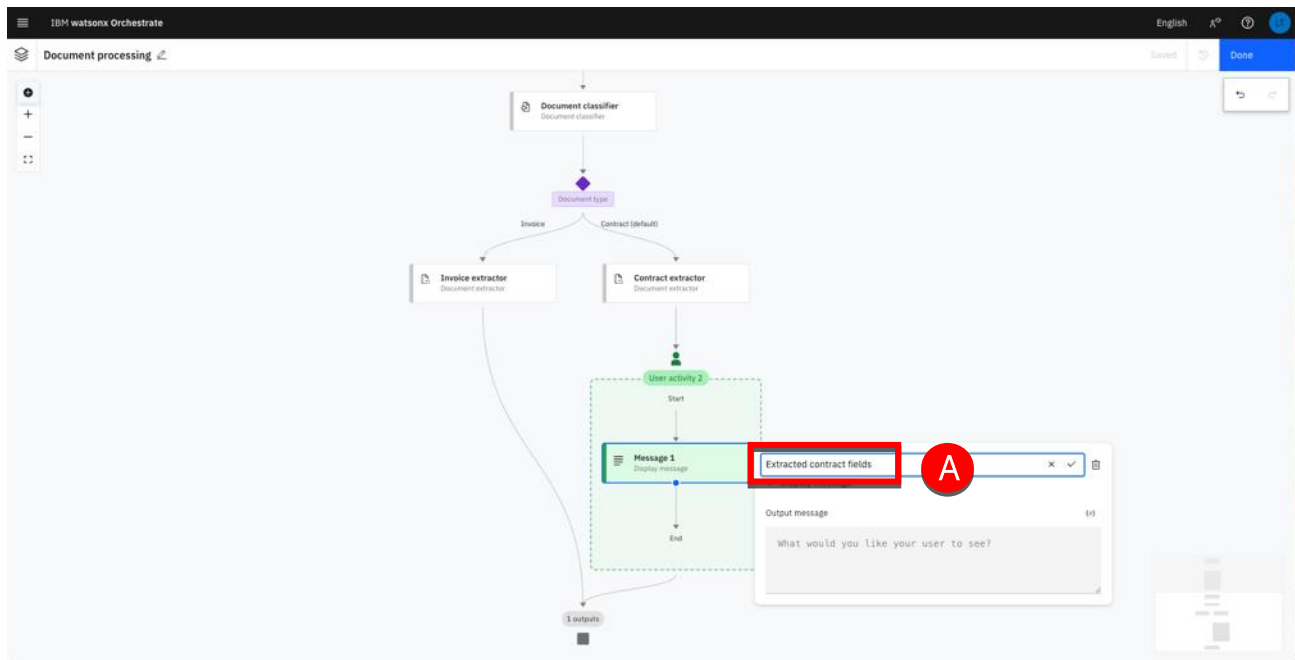
6. Click the **Message** box to edit its content (A).



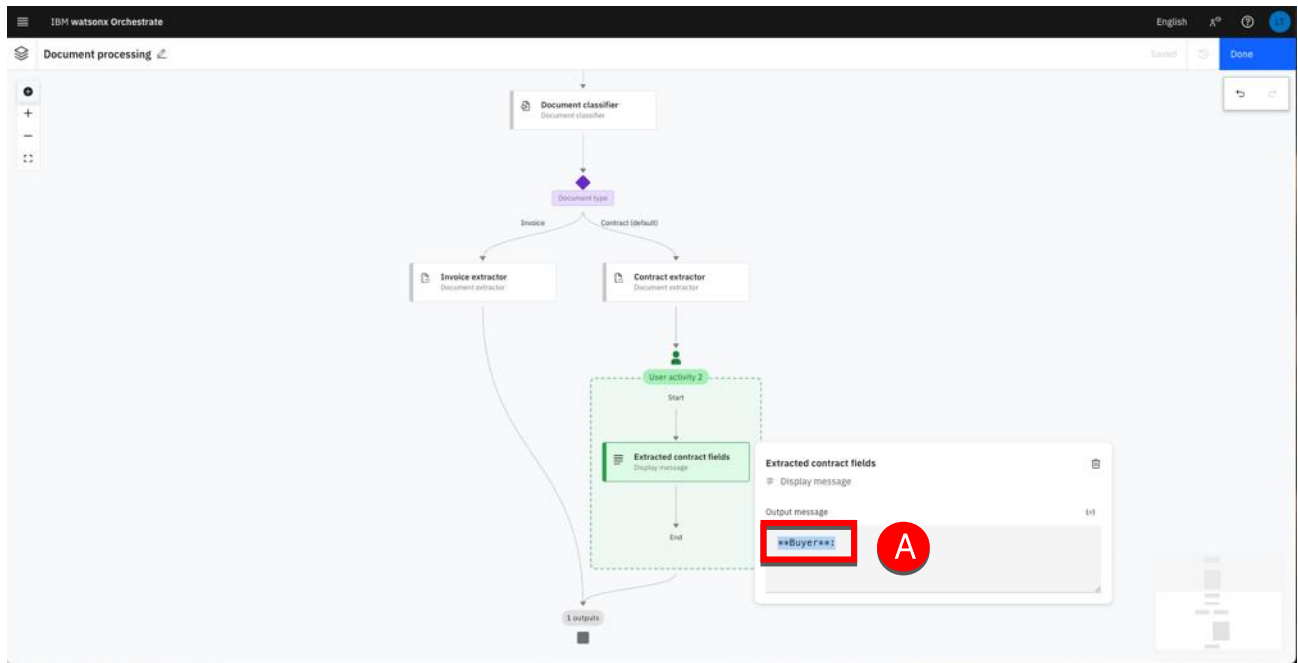
7. Hover the **Message 1** name (A) then click the edit button (B) to rename it.



8. Type 'Extracted contract fields' (A) and hit Enter.

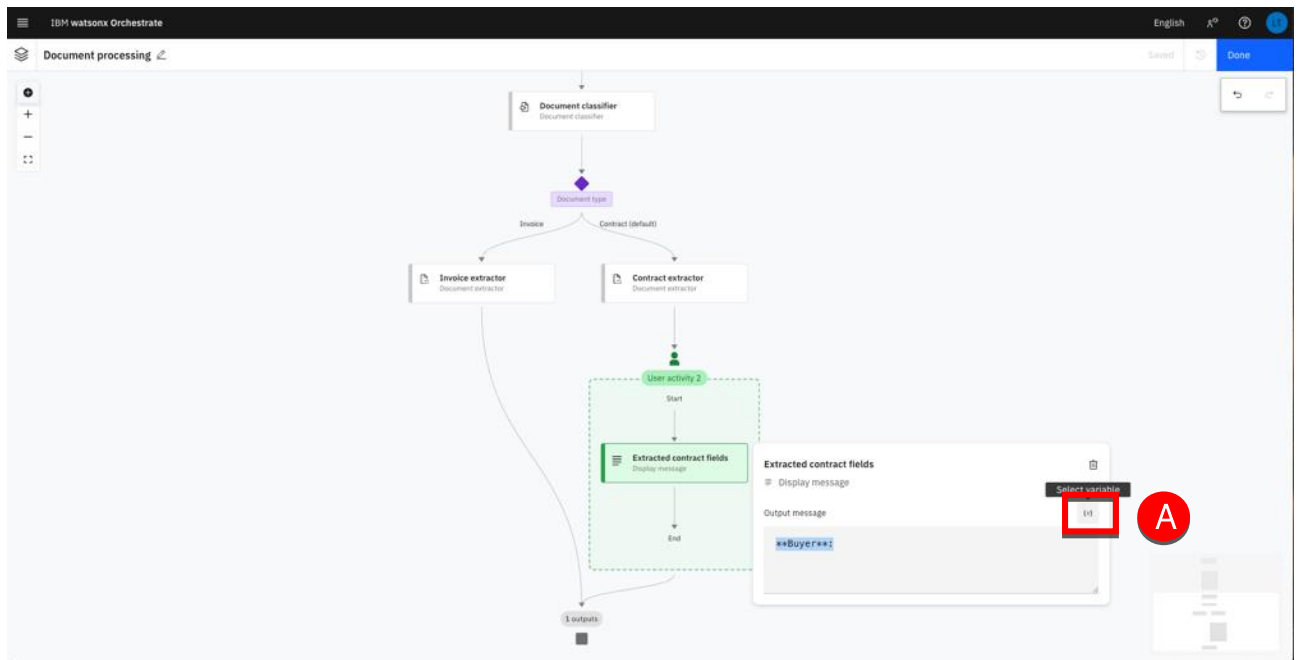


9. Type “**Buyer**:” in the Output message field (A)

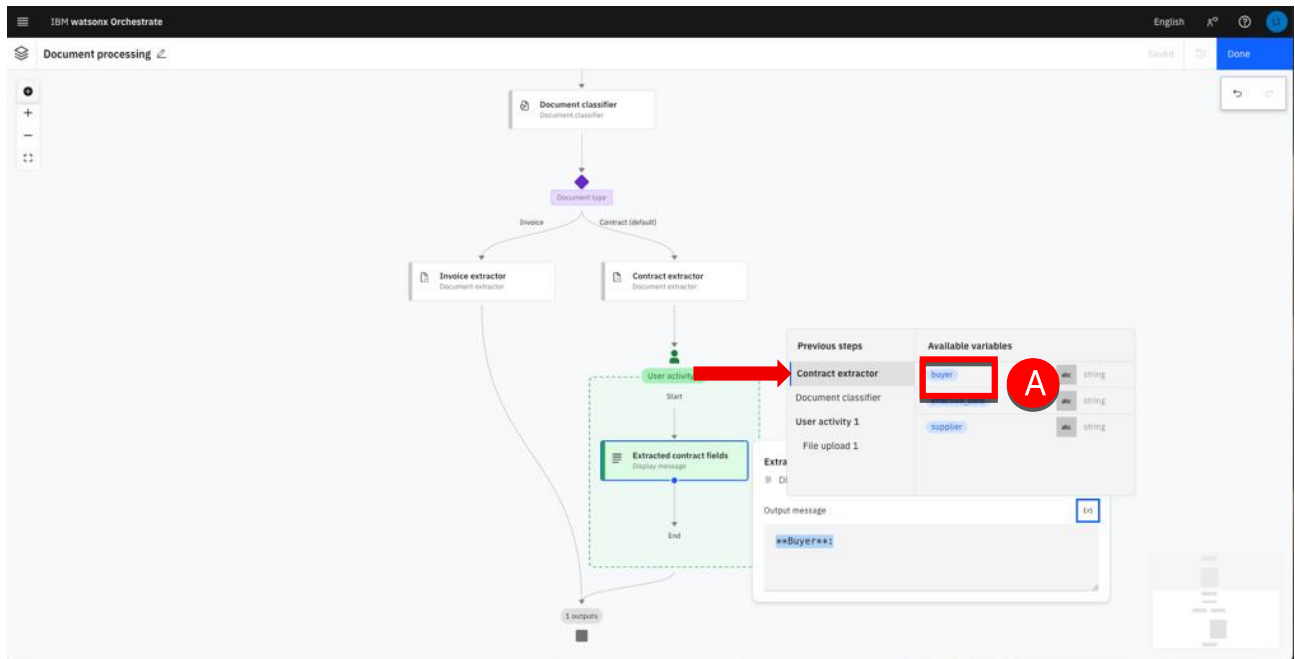


Note: the ‘**’ are just here to specify that the label must be displayed in Bold.

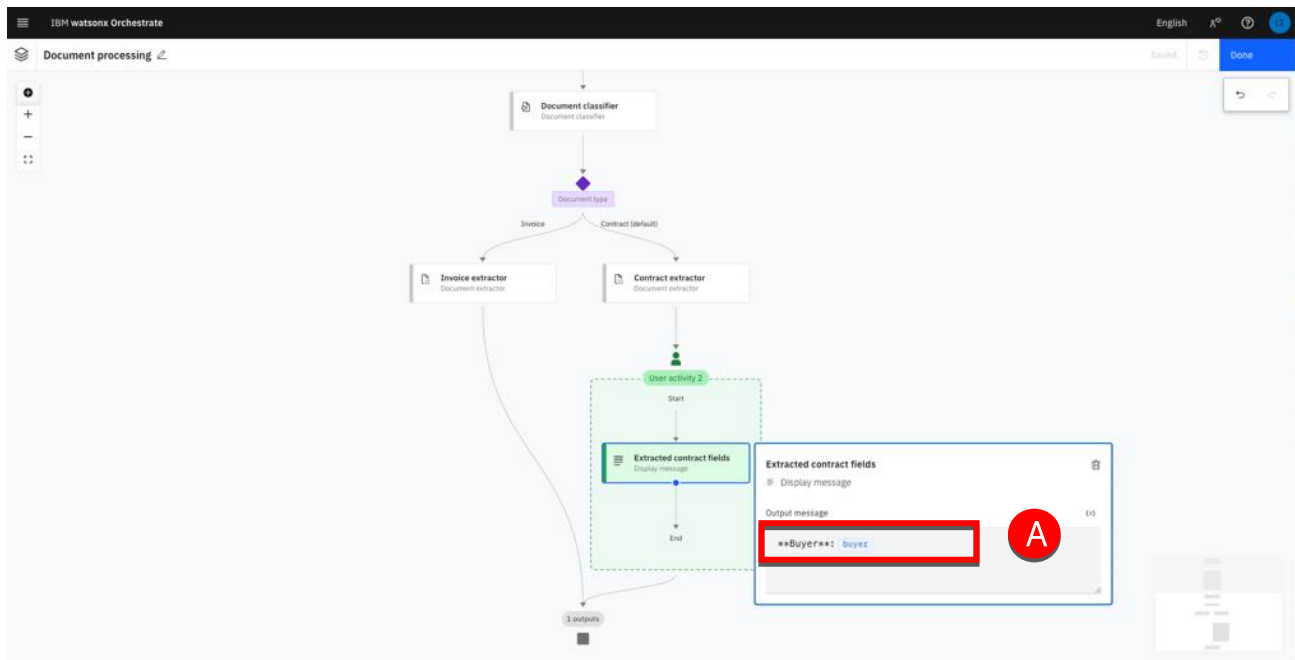
10. To display the results for the field, ‘Buyer’ you must assign a variable to it. Click on the Select variable ‘[x]’ button.



11. Under the 'Contract extractor' menu click on the **buyer** variable (A).



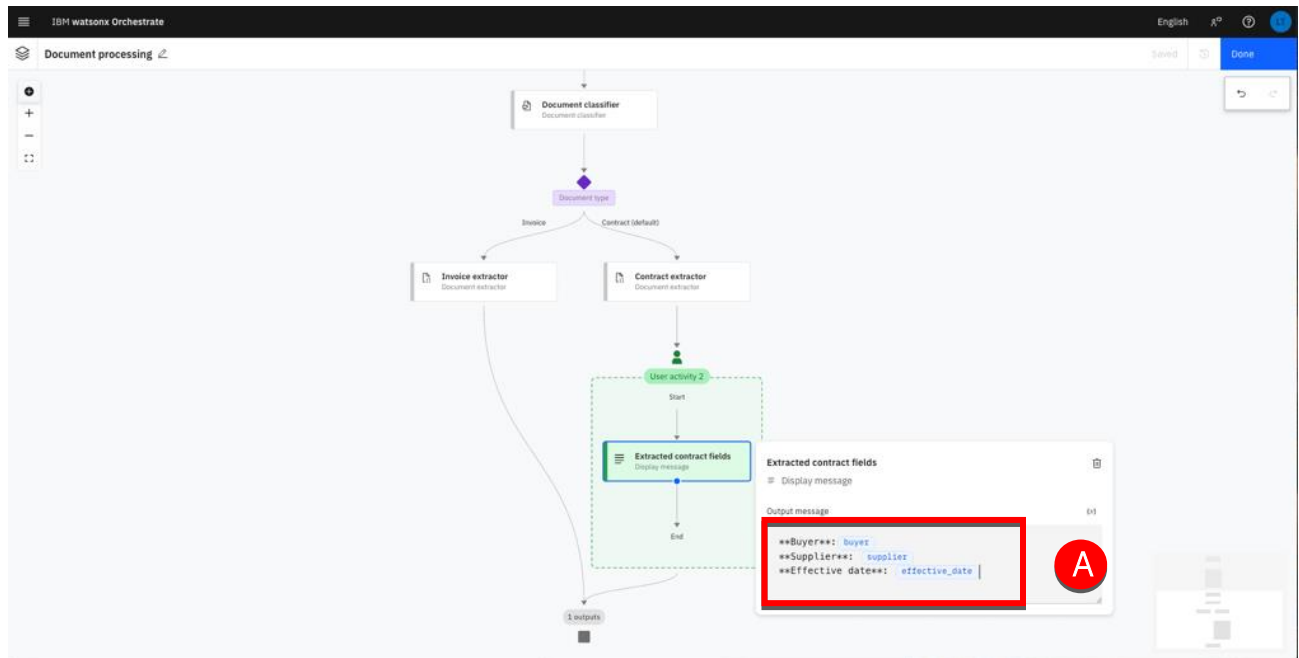
12. Your screen should look like (A):



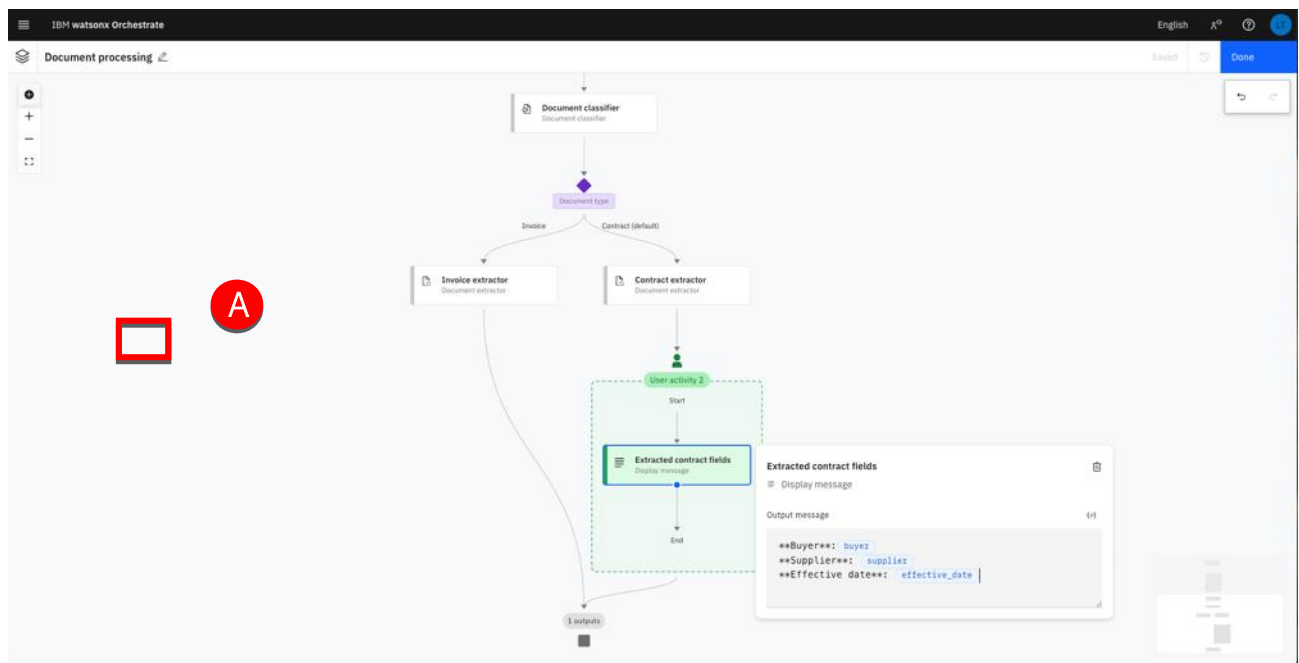
13. Repeat from step 9 to add the remaining fields defined in the Contract extractor:

- Supplier
- Effective date

Your screen should look like (A):

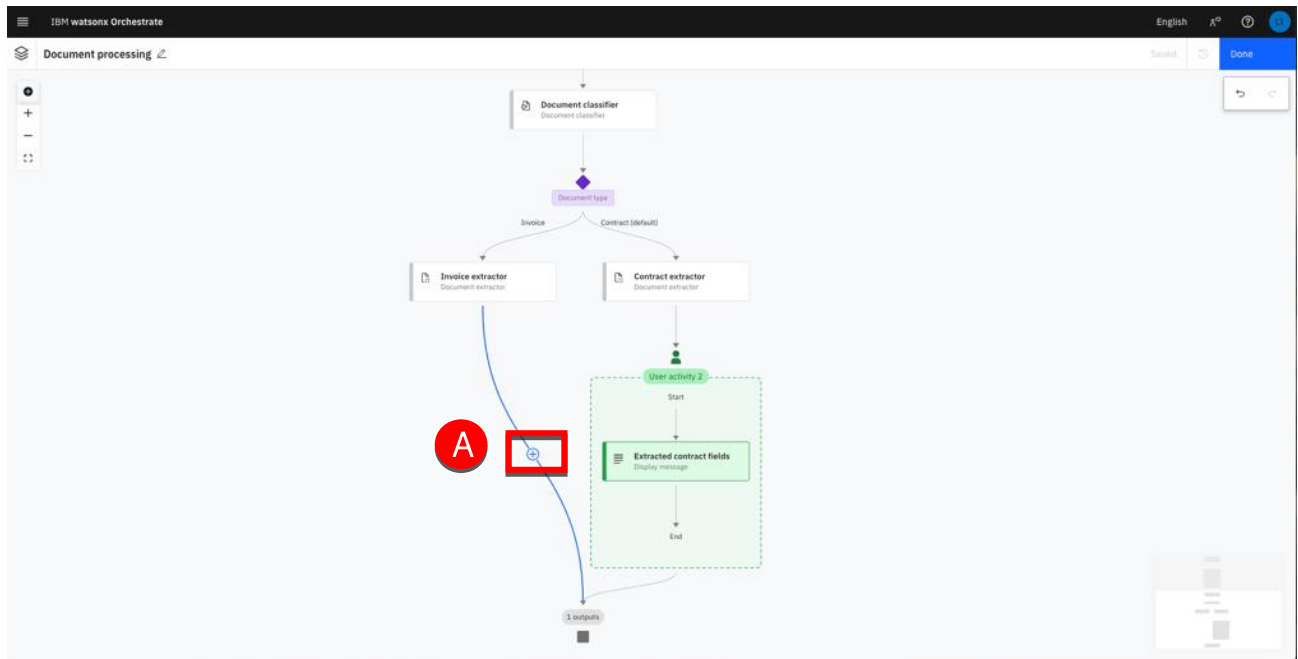


14. Click the background to close the **Extracted contract fields** panel (A).



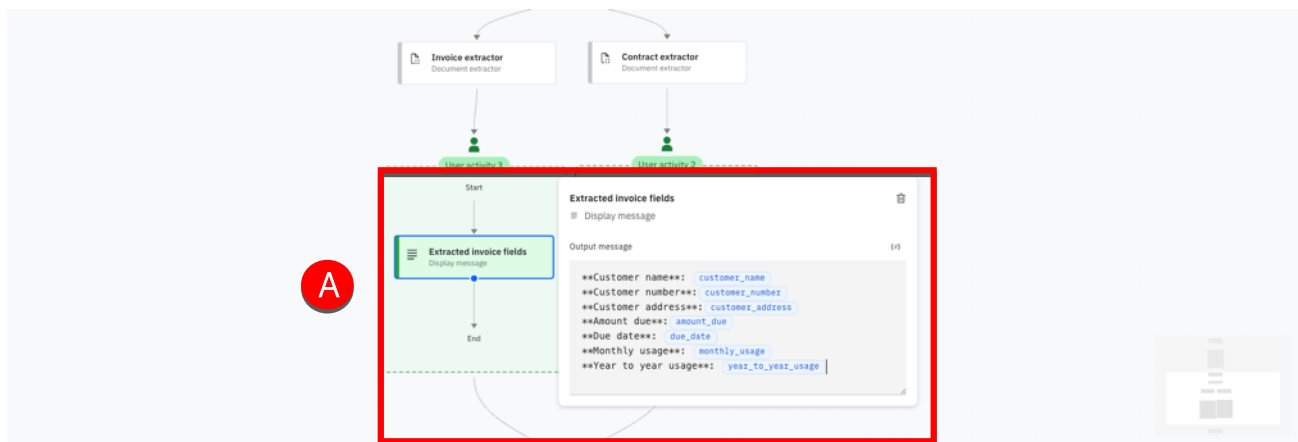
15. Repeating from Step 1, create a user activity in the **Invoice** branch (A) to display the following fields:

- Customer name
- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage



Note: The invoice variables will be under the Invoice extractor folder.

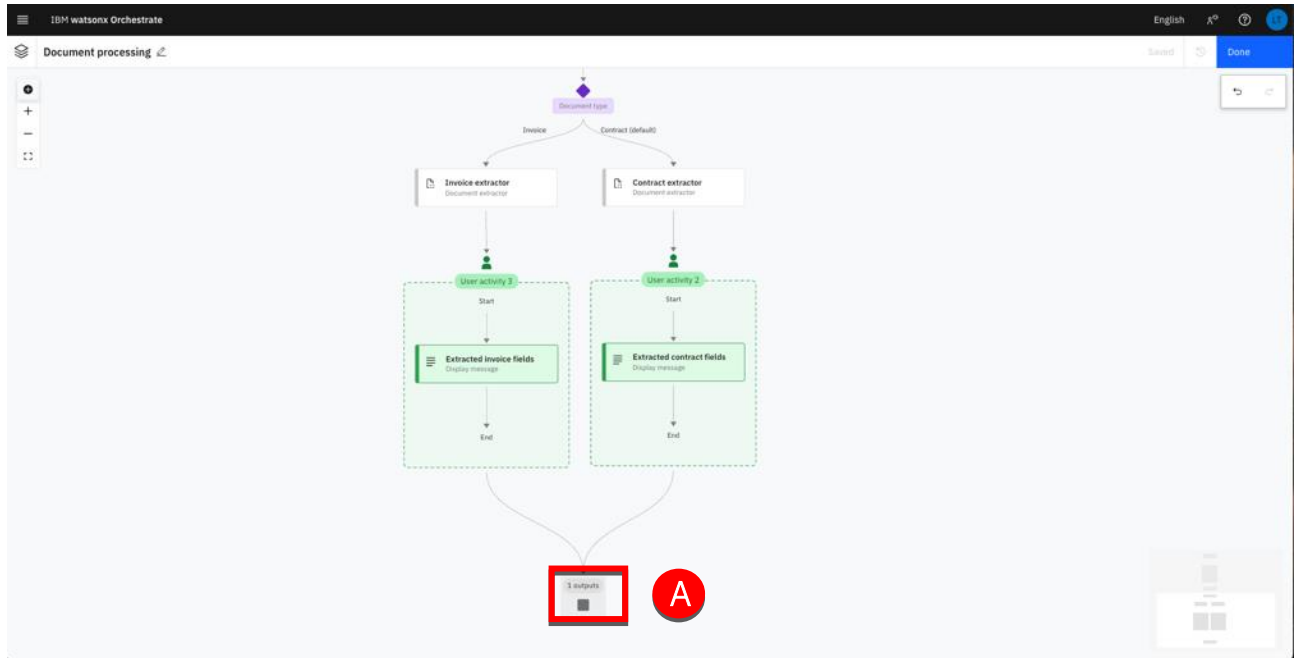
You should get the following result:



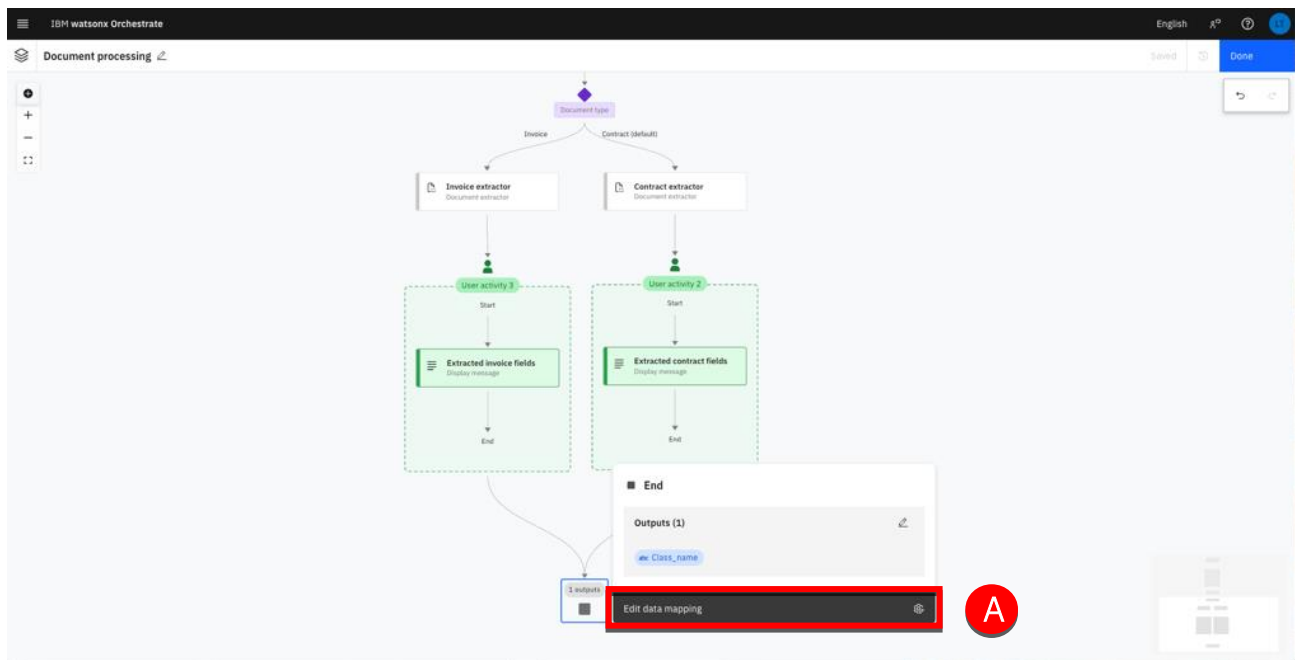
Define the end flow

Now that the output message has been defined for both file types, we can now conclude the flow by updating the end node. The end node will just return the Document type (i.e. Contract/Invoice).

1. Click the **End** node (A).



2. Click **Edit data mapping** (A).



Note that the output, 'class_name' is auto-mapped by the LLM (A). This means that the LLM will attempt to automatically link the input/outputs.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram for 'Document processing' is visible, featuring a 'Document type' input, an 'Invoice extractor' activity, and a 'User activity 2' block containing 'Start', 'Extracted invoice fields', and 'End' steps. On the right, the 'Map data for End' panel is open. It displays 'Autonomous data mapping enabled' and a table of outputs. The 'Class_name' output is highlighted with a red box, and a red circle with the letter 'A' points to the 'Auto-map' button next to it.

Note: The auto-map feature works effectively for those flows that are less complex. But for the more complex flows, (for instance if you had two 'class_name' outputs) then explicit mapping performs best.

3. Hover the variable row and click on the **Variable** icon (A).

This screenshot is similar to the previous one, showing the same flow diagram and 'Map data for End' panel. However, in this view, the 'Auto-map' button for 'Class_name' is no longer highlighted. Instead, a red circle with the letter 'A' points to a 'Variable' icon (a small square with a red border) located to the right of the 'Auto-map' button in the output row.

- Click the Document classifier (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram is visible with a 'Document type' input branching into 'Invoice' and 'Contract details'. The 'Invoice' path leads to an 'Invoice extractor' activity, followed by a 'User activity 3' block containing an 'Extracted invoice fields' activity. The 'Contract details' path leads to a 'Contract extractor' activity. Both paths converge at an 'End' node. On the right, the 'Map data for End' panel is open. It displays 'Autonomous data mapping enabled' and a table of 'Outputs'. The 'Outputs' table has columns for 'Class_name', 'Previous steps', and 'Available variables'. The 'Previous steps' column lists 'Contract extractor', 'Document classifier', 'Invoice extractor', and 'User activity 1'. The 'Available variables' column lists 'buyer', 'effective_date', and 'supplier'. The 'Document classifier' step is highlighted with a red box and a red circle labeled 'A'.

- Select class_name (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the flow diagram is the same as in the previous screenshot. On the right, the 'Map data for End' panel is open. The 'Outputs' table is the same, but the 'class_name' variable is now selected in the 'Available variables' column, highlighted with a red box and a red circle labeled 'A'.

6. The output 'class_name' is now assigned to the 'class_name' variable (A). Click x (B) to close the window.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a workflow diagram is visible, featuring a 'Document type' input branching into 'Invoice' and 'Contract (default)'. These lead to 'Invoice extractor' and 'Contract extractor' respectively. Below each extractor is a 'User activity 2' block containing 'Extracted invoice fields' and 'Extracted contract fields'. The workflow concludes with a '1 outputs' block. On the right, a 'Map data for End' dialog box is open. It includes a section for 'Outputs' with a table containing the following data:

Output	Variable
Class_name	class_name

Red circle 'A' highlights the 'class_name' variable in the table. Red circle 'B' highlights the 'x' button in the top right corner of the dialog box, used to close it.

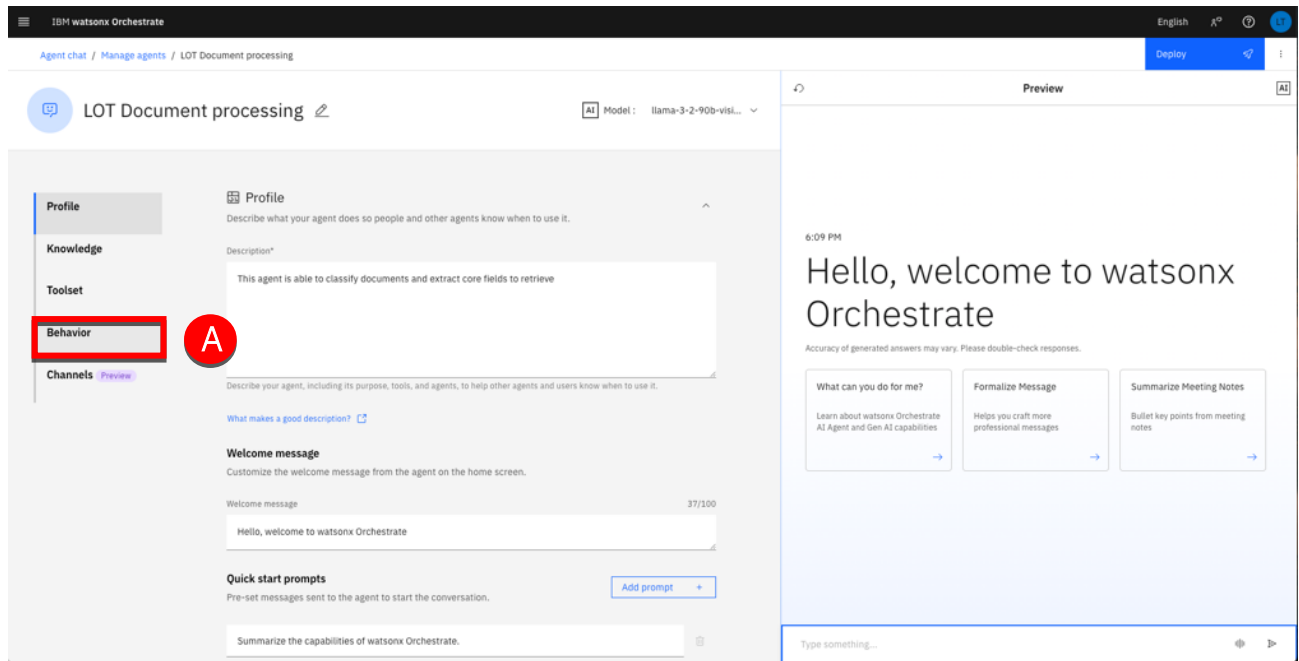
7. Click Done (A).

This screenshot shows the same IBM watsonx Orchestrate interface after the 'Map data for End' dialog box has been closed. The workflow diagram is now fully visible, showing the parallel processing paths for 'Invoice' and 'Contract' documents, each passing through an extractor and a user activity before reaching the final '1 outputs' block. In the top right corner, a red circle 'A' highlights the 'Done' button, which is used to save the changes to the workflow.

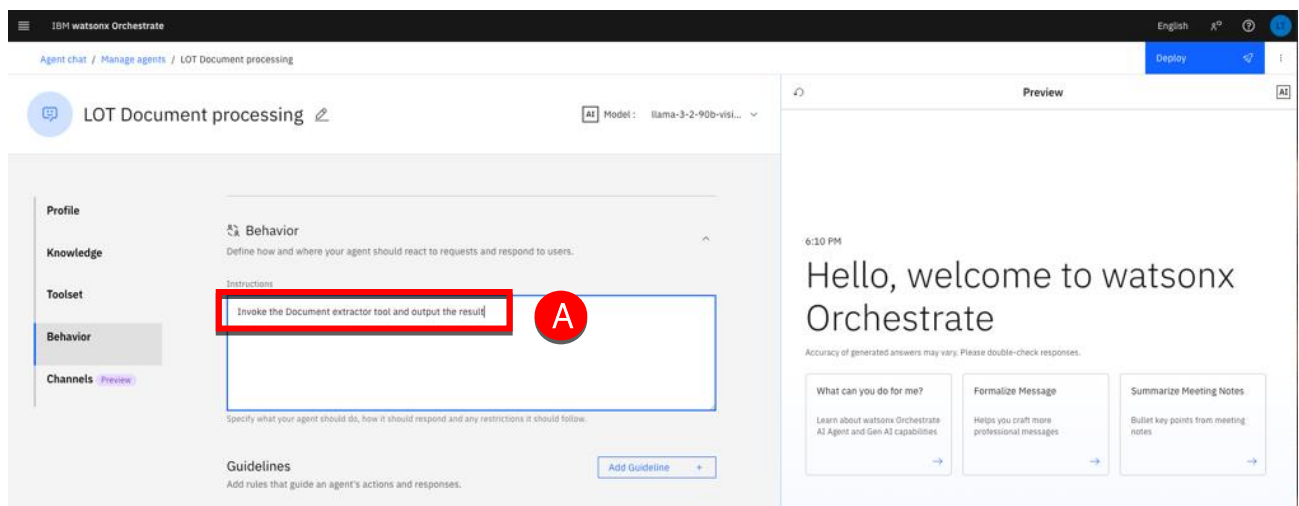
Add a behavior

To trigger the tool you just created, you must update the agent behavior.

1. Click **Behavior** (A).



2. Type 'Invoke the Document extractor tool and output the result' (A) in the Instructions.



Your agent is ready to be tested.

Test your agent

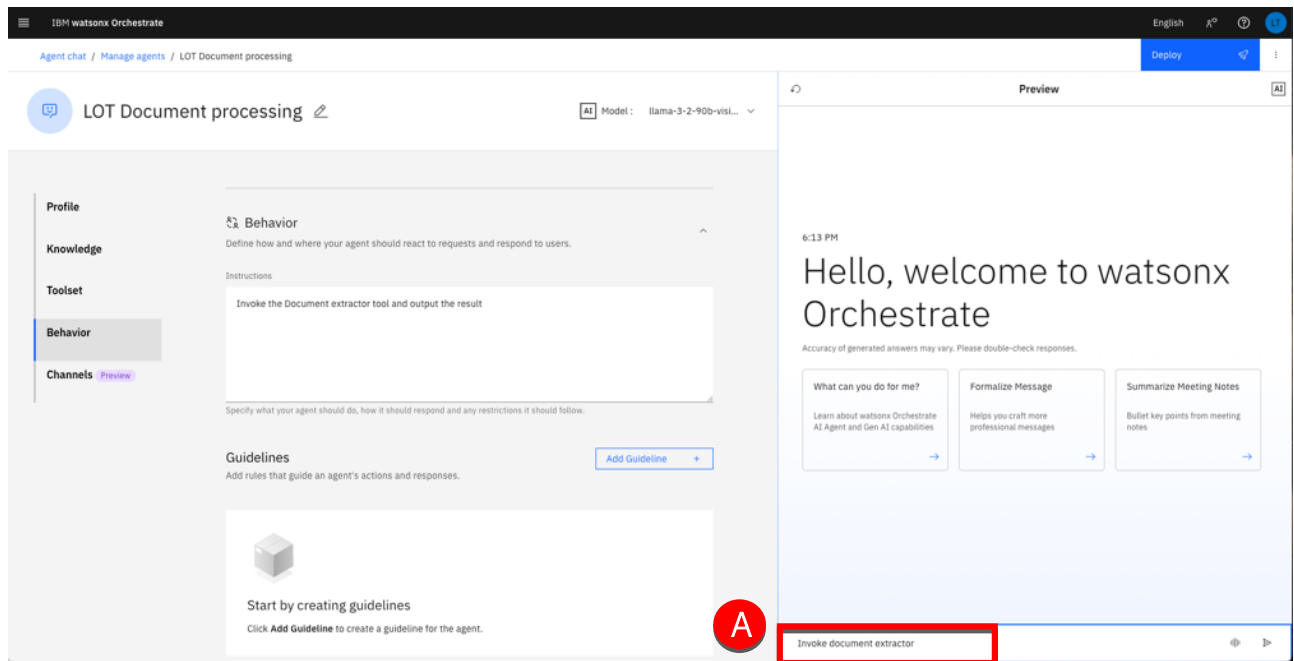
In this section, you will use the Agent preview to test your agent behavior. You will first use the contract.pdf to first test the workflow behavior. As this document has been used to train the contract extractor, there should be no extraction problem.

In a second test, you will use the Utility_Bill_5.pdf to test the agent. As this document has never been used by the invoice extractor, you may have to confirm some extracted value when the extraction confidence score is too low.

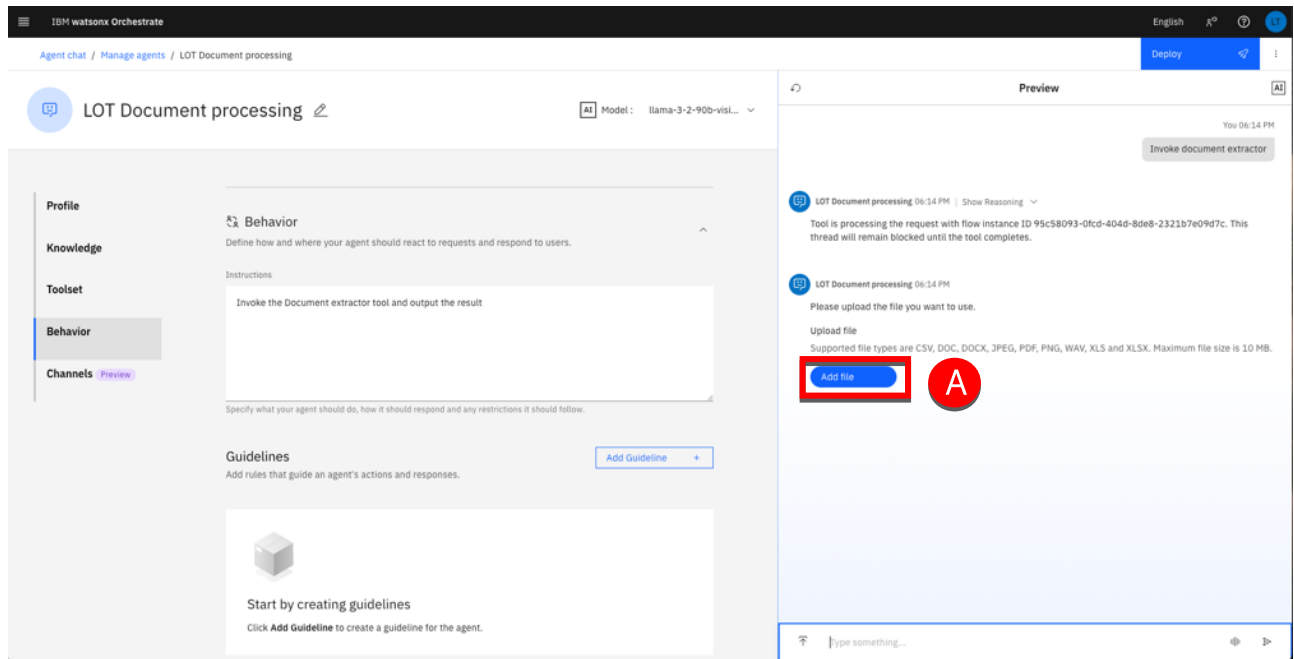
Simple workflow test

Run a simple test for contract extraction.

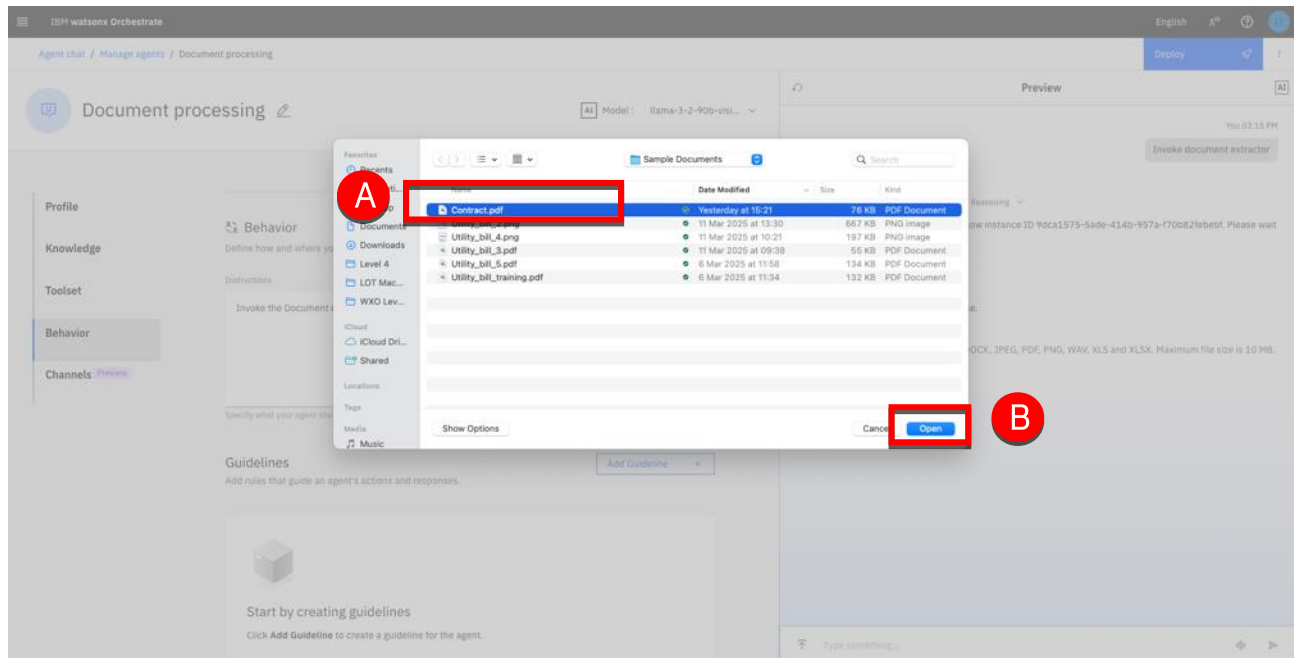
1. Type 'Invoke document extractor' (A) in the Preview instructions area.



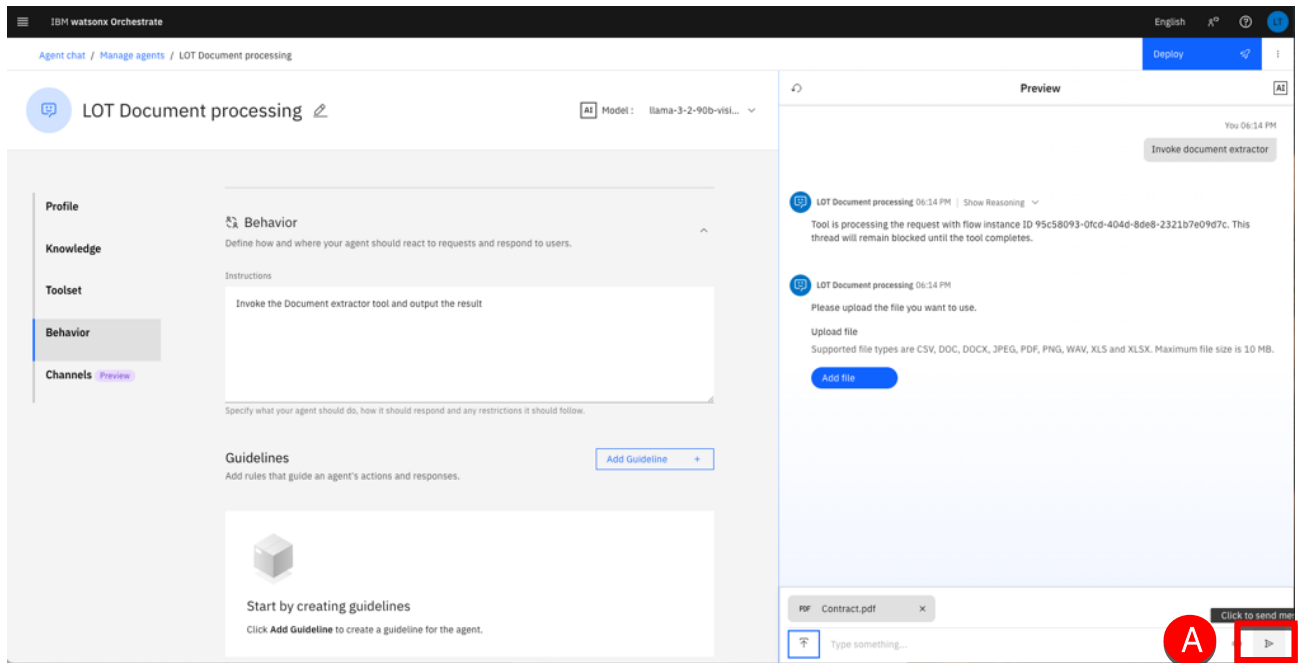
- Click **Add file** (A).



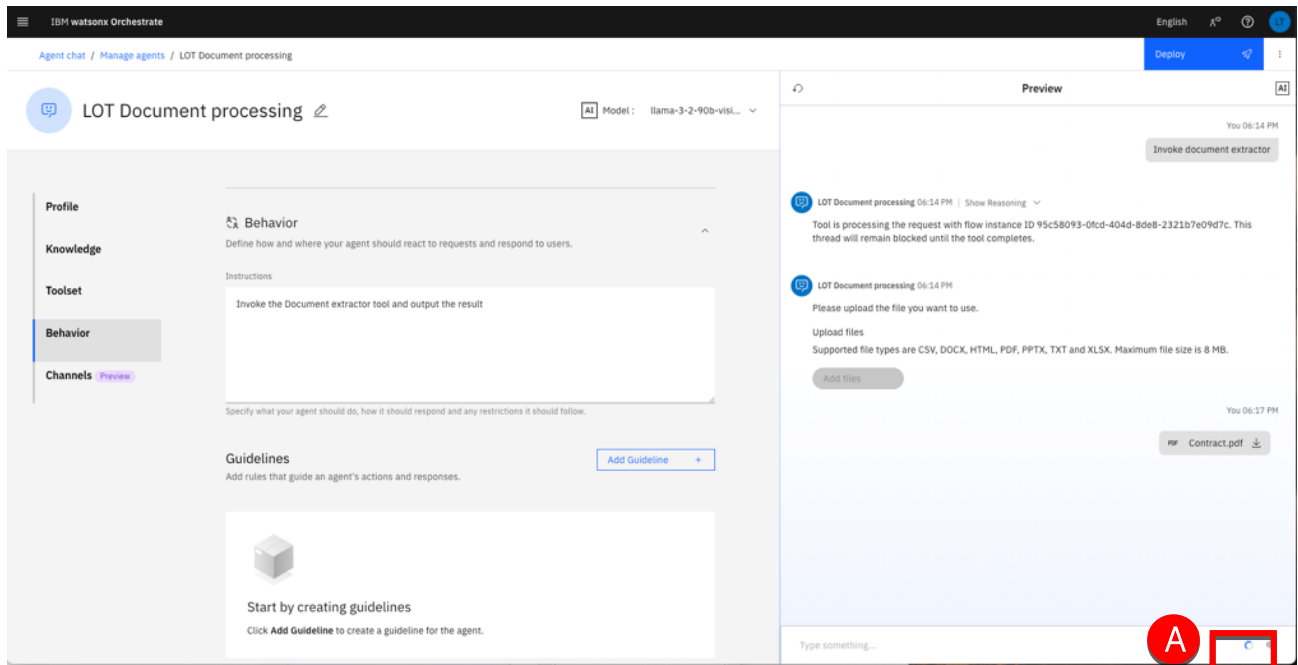
- Navigate to your **contract.pdf** location and select it (A). Click **Open** (B).



- Click the send button (A).

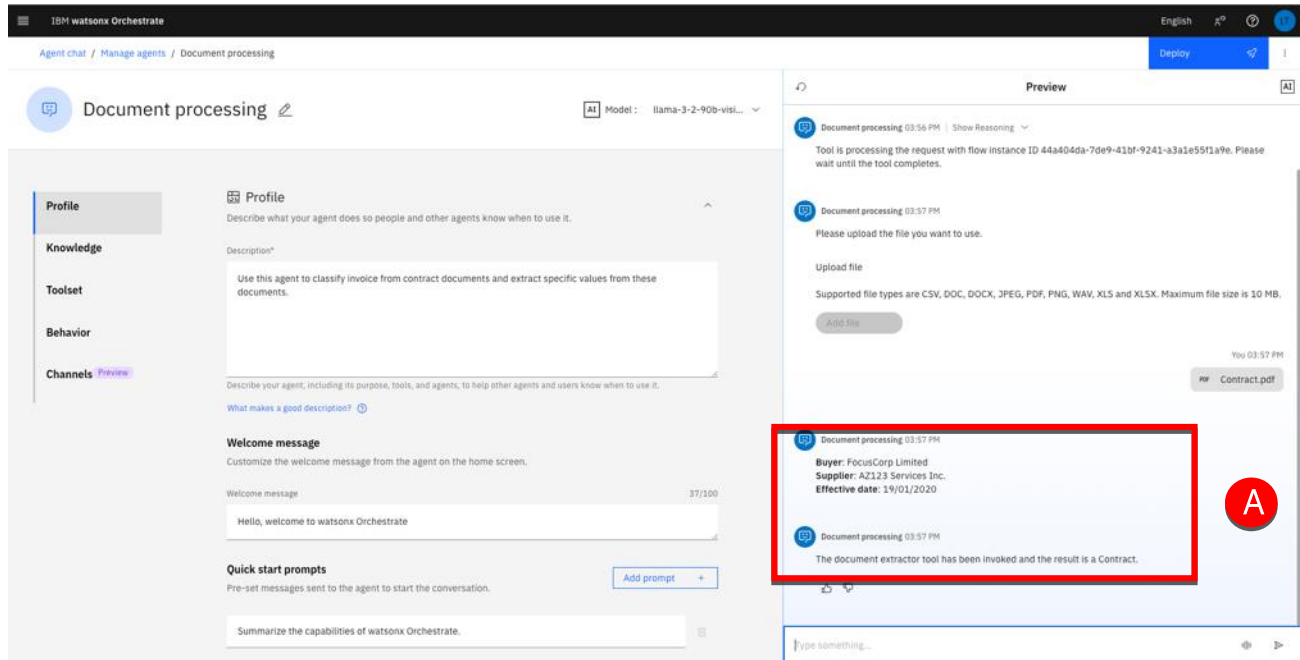


- Wait for the agent to respond (A). This will take a few seconds.



After uploading your document, the agent will automatically execute the predefined workflow steps in the background. It will extract relevant data, focusing on the fields it has been trained on, and generate a response with the extracted information. Additionally, it will provide insights into the document classification.

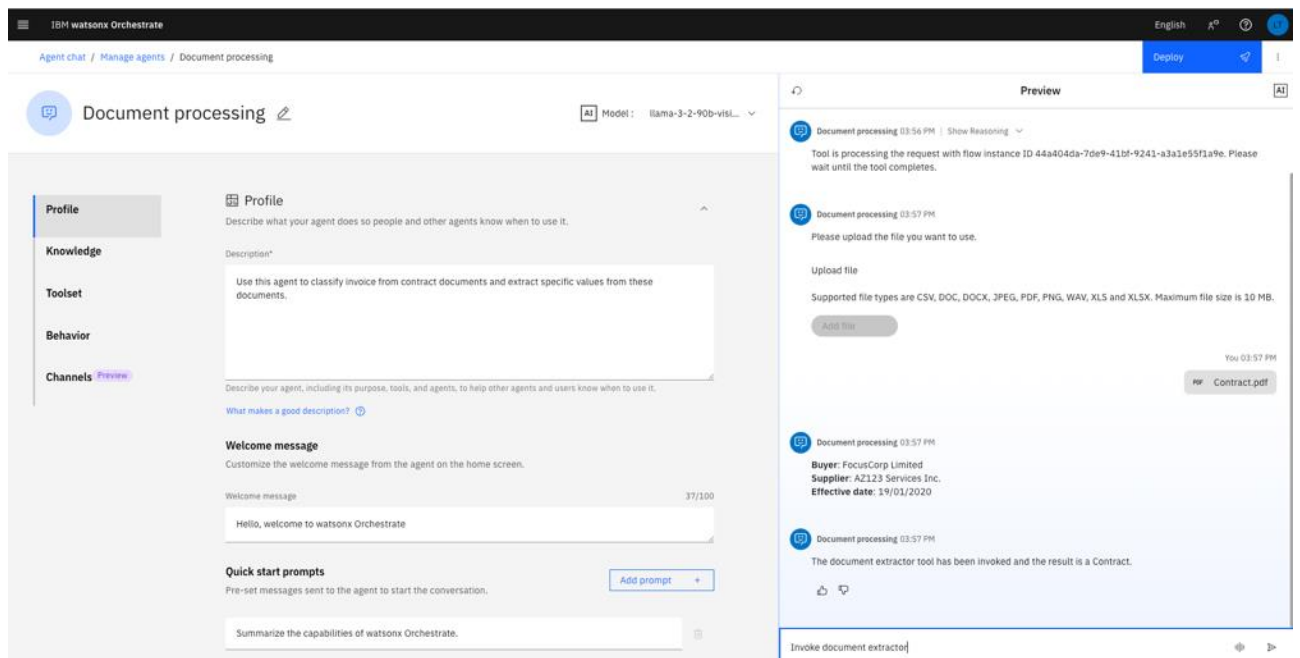
- After a few seconds the agent will reply with the extracted data and the document type (A).



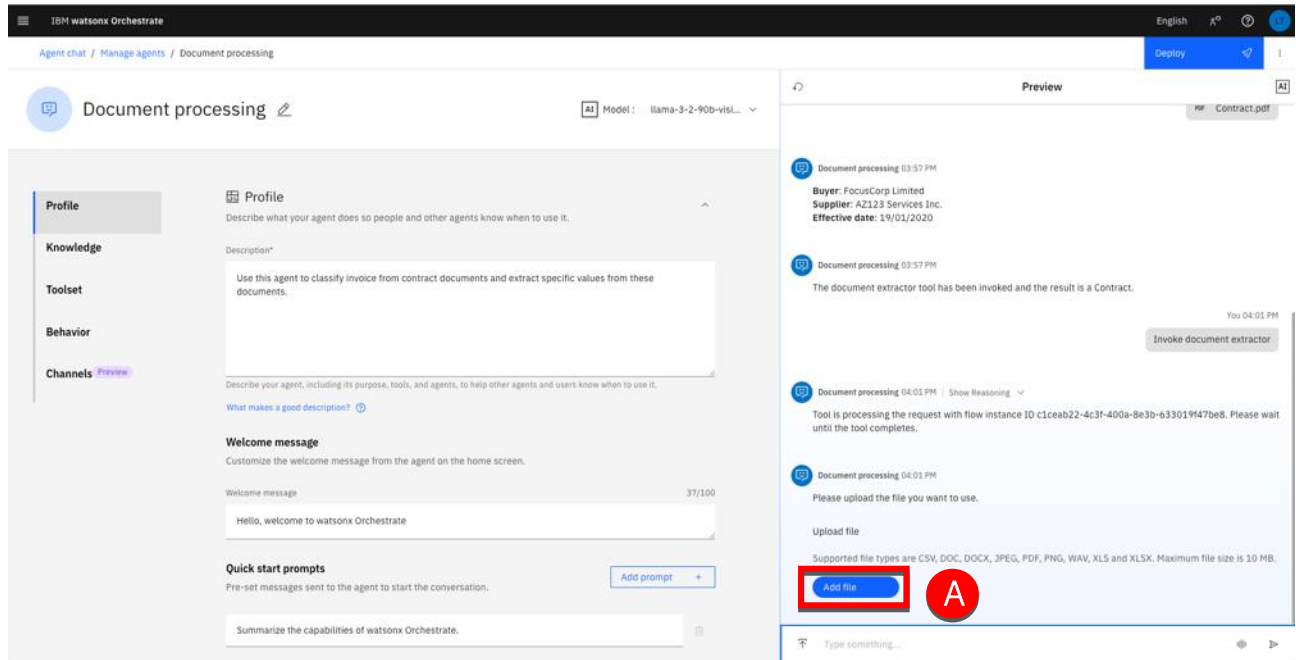
Test with possible extraction validation

Run a more complex test that may require user validation.

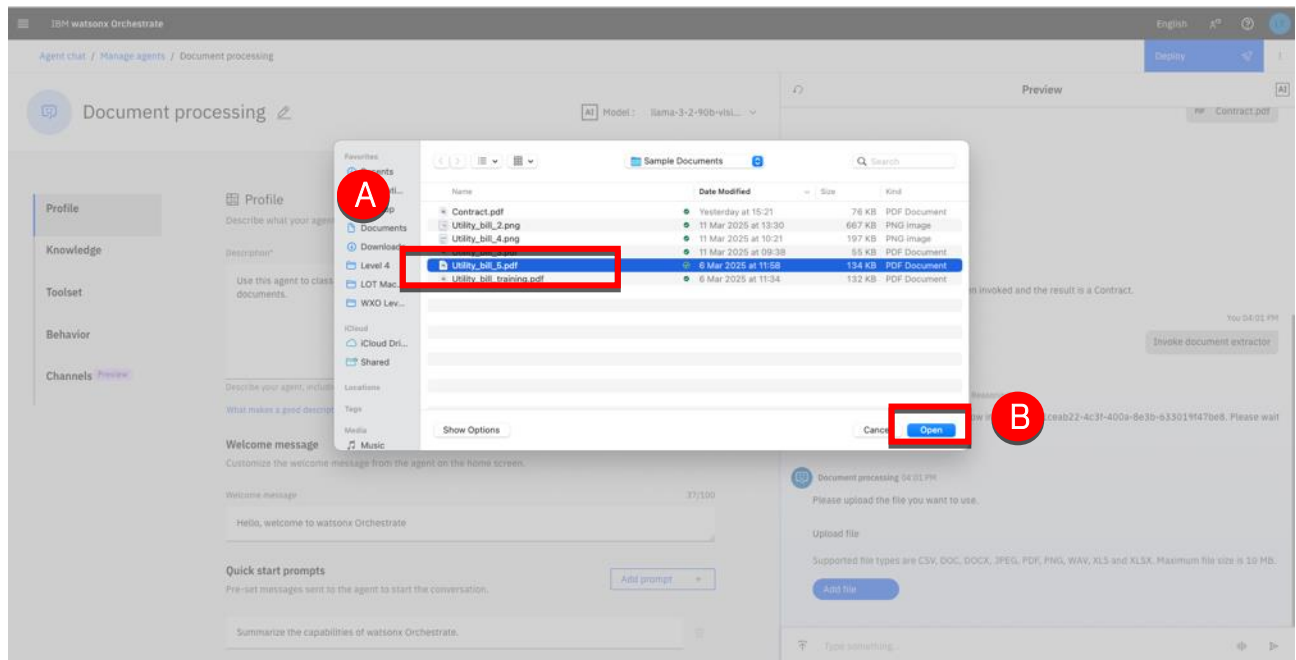
- Type 'Invoke document extractor' (A) in the Preview instructions area.



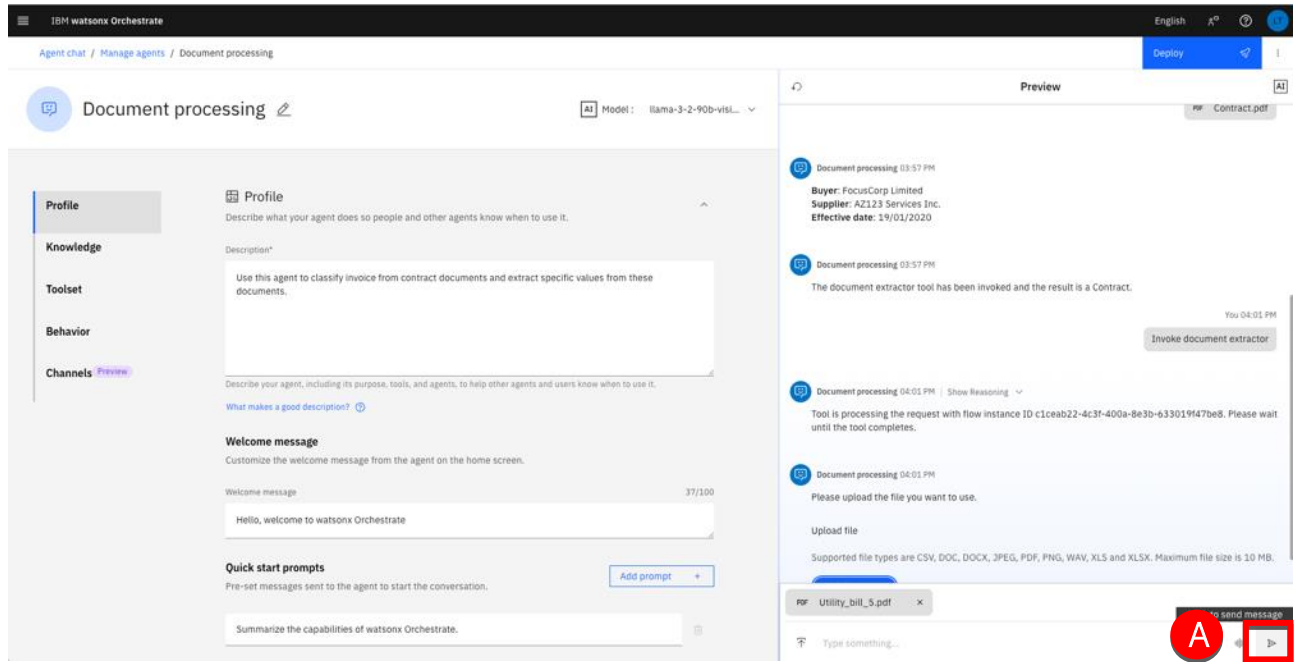
2. Click **Add file** (A).



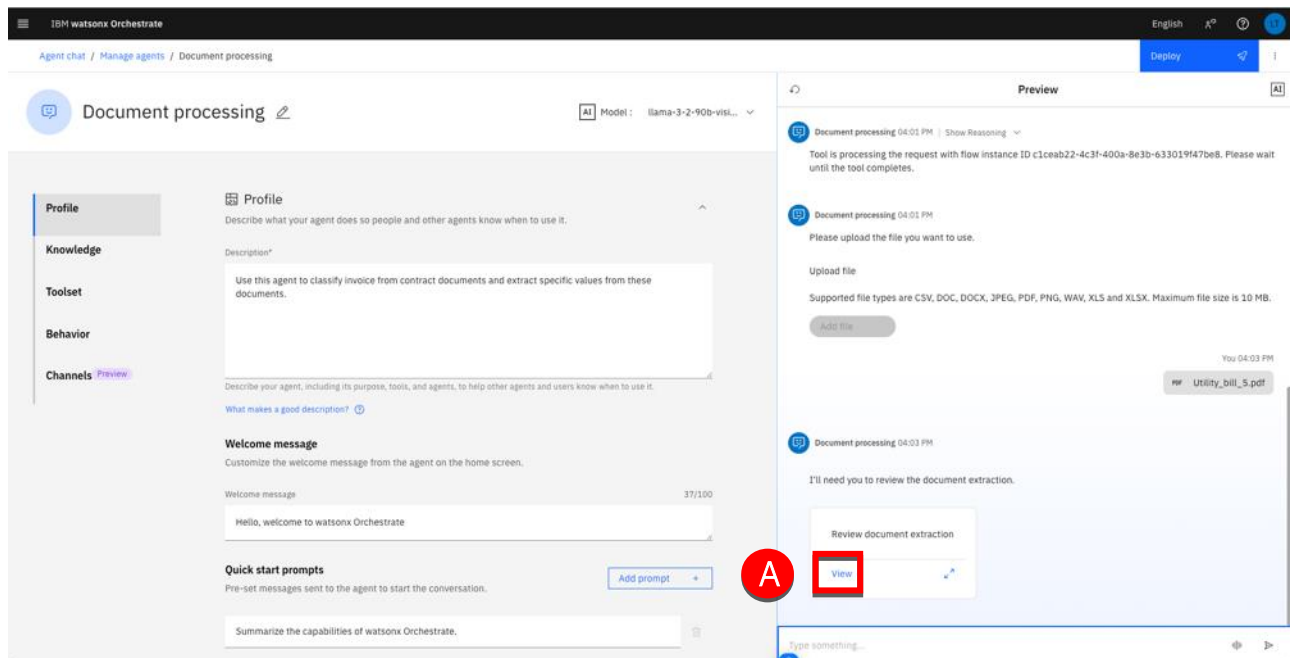
3. Navigate to your **Utility_bill_5.pdf** location and select it (A). Click **Open** (B).



4. Click the send message icon (A).



5. For this document the confidence score is too low for automatic extraction. The agent will ask you to validate some of the extracted fields. Click **View** (A).



6. In this example, 2 fields require a review:

- Customer address (A)
- Customer number (B)

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Document processing' agent configuration is visible, including a profile description and a welcome message. On the right, the 'Preview' tab shows a 'Review document extraction' window. This window displays a sample document (a 'Monthly Invoice' from Constellation) and a table of extracted fields. Two fields are highlighted with red boxes and labeled A and B:

Field	Value	Confidence
Customer Address	145 Andover Street Norwich, CT 06350	Low confidence extraction, Confirm value.
Customer Number	6696913-3	Low confidence extraction, Confirm value.

The extraction looks good despite the low confidence score. Should you need to correct an extracted value, follow the following step:

- a. Click the value you need to correct (A).

This screenshot shows the same 'Review document extraction' window as the previous one. A red box labeled 'A' is now highlighting the 'Customer Address' field, indicating that the user has selected it for correction. The 'Customer Number' field remains highlighted with a red box labeled 'B'.

- b. Draw a rectangle around the value in the document image (A) to capture the value.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Document processing' configuration panel is visible, with the 'Profile' tab selected. The 'Description' field contains the text: 'Use this agent to classify invoice from contract documents and extract specific values from these documents.' The 'Welcome message' field contains: 'Hello, welcome to watsonx Orchestrate'. The 'Quick start prompts' field contains: 'Summarize the capabilities of watsonx Orchestrate.' On the right, the 'Preview' panel shows a document titled 'Constellation Monthly Invoice'. A red circle labeled 'A' highlights a value in the document, which is '145 Andover Street'. The 'Needs review (2)' panel on the far right shows the extracted information, including 'Customer Address' (145 Andover Street, Norwich, CT 06350) and 'Customer Number' (6696913-3). The 'Submit' button is highlighted in blue.

Note: You can also directly type in the value in the field.

- c. Click **Submit** (A) when all the fields have been reviewed.

This screenshot is identical to the one above, showing the IBM watsonx Orchestrate interface with the 'Document processing' configuration and the 'Preview' panel. The 'Needs review (2)' panel shows the extracted information. The 'Submit' button is highlighted with a red circle labeled 'A', indicating the next step in the process.

d. Wait for the process to complete and display the final result (A).

The screenshot displays the IBM watsonx Orchestrate interface. The top navigation bar includes the IBM logo, the text 'watsonx Orchestrate', and a 'Deploy' button. Below the navigation bar, the breadcrumb trail reads 'Agent chat / Manage agents / Document processing'. The main content area is divided into two panels. The left panel, titled 'Document processing', contains a sidebar with 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels'. The 'Profile' section is active, showing a 'Description' field with the text 'Use this agent to classify invoice from contract documents and extract specific values from these documents.' and a 'Welcome message' field with the text 'Hello, welcome to watsonx Orchestrate'. The right panel, titled 'Preview', shows a chat conversation. The first message is a system message: 'Document processing 04:15 PM' followed by extracted data: 'Customer name: LOT Corporation', 'Customer number: 6696913-3', 'Customer address: 145 Andover Street Norwich, CT 06350', 'Amount due: \$13.07', 'Due date: 10/01/2021', 'Monthly usage: 98 kWh', and 'Year to year usage: 0.0%'. The second message is a user message: 'The document document extractor extractor tool tool has has been invoked invoked and and the the result result is is an an Invoice Invoice..'. A red circle with the letter 'A' is placed over the second message in the chat preview.

Note: At the time we are writing this guide, a reported defect causes the result to be displayed twice when performing manual validation.

